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Well modeling

This chapter describes well modeling in Eclipse.

- The Well inflow performance chapter describes the equations governing flows between the reservoir and the wellbore, including effects such as flow-dependent skin factors and special inflow equations such as generalized pseudo-pressure.
- The Well modeling facilities chapter details the wide range of options for controlling wells in Eclipse.
- The Multisegment wells chapter explains this advanced well modeling option. Multisegment wells have many advantages, such as more accurate modeling of wellbore friction effects and the ability to model well engineering from a simple annulus and tubing to multi-lateral horizontal wells with inflow control devices.
- The Wellbore friction option is a pre-cursor of the multisegment well model; it is included for legacy users.

Well inflow performance

x	Eclipse 100
x	Eclipse 300

This chapter outlines the relationships that Eclipse uses to calculate well inflow performance.

We define the flow path between the well bore and a single reservoir grid block as a “connection”. The flow rate of a phase p (oil, water or gas) across a connection j is given by the inflow performance relationship.

Inflow performance relationship in Eclipse 100

In Eclipse 100 the inflow performance relationship is written in terms of the volumetric production rate of each phase at stock tank conditions:

$$q_{p,j} = T_{wj} M_{p,j} \left(P_j - P_w - H_{wj} \right) \qquad \text{Eq. 8.1}$$

where

$q_{p,j}$ is the volumetric flow rate of phase p in connection j at stock tank conditions. The flow is taken as positive from the formation into the well, and negative from the well into the formation.

T_{wj} is the connection transmissibility factor, defined below.

$M_{p,j}$ is the phase mobility at the connection.

P_j is the nodal pressure in the grid block containing the connection.

P_w is the bottom hole pressure of the well.

H_{wj} is the well bore pressure head between the connection and the well’s bottom hole datum depth. $P_w + H_{wj}$ is thus the pressure in the well at the connection j, which we call the “connection pressure”.

The calculation of the terms in this relationship is described in the following sections of this chapter.

Inflow performance relationship in Eclipse 300

In Eclipse 300 the inflow performance relationship is written in terms of the volumetric production rate of each phase at reservoir conditions:

$$\hat{q}_{p,j} = T_{w,j} \left(\frac{K_{rp,j}}{\mu_{p,j}} \right) \left(P_j - P_w - H_{wj} \right) \qquad \text{Eq. 8.2}$$

where

$\hat{q}_{p,j}$ is the volumetric flow rate of phase p in connection j at reservoir conditions.

$\mu_{p,j}$ is the phase viscosity at the connection.

$k_{rp,j}$ is the phase relative permeability at the connection.

The surface volumetric rate is

$$q_{p,j} = \hat{q}_{p,j} b_{p,j}$$ Eq. 8.3

where $b_{p,j} = \dfrac{1}{B_{p,j}}$ and $B_{p,j}$ is the phase formation volume factor.

The molar rate is obtained by multiplying by the phase reservoir molar densities, $b^m_{o,j}$. The molar rate of component i is then:

$$q_{i,j} = x_{i,j} b^m_{o,j} q_{o,j} + y_{i,j} b^m_{g,j} q_{g,j}$$ Eq. 8.4

where:

q_i is the component i molar rate

x_i is the liquid mole fractions

y_i is the vapor mole fractions

The connection transmissibility factor

For brevity, we refer to this quantity as the “connection factor”. It depends on the geometry of the connecting grid block, the well bore radius, and the rock permeability. Its value may be specified directly by the engineer, or it can be calculated by the program using one of the formulae below.

Cartesian grids

In a Cartesian grid, Eclipse uses the relationship

$$T_{wj} = \frac{c\theta Kh}{\ln(r_o/r_w) + S}$$ Eq. 8.5

where

c is a unit conversion factor (0.001127 in field units, 0.008527 in metric units, 3.6 in lab units).

θ is the angle of the segment connecting with the well, in radians. In a Cartesian grid its value is 6.2832 ($= 2\pi$), as the connection is assumed to be in the center of the grid block. For wells located on an edge (or a corner) of a Cartesian grid, for example in a sector model or symmetry element, use keyword `WPIMULT` after `COMPDAT` to scale the resulting connection factors by 0.5 (or 0.25).

Kh is the effective permeability times net thickness of the connection. For a vertical well the permeability used here is the geometric mean of the x- and y-direction permeabilities, $K = \left(K_x K_y\right)^{1/2}$.

r_o is the “pressure equivalent radius” of the grid block, defined below.

r_w is the well bore radius.

S is the skin factor.

The pressure equivalent radius of the grid block is defined as the distance from the well at which the local pressure is equal to the nodal average pressure of the block. In a Cartesian grid we use Peaceman’s formula, which is applicable to rectangular grid blocks in which the permeability may be anisotropic. The well is assumed to penetrate the full thickness of the block, through its center, perpendicularly to two of its faces.

$$r_o = 0.28 \frac{\left[D_x^2 \left(\frac{K_y}{K_x} \right)^{1/2} + D_y^2 \left(\frac{K_x}{K_y} \right)^{1/2} \right]^{1/2}}{\left(\frac{K_y}{K_x} \right)^{1/4} + \left(\frac{K_x}{K_y} \right)^{1/4}} \qquad \text{Eq. 8.6}$$

where

D_x and D_y are the x- and y- dimensions of the grid block

K_x and K_y are the x- and y- direction permeabilities.

Equations 8.5 and 8.6 are intended for vertical wells. Horizontal wells may penetrate the block in either the x- or y- direction, and the appropriate components of permeability and block dimensions are substituted in these equations. For a well penetrating in the x-direction, for example, the quantities K_y, K_z, D_y, D_z will be used in equation 8.6, and in equation 8.5 the expression $Kh = D_x \left(K_y K_z \right)^{1/2}$ will be used.

Eclipse 300 requires the `HWELLS` keyword to be entered in the `RUNSPEC` section if there are well connections penetrating in a direction other than z; this is not necessary in Eclipse 100.

A net-to-gross ratio applied to a grid block containing a horizontal well will not affect Kh, but will still affect r_o through the reduction in the value of D_z.

Radial grids

In radial geometry, the formula used to calculate T_{wj} depends on the location of the connecting grid block.

When the connection is situated in one of the innermost grid blocks (i = 1), it is assumed to lie on the inner boundary of the block. The connection factor is

$$T_{wj} = \frac{c \theta K h}{\frac{r_2^2}{r_2^2 - r_w^2} \ln\left(r_2 / r_w \right) - 0.5 + S} \qquad \text{Eq. 8.7}$$

where

θ is the segment angle of the grid block in radians,

r_2 is the block's outer radius.

If a connection is situated in one of the outermost blocks (i = `NDIVIX`), it is assumed to occupy the outer boundary of the block for the purpose of applying boundary conditions. The corresponding connection factor is

$$T_{wj} = \frac{c \theta K h}{\frac{r_1^2}{r_2^2 - r_1^2} \ln\left(r_1 / r_2 \right) + 0.5 + S} \qquad \text{Eq. 8.8}$$

where r_1 and r_2 are the inner and outer radii of the block.

This enables you to use a BHP-controlled well to apply constant pressure outer boundary conditions. If on the other hand you wish to model a real well in an outer block, the calculated connection factor will be much too large. In this case, you should either supply an appropriate connection factor value in item 8 of the `COMPDAT` keyword, or add an extra ring of inactive cells so that the well is no longer in an outer block.

If a well is completed in any other block (1 < i < NDIVIX), its connection factor is calculated as if it were in a Cartesian grid block with $D_x = r_2 - r_1$ and $D_y = \theta(r_1 + r_2)/2$

Fractured completions

Eclipse 300

When using one of the built-in formulae above to calculate the connection factor, the engineer can supply a constant value for the skin factor S, to represent the effects of, for example, formation damage or stimulation, partial penetration and well bore deviation. However, Eclipse 300 provides an additional formula to calculate the connection factor for cases where a vertical fracture penetrates the full width of the cell in the x- or y- direction.

The fractured completion formula is selected by entering `FX` or `FY` in item 13 of `COMPDAT`. The `FWELLS` keyword must be entered in the `RUNSPEC` section if any wells use this option.

In the case of FX, the completion connection factor is

$$T_{wj} = \frac{8cK_y D_x D_z}{D_y + SD_x/\pi} \tag{Eq. 8.9}$$

In the case of FY, the completion connection factor is

$$T_{wj} = \frac{8cK_x D_y D_z}{D_x + SD_y/\pi} \tag{Eq. 8.10}$$

In both cases we assume uniform flow into the fracture at the center of the cell. The mean distance of flow is $D_y/4$ or $D_x/4$, over an area of $2D_x D_z$ or $2D_y D_z$.

Rock crushing modifications

If a transmissibility multiplier is specified with the `ROCKTAB` or `ROCKTABH` keywords, the well connection transmissibility factor, grid block transmissibilities and NNCs are modified accordingly.

Eclipse 300

Eclipse 300 also updates the generalized mobility terms for grid cells and well connections. Similar functionality, but based on the corresponding changes in porosity, is also available by specifying the keywords `ROCKTRMX`, `ROCKTRMY` and `ROCKTRMZ`.

If the `ROCKTAB` or `ROCKTABH` table contains directional transmissibility multipliers, as specified in keyword `RKTRMDIR`, the well connection transmissibility factor is modified according to the formula

$$\widetilde{T}_{wj} = T_{wj}\left(m_x m_y\right)^{1/2} \tag{Eq. 8.11}$$

where

$\widetilde{T}_{wj}$ is the pressure-adjusted connection factor,

m_x, m_y are the transmissibility multipliers interpolated from the rock table at the grid block pressure P_j. The formula assumes a vertically completed well within the grid block j. For other well orientations, permute the directions accordingly.

In Cartesian grids, this transmissibility update is in accordance with modifying the effective Kh and with keeping the Peaceman's formula term $\ln(r_o/r_w)$ unaffected.

The flow-dependent skin factor

When Eclipse uses one of the built-in formulae above to calculate the connection factor, you can supply a constant value for the skin factor S, to represent the effects of, for example, formation damage or stimulation, partial penetration and well bore deviation. In addition, a flow-dependent contribution can be included to model non-Darcy effects in gas flow near the well,

$$S \rightarrow S + D\left|q_{fg}\right| \quad \text{Eq. 8.12}$$

q_{fg} represents the flow rate of free gas through the connection. The D-factor is usually obtained from measurements on the well, and is multiplied by the well's free gas flow rate Q_{fg} to give the non-Darcy contribution to the skin factor. But in reality the effect of non-Darcy flow at a connection depends on the free gas flow rate through the connection itself rather than the well as a whole. Accordingly Eclipse calculates the non-Darcy skin from the connection flows, as in equation 8.12, rather than the well flow. If the well has more than one connection, the D-factor must be scaled to apply to the connection flows instead of the well flow. The scaling is performed by the program; the engineer specifies the measured D-factor for the well (with keyword WDFAC) and the program will scale it for each open connection. The scaling process aims to give each connection initially the same non-Darcy skin, by setting the D-factors in inverse proportion to the gas flow rate. The calculation assumes that initially the free gas mobility and the drawdown are the same in each connection, so that

$$D_j = D_w \frac{\underset{i}{\Sigma} T_{wi}}{T_{wj}} \quad \text{Eq. 8.13}$$

where

D_j is the scaled D-factor for the connection j,

D_w is the measured D-factor for the well, and the

$\underset{i}{\Sigma} T_{wi}$ term denotes the sum of the connection transmissibility factors for all the connections in the well that are open at the time the D-factor is specified.

If you do not wish the D-factors to be scaled, you may enter them **with a negative sign** in item 12 of the COMPDAT keyword. The program will simply reverse the sign without applying scaling. (Entering a positive value in this item was a former method of specifying the D-factor for the whole well in Eclipse 100, before the WDFAC keyword was implemented; so positive values are scaled as if they were entered with WDFAC.)

Eclipse 300

D-factors may also be calculated by the simulator for each connection in a well using an expression defined in the WDFACCOR keyword. This expression is based on a correlation for the coefficient of inertial resistance, β, in terms of the permeability and porosity of the connected grid blocks. Please refer to the keyword for more details.

The program calculates the connection factor without the contribution from the D-factor, and the D-factor is transformed into a multiplying factor D_m to be applied to the free gas mobility

$$D_m = \frac{G}{G + D\left|q_{fg}\right|} \quad \text{Eq. 8.14}$$

where G is a geometric factor:

For the Cartesian case:

$$G = \ln\left(\frac{r_o}{r_w}\right) + S \qquad \text{Eq. 8.15}$$

For the inner cell radial case:

$$G = \frac{r_2^2}{r_2^2 - r_w^2}\ln\left(\frac{r_2}{r_w}\right) - 0.5 + S \qquad \text{Eq. 8.16}$$

For the outer cell radial case:

$$G = \frac{r_1^2}{r_2^2 - r_1^2}\ln\left(\frac{r_1}{r_2}\right) + 0.5 + S \qquad \text{Eq. 8.17}$$

For the fractured completion cases:

- Fracture in x-direction:

$$G = \pi\frac{\Delta_y}{\Delta_x} + S \qquad \text{Eq. 8.18}$$

- Fracture in y-direction:

$$G = \pi\frac{\Delta_x}{\Delta_y} + S \qquad \text{Eq. 8.19}$$

Since the multiplying factor itself depends upon the free gas flow rate, Eclipse uses a calculation that provides a completely self-consistent pair of values for these two quantities. The modified inflow performance relationship for the well connection can be written as

$$q_{fg} = T_w D_m M_{fg} \Delta P \qquad \text{Eq. 8.20}$$

where ΔP is the drawdown into the connection.

Substituting equation 8.14 for D_m gives the quadratic equation

$$T_w M_{fg} \Delta P = q_{fg} + \left(\frac{D}{G}\right) q_{fg} \left|q_{fg}\right| \qquad \text{Eq. 8.21}$$

that may readily be solved for $\left|q_{fg}\right|$, thus enabling D_m to be obtained.

Eclipse 300

Note that Eclipse 300 uses the reservoir gas phase flow rate converted to surface volume, rather than the stock tank gas rate for the completion. The latter would involve a separator flash within the above calculations, and be dubious physically - the flow into the well should not depend on the subsequent separation process. At near critical conditions the oil phase may be sufficiently close to the gas phase that a D-factor should also be applied to oil inflow - this is not currently done in Eclipse 300 as oil does not generally show non-Darcy flow effects.

The phase mobilities

The term $M_{p,j}$ in equation 8.1 represents the mobility of the phase p at the connection. In producing connections, where the flow is from the formation into the well bore, the mobility depends on the conditions in the grid block containing the connection. The mobility of a free phase (free oil, water, or free gas) is given by

$$M_{fp,j} = k_{p,j}\,\lambda_{p,j} \qquad \text{Eq. 8.22}$$

where

$k_{p,j}$ is the relative permeability of the phase,

$\lambda_{p,j}$ is defined by

$$\lambda_{p,j} = \frac{1}{B_{p,j}\,\mu_{p,j}} \quad \text{Eq. 8.23}$$

$\mu_{p,j}$ is the phase viscosity, and

$B_{p,j}$ is the phase formation volume factor.

The quantities in equation 8.22 are determined from the pressure and saturations in the grid block. However, the relative permeability can be obtained using a separate saturation functions table that the engineer can supply to take account of the distribution of fluids and the location of the perforations within the grid block.

Eclipse 100

The total gas and total oil mobilities are obtained by including the dissolved gas and vaporized oil content of the phases

$$\begin{aligned} M_{o,j} &= M_{fo,j} + R_{v,j} M_{fg,j} \\ M_{g,j} &= M_{fg,j} + R_{s,j} M_{fo,j} \end{aligned} \quad \text{Eq. 8.24}$$

If a flow-dependent skin factor is used, the free gas mobility is multiplied by the factor D_m defined in equation 8.14.

$$M_{fg,j} \rightarrow D_m M_{fg,j} \quad \text{Eq. 8.25}$$

Special inflow equations

Eclipse 100

There is an option to use one of three special inflow equations to provide a more accurate model of the flow of gas into the well: the Russell-Goodrich equation, the dry gas pseudo-pressure equation and the generalized pseudo-pressure method. These are all methods of taking into account the pressure-dependence of $\lambda_{g,j}$ between the grid block pressure and the well bore pressure. The generalized pseudo-pressure equation alters both the gas and oil mobilities, and takes account also of the effects of condensate dropout. Item 8 of keyword `WELSPECS` selects the method to use for each well.

Russell-Goodrich

Eclipse 100

The Russell-Goodrich equation (*Russell et al, 1966*) shows the flow rate to vary as the difference between the squares of the grid block and well bore pressures. But it can be written in the form of equation 8.1 if the term $\lambda_{g,j}$ defined in 8.23 is evaluated at the average of the grid block pressure and the well bore pressure at the connection, that is $\left(P_j + P_w + H_{wj}\right)/2$. The relative permeability is evaluated at the grid block conditions.

Dry gas pseudo-pressure

Eclipse 100

In the pseudo-pressure equation, the gas flow rate varies as the difference between the “real gas pseudo-pressure” values at the grid block and well bore pressures. This also can be written in the form of equation 8.1 by setting $\lambda_{g,j}$ equal to its integrated average value between the pressures P_j and $P_w + H_{wj}$. This representation avoids the need to calculate tables of pseudo-pressure vs. pressure. The integration is performed by applying the trapezium rule between each intermediate pair of nodal pressure values in the gas PVT table. This essentially treats λ_g as a linear function of P between adjacent nodal pressure values. The relative permeability is evaluated at the grid block conditions.

At low values of the drawdown the Russell-Goodrich and pseudo-pressure equations will give approximately the same results. When the drawdown is large enough to span a significantly non-linear segment of the graph of λ_g versus P, the pseudo-pressure equation will give more accurate results than the Russell-Goodrich equation.

This option is not available in gas condensate runs. In these cases the generalized pseudo-pressure method may be used instead.

Generalized pseudo-pressure in Eclipse 100

The generalized pseudo-pressure method is intended for use by gas condensate producers. It provides a means of taking account of condensate dropout, as well as compressibility, in the calculation of the mobility integral. It is based on the method described by *Whitson and Fevang (1997)*.

At the beginning of each timestep the integral of the total oil and gas mobility is evaluated between the grid block pressure and the well bore pressure at the connection. This is compared with the total oil and gas mobility at grid block conditions multiplied by the drawdown, and the ratio of the two quantities is stored as a "blocking factor" for each grid block connection in the well,

$$\beta_j = \frac{\int_{(P_w + H_{wj})}^{P_j} (M_g + M_o) dp}{(M_{g,j} + M_{o,j})(P_j - (P_w + H_{wj}))} \qquad \text{Eq. 8.26}$$

The integrand is fundamentally a function of two independent variables: the pressure P and the gas saturation S_g. (The oil saturation is equal to $1-(S_g + S_w)$ and S_w is regarded as fixed at the grid block value.) However, S_g is eliminated as an independent variable, making it a function of P at pressures below the dew point by requiring that the local total mobility ratio should be the same as the total mobility ratio at grid block conditions:

$$M_o / M_g = M_{o,j} / M_{g,j} \qquad \text{Eq. 8.27}$$

This requirement assumes that within the grid block the flows are in steady state. Only the mobile oil is included when calculating this ratio. Immobile dropped-out oil is not included because the oil relative permeability will be zero until the oil saturation exceeds the critical oil saturation.

The integral in equation 8.26 is evaluated by applying the trapezium rule to a set of pressure values between the grid block and connection pressures. You can control the distribution of these pressure values, and hence the accuracy of the integration, using keyword `PICOND`. At each pressure value below the dew point, the gas saturation is determined by solving equation 8.27 using Newton's method.

The blocking factor is then used to multiply both the oil and gas mobilities in the inflow performance relationship, equation 8.1. Note that the free oil mobility is modified by this treatment, whereas the Russell-Goodrich and dry gas pseudo-pressure treatments leave this unchanged. At present this calculation is not performed in injecting connections; the blocking factor is set equal to 1.0. The value of the blocking factor can be written to the Summary file by using the `SUMMARY` section keyword `CDBF`.

The blocking factor for each connection is retained for the duration of the timestep, and recalculated at the beginning of each subsequent timestep. In case this degree of explicitness should cause oscillations, these may be damped by averaging the calculated blocking factor with its value at the previous timestep,

$$\beta_{\text{used}} = f \beta_{\text{new}} + (1-f) \beta_{\text{previous}} \qquad \text{Eq. 8.28}$$

where the weighting factor f is specified in the `PICOND` keyword.

Generalized pseudo-pressure in Eclipse 300

The generalized pseudo-pressure method of *Whitson and Fevang (1997)* is also available in Eclipse 300.

For the current version of the generalized pseudo-pressure (GPP) model, which is a revised and extended version of the previous model, the entire specification and control of the calculation can be handled by the `WPICOND` keyword. This provides an alternative to the use of the `WELSPECS`, `PSEUPRES` and `PICOND` keywords whilst additionally providing individual control over the calculation parameters for each well together with the facility to turn both on and off the generalized pseudo-pressure calculation arbitrarily during the course of the simulation. The `WELSPECS`, `PSEUPRES` and `PICOND` keywords can be still be used without any changes to their previous usage interpretations (except that items 7 and 8 of the `PICOND` keyword are interpreted as absolute rather than fractional changes unless item 330 of the `OPTIONS3` keyword is set to 2) and can be interspersed with the `WPICOND` keyword. The `PSEUPRES` keyword is no longer required to be specified early in the simulation and will simply have the effect of turning the generalized pseudo-pressure option on for all wells in the simulation. The `PICOND` keyword will have the effect of specifying the calculation control parameters for all wells currently in the simulation and updating the default calculation control parameters for all wells defined subsequently. The `PICOND` keyword does not select or deselect the generalized pseudo-pressure calculation.

For the generalized pseudo-pressure model prior to the 2015.1 version, the generalized pseudo-pressure option can be requested for all wells with the `PSEUPRES` keyword, which should be entered early in the `SCHEDULE` section before any well operations or time stepping keywords are specified. Alternatively this option can be activated for individual wells by entering GPP in item 8 of keyword `WELSPECS`. Calculation controls are specified for all wells via the keyword `PICOND`. The pre-2015.1 version may be selected by setting item 330 of the `OPTIONS3` keyword to 1. This version does not support the `WPICOND` keyword.

The calculation and implementation of the generalized pseudo-pressure method is described below.

Consider the flow of phase j = (oil, gas), in a radial homogenous media of height h and permeability K. Darcy's Law describes the velocity of the fluid at radius r in the equation:

$$v_j = \frac{Q_j}{2\pi rh} = \frac{KK_{rj}}{\mu_j}\frac{dP}{dr} \qquad \text{Eq. 8.29}$$

where

Q_j	is the in-situ volumetric flow rate,
K_{rj} and μ_j	are the phase relative permeability and viscosity and
dP/dr	is the pressure gradient.

The phase volumetric flow rate Q_j is related to phase molar flow rate n_j by:

$$Q_j = n_j V_{mj} \qquad \text{Eq. 8.30}$$

where V_{mj} is phase molar volume.

Combining equations 8.29 and 8.30 and rearranging gives:

$$n_j\frac{dr}{r} = 2\pi Kh\frac{K_{rj}b_j}{\mu_j}dp \qquad \text{Eq. 8.31}$$

where $b_j = 1/V_{mj}$, the phase molar density.

For component, i, the component phase molar rate will be:

$$n_{ij}\frac{dr}{r} = 2\pi Khx_{ij}\frac{K_{rj}b_j}{\mu_j}dp \qquad \text{Eq. 8.32}$$

where x_{ij} is the mole fraction of the i[th] component in the j[th] phase.

Integrating equation 8.32 between the Peaceman pressure equivalent radius r_B and the well radius r_W gives:

$$n_{ij}\int_{r_W}^{r_B}\frac{dr}{r} = 2\pi Kh\int_{p_W}^{p_B}x_{ij}\frac{K_{rj}b_j}{\mu_j}dp \qquad \text{Eq. 8.33}$$

Summing component phase molar rates in the oil and gas phases for component i:

$$\left(n_{io} + n_{ig}\right)\left[\ln\left(\frac{r_B}{r_W}\right) + S\right] = 2\pi Kh\int_{p_w}^{P_B}\left[x_i\frac{K_{ro}b_o}{\mu_o} + y_i\frac{K_{rg}b_g}{\mu_g}\right]dp \qquad \text{Eq. 8.34}$$

or:

$$n_i = T\int_{p_W}^{p_B}M_i(p)dp \qquad \text{Eq. 8.35}$$

where the well connection factor, T, including the skin factor, S, is given by:

$$T = \frac{2\pi Kh}{\ln\left(\frac{r_B}{r_W}\right) + S} \qquad \text{Eq. 8.36}$$

and the Component Generalized Molar Mobility (CGMM) is given by:

$$M_i = x_i\frac{K_{ro}b_o}{\mu_o} + y_i\frac{K_{rg}b_g}{\mu_g} \qquad \text{Eq. 8.37}$$

Normally, we do not consider the pressure dependency in equation 8.35. Instead, we compute the CGMM at the block pressure:

$$\int_{p_W}^{p_B}M_i(p)dp \rightarrow M_i(p_B)[p_B - p_W] \qquad \text{Eq. 8.38}$$

that we know can be a poor approximation for a gas condensate where, although the block pressure may be well above the dew point and hence the gas relative permeability is high, the well pressure may be well below the dew point in which case the gas relative permeability can be substantially reduced. In this case, we want perform the integral in equation 8.34.

To prevent accumulation of moles in the completion cell, the ratio of CGMM for component i to the Total Generalized Molar Mobility (TGMM) must be a constant:

$$z_{pi} = \frac{M_i}{M_T} \qquad \text{Eq. 8.39}$$

where:

$$M_T = \sum_i M_i \qquad \text{Eq. 8.40}$$

Therefore, equation 8.35 can be written:

$$n_i = Tz_{pi}\int_{p_W}^{p_B}M_T(p)dp \qquad \text{Eq. 8.41}$$

where:

$$M_T = \frac{K_{ro} b_o}{\mu_o} + \frac{K_{rg} b_g}{\mu_g} \qquad \text{Eq. 8.42}$$

If the block pressure p_B exceeds the dew point pressure, the production composition z_{pi} is simply the total hydrocarbon composition in the block. If the block pressure is less than the dew point pressure, the production composition is computed from CGMM for oil and gas.

When the Generalized Pseudo-Pressure (GPP) model is applied in Eclipse 300, rather than rewriting the appropriate parts of the well model, we simply modify the conventional well inflow equation through the addition of the component dimensionless *flow blocking* factor, F_{Bi}, to give:

$$n_i = T F_{Bi} M_i(p) [p_B - p_W] \qquad \text{Eq. 8.43}$$

where:

$$F_{Bi} = \frac{1}{M_i(p_B)} \frac{1}{(p_B - p_W)} z_{pi} \int_{p_W}^{p_B} M_T(p) dp \qquad \text{Eq. 8.44}$$

Because of equation 8.39, we can write:

$$M_i(p_B) = z_{pi} M_T(p_B) \qquad \text{Eq. 8.45}$$

therefore, we define the total dimensionless flow blocking factor, F_B, (DFBF)

$$F_B = \frac{1}{M_T(p_B)} \frac{1}{(p_B - p_W)} \int_{p_W}^{p_B} M_T(p) dp \qquad \text{Eq. 8.46}$$

and hence the modified inflow equation becomes:

$$n_i = T F_B M_i(p) [p_B - p_W] \qquad \text{Eq. 8.47}$$

Note: Item 121 of the `OPTIONS3` keyword can be used to impose an upper limit (greater than or equal to unity) on the value of the dimensionless blocking factor in order to prevent convergence issues in the simulation.

The final form of the GPP inflow equation, 8.47, differs only from the normal inflow equation by the additional multiplication factor, F_B, which we calculate from equation 8.46. The block pressure, p_B and the TGMM at block conditions, $M_T(p_B)$ is already known prior to entering the well routines. The integral in equation 8.46 appears to be a function of two variables:

$$Q(p_B, p_W) = \int_{p_W}^{p_B} M_T(p) dp \qquad \text{Eq. 8.48}$$

but in fact it can be constructed as the difference of two integrals, namely:

$$Q(p_B, p_W) = R(p_B) - R(p_W) = \int_{p_L}^{p_B} M_T\left(p\right) dp - \int_{p_L}^{p_W} M_T(p) dp \qquad \text{Eq. 8.49}$$

where p_L is some suitable *lower* or base pressure, such as zero or other user-specified minimum pressure.

The contributions of the oil and gas phases to the TGMM are clearly saturation dependent through the respective relative permeability functions, $\left(K_{ro}, K_{rg}\right)$. The requirement that the produced composition be constant is equivalent to saying that the produced Gas-Oil-Ratio (GOR) or the produced gas-oil molar ratio be constant. Therefore:

$$\frac{M_o(S_o)}{M_g(S_g)} = \frac{L}{V} \quad \text{Eq. 8.50}$$

where

L and V are the total liquid and vapor moles at the pressure $p_L \le p \le p_B$ such that $L + V = 1$ and the gas saturation $S_g = 1 - S_w - S_o$

where (S_w, S_o) are the water and oil saturations, respectively.

The phase molar mobility, M_j is simply:

$$M_j = \frac{K_{rj} b_j}{\mu_j} \quad \text{Eq. 8.51}$$

Equation 8.50 is a simple non-linear equation in one variable, the oil saturation. The single integral in equation 8.49 is constructed using the trapezium rule, following the suggestions made by *Whitson and Fevang (1997)*. In practice, the integral is constructed at the beginning of each time step in the form of a table of pressure versus total generalized molar mobilities but only when the grid block conditions have changed significantly. Control over the number of pressure nodes in this table together with the pressure spacing and extent can be specified via either the WPICOND or PICOND keywords.

The table of pressure versus total generalized molar mobilities is constructed for each well connection over a pressure range which is defined by the lower and upper pressure bounds specified by item 7 and item 8 respectively of the WPICOND keyword (item 5 and item 6 of the PICOND keyword). These bounds are defined in terms of multiplicative factors of the current pressure of the grid block containing the connection. The table pressures should typically span the range from zero to the highest point above the current grid block pressure for which an instantaneous pressure may be required when finding a future well solution at the prevailing reservoir conditions such that the table may be potentially reused at a future time without recalculation. The lower pressure limit can be increased up to a maximum value of 0.95 p_B, but is constrained to the minimum of the specified limit and the dew-point pressure. If the lower limit is too high, this may impair the accuracy of the integrals which can be constructed and hence limit the accuracy of the calculated blocking factor, in particular, if the total generalized molar mobility is of significant value at pressures below this lower limit. Similarly, the upper pressure limit can be specified to be as low as 1.05 p_B; however, this may not be sufficient to encompass the range of grid block pressures involved at a future time. If, when using the table to calculate the blocking factor, the completion or grid block pressures are outside the pressure range of the table a warning message is issued. However, this will potentially result in inaccuracies in the calculated blocking factor.

The table is constructed with two sets of pressure steps; the sizes of the steps below and above the dew point are specified by item 3 and item 4 of the WPICOND keyword (item 1 and item 2 of the PICOND keyword). The rate of change of total generalized molar mobility with pressure is typically greatest in the region close to the dew-point pressure. Ideally, therefore, smaller pressure steps should be selected in the vicinity of the dew-point pressure. This may be accomplished via the adaptive ordinate control using item 13 and item 14 of the WPICOND keyword (item 11 and item 12 of the PICOND keyword). This simply involves the automatic addition of extra pressure steps in the regions between the last pressure step before the dew-point pressure and the first pressure step after the dew-point pressure, such that these pressure step sizes become smaller as they approach the dew point (The intervals are successively halved until the minimum value specified by item 14 of the WPICOND keyword (item 12 of the PICOND keyword) is attained.)

The table of pressure versus total generalized molar mobilities for each connection for each well (for which the generalized pseudo-pressure option has been specified) is recalculated at the start of each time step when one or more of the conditions for recalculation are triggered. These conditions are specified in terms of four control parameters:

- Change in water saturation — Item 9 of WPICOND or item 7 of PICOND.
- Change in saturation pressure (dew-point pressure) or grid block pressure — Item 10 of WPICOND or item 8 of PICOND.
- Fractional change in composition of any component — Item 11 of WPICOND or item 9 of PICOND.
- Change in simulation time — Item 12 of WPICOND or item 10 of PICOND.

The checks for the conditions which can trigger a recalculation are performed in the order list above; that is, if the change in water saturation is sufficient to trigger a recalculation, the three subsequent checks are not performed.

Each of these of these controls can be disabled, that is, not used to trigger a recalculation, by specifying a negative value. Conversely, if one or more of these controls are set to zero, this will have the effect of requiring a recalculation at each time step.

By default, the changes in water saturation and saturation or grid block pressure are interpreted as the magnitudes of the differences between the values in force at the simulation time when the table was previously calculated and the values at the current simulation time. For example, if the water saturation has changed from 0.10 to 0.12, this corresponds to a change of 0.02 which would trigger a recalculation for the default change in water saturation of 0.01. If required, changes in water saturation and saturation or grid block pressure can be treated as fractional changes by setting item 330 of the OPTIONS3 keyword to 2. This corresponds to the magnitude of the difference between the current and previous values normalized by the previous value. Changes in composition are always treated as fractional.

In addition to changes specified by the control parameters described above, a recalculation will also be triggered each time a WPICOND keyword is encountered in the SCHEDULE section for all the connections for those wells matching the well specification defined in item 1, if one or more of the items specified by the keyword have changed compared to those currently in force. Similarly, when a PICOND keyword is encountered, a recalculation will be triggered for all connections in all wells for which one or more of the items specified by the keyword have changed compared to those currently in force. If the generalized pseudo-pressure option is deselected for a well by setting item 2 of the WPICOND keyword to NO, the complete set of tables for this well will be deleted. These will be recalculated at a later time if item 2 of the WPICOND keyword is subsequently set to YES.

The number of pressure entries in the table of pressure versus total generalized molar mobilities for a given well connection will depend upon the following:

- The current grid block pressure.
- The lower and upper pressure table bounds.
- The step sizes in pressure below and above the dew point.
- The location of the dew-point pressure relative to the lower and upper table bounds.
- The optional inclusion of additional pressure steps in the vicinity of the dew-point pressure.

The maximum number of rows per table is specified via item 14 of the WELLDIMS keyword. This has a default value of 201 and this may be increased or decreased as required but is expected to be sufficient for most modeling purposes. The minimum number of rows which may be specified is 21. The maximum number of rows actually used (over all tables of all connections in all wells) is reported at the end of the simulation. If the maximum available number of rows is less than the number required for a given well connection, the sizes of the pressure steps are widened to reduce the number of rows to the available maximum and an appropriate warning message is used. In this case, the actual number of rows allocated may vary slightly from the specified maximum because of integer rounding.

The calculation time required to construct these tables increases with the number of rows (pressure entries) required and the total number of well connections for which tables are required. The status of the tables can be reported to the PRT file by setting item 19 of the RPTPRINT keyword to 1 or 3. This provides information such as the time when the table was last updated, the reason for the update and the values of the control parameters at the time of the update. If item 19 of the RPTPRINT keyword is set to 2 or 4, the actual table of pressure versus total generalized molar mobilities is also reported. Values of 1 and 2 will only provide a report at the report step for a table which has been updated since the previous report step, whereas values of 3 and 4 will provide full reports at each report step irrespective of when the table was last updated.

The number of pressure table update calculations performed for a specified well connection may be inspected via the connection summary vector CGPPTN. This is essentially a counter which is incremented by one each time the pressure table is recalculated. If the GPP option is deselected for a well, the counters associated with all the well connections will be reset to zero. The status of the pressure table update calculation for a specified well connection, that is, the reason for the update, may be inspected via the companion connection summary vector CGPPTS. This has one of the following values:

- -3 Recalculation triggered by WPICOND keyword.
- -2 Recalculation triggered by PICOND keyword.
- -1 Recalculation triggered by empty table.
- 0 No table calculation yet performed or GPP inflow model not applied to well.
- 1 Recalculation triggered by change in water saturation.
- 2 Recalculation triggered by change in saturation pressure.
- 3 Recalculation triggered by change in composition.
- 4 Recalculation triggered by change in simulation time.
- 5 Recalculation triggered by change in grid block pressure.

Typically this summary vector will commence with a value of zero for any initial period of simulation for which the GPP inflow model is not applied to the well connection. Upon initial specification of the GPP inflow model for the well via one or more of the keywords described above, the value will initially change to -1 indicating that a calculation has been triggered by the absence of an existing table and will subsequently fluctuate between values of 1 and 5 according to which condition has triggered a recalculation of the pressure table. If subsequently a PICOND or WPICOND keyword is specified which alters one or more of the keyword items, this will also trigger a recalculation resulting in an instantaneous status value of -2 or -3 respectively. If at any point during the simulation the GPP inflow model is subsequently deselected, the status value will be reset to zero. For periods during which the table is not updated, the status value will remain fixed at the value corresponding to the reason for the most recent update.

Notes

- As described on page 107 of *Whitson and Fevang (1996)*, and also illustrated in *Singh and Whitson (2008)*, an over-coarse grid will lead potentially to an underestimate of well deliverability. It is important that the grid size containing the well should be small enough to model the region 1 steady-state flow using GPP. This will be of the order of 50-100 m for lean gas condensate and 100-200 m for rich gas condensate reservoirs, although sensitivity studies should be carried out to determine the required well grid.

- If the rock compaction option has been specified via the ROCKCOMP keyword, the values of oil and gas relative permeability used in the above equations will be modified by the transmissibility multipliers specified via the ROCKTAB keyword. This can be disabled by setting item 299 of the OPTIONS3 keyword to 1.
- The calculated blocking factor will depend upon the calculation control criteria specified via either the WPICOND or the PICOND keywords. It is recommended, therefore, that a sensitivity analysis is performed in order to establish the dependence of the calculated blocking factor upon the spacing and extent of the pressure nodes in the total generalized molar mobility table and upon the frequency at which the table is updated.
- An example table is shown below in figure 8.1 for which a fixed spacing of 20 and 40 bars below and above the dew point respectively without the adaptive spacing option selected (red curve) is compared with a fixed spacing of 40 and 80 bars with the adaptive spacing option selected with a minimum spacing of 0.1 bar (blue curve). The adaptive option adds additional points by applying successive halving of the final step below and above the dew point thereby constructing progressively smaller pressure steps on the approach to the dew point until the minimum specified step size is reached. This provides considerably improved resolution in the vicinity of the dew point for a comparatively small increase in table points and will improve the accuracy of the integration applied via the trapezium rule, in particular, when the connection pressure is close to the dew-point pressure.

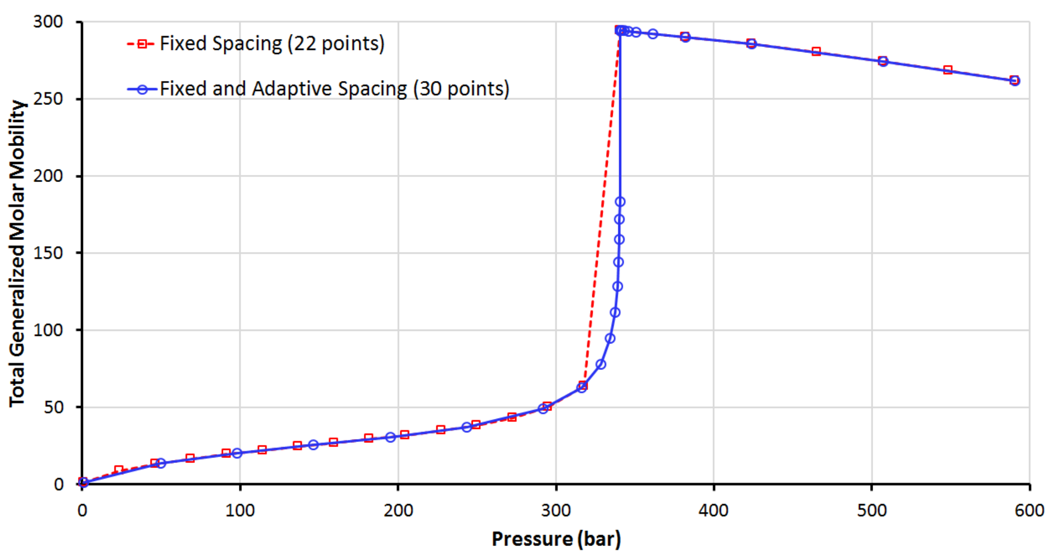

Figure 8.1. Example Table for Pressure versus Total Generalized Molar Mobility

Velocity dependent relative permeability effects

The generalized pseudo-pressure model may optionally include the effects of velocity dependent relative permeabilities (see item 4 of the PICOND keyword and item 6 of the WPICOND keyword) if the VELDEP keyword has also been specified. In this case, the pseudo-pressure integral defined by equation 8.46 and the phase molar mobility M_j defined by equation 8.51 require the calculation of the oil and gas capillary numbers in order to calculate the velocity dependent relative permeabilities. This involves the calculation of the gas velocity at each pressure used in the pseudo-pressure integration which in turn requires the equivalent radius r corresponding to each of these pressures which is estimated using the following equation:

$$\frac{\ln r - \ln r_w}{\ln r_o - \ln r_w} = \frac{m(p) - m(p_w)}{m(p_B) - m(p_w)} \quad \text{Eq. 8.52}$$

Given the radius at each pressure in the pseudo-pressure integration, the corresponding *pipe-equivalent* gas phase velocity v_g in the vicinity of the well completion can be calculated according to:

$$v_g = \frac{n_g}{2\pi r h \phi (1 - S_w) b_g} \quad \text{Eq. 8.53}$$

where h is the effective completion thickness, ϕ is the porosity and S_w is the water saturation. The alternative models for the calculation of capillary number are described in "Velocity-dependent relative permeabilities". The choice of oil and gas capillary number models used here are specified via items 1 and 2 of the `VELDEP` keyword. However, only capillary number models 1 and 3 are used here; that is, model 1 will be substituted if model 2 is specified. Capillary number model 3 includes a permeability term which, for the 2015.1 version, is calculated according to:

$$K = \sqrt{K_h (K_m / h)}$$

where K_h is the permeability thickness and K_m is the mean permeability in the plane perpendicular to the plane of the completion direction. For example, for a vertical completion:

$$K_m = \sqrt{K_x K_y}$$

where K_x and K_y are the x and y directed grid block permeabilities respectively. The permeability thickness K_h can be specified directly or calculated internally according to the items specified via the `COMPDAT` keyword but in its simplest form will default to the product $K_m h$. Prior to 2015.1, the permeability value was calculated using the interblock link permeabilities in the plane perpendicular to the direction of the completion. For example, for a vertical completion:

$$K = \sqrt{0.25 (K_x + K_{x+1})(K_y + K_{y+1})}$$

where the subscript +1 signifies the neighboring cell. Pre-2015.1 behavior may be restored by setting item 332 of the `OPTIONS3` keyword to 1.

The capillary number relative permeabilities may be further modified by a turbulent, Forchheimer effect. A description of the models available for the Forchheimer parameter is presented in "Forchheimer models". This effect is modeled as a non-Darcy flow factor as described in "Non-Darcy flow".

Note: If a cell containing a well connection drops below dew point, and velocity dependent effects are strong, rapid variations in PI are occasionally observed. These effects are caused by gas condensate effects being modeled by both the GPP algorithm and the grid block discretization simultaneously. If the connected cell is coarse enough that all of regions 1 and 2 (as described in *Whitson and Fevang (1997)* and *Whitson and Fevang (1995)*) are contained within the pressure-equivalent radius then these anomalies will be eliminated.

Injecting connections

In injecting connections, an alternative expression for the mobility is required to reflect the upstream saturation conditions in the well bore. In the absence of well bore crossflow, there would be single-phase conditions at all injecting connections. In these circumstances it is standard practice to replace the term $k_{p,j} / \mu_{p,j}$ in equation 8.22 with its sum over all phases, to give in Eclipse 100

$$M_{p,j} = \frac{\frac{k_{o,j}}{\mu_{o,j}} + \frac{k_{w,j}}{\mu_{w,j}} + \frac{k_{g,j}}{\mu_{g,j}}}{B_{p,j}} \quad \text{Eq. 8.54}$$

and in Eclipse 300

$$M_{p,j} = \frac{k_{o,j}}{\mu_{o,j}} + \frac{k_{w,j}}{\mu_{w,j}} + \frac{k_{g,j}}{\mu_{g,j}} \quad \text{Eq. 8.55}$$

where p here refers to the injected phase. The connection injectivity is thus a function of the total fluid mobility in the grid block. In Eclipse 300, the fluid in the wellbore is flashed at the bottom hole pressure to determine the wellbore fluid molar density. For each connection, this wellbore molar density is used with the mobility in equation 8.55, the pressure drawdown and the connection factor to form connection molar injection rates. (These are summed to give the total molar injection rate of the well, which is multiplied by a surface volume per mole factor to give the reported injection volume rate.) The Eclipse 300 calculation thus relies on quantities at both wellbore conditions (fluid molar density) and grid block conditions (equation 8.55) to determine the injection flow for a given drawdown. Eclipse 100, in contrast, only relies on quantities at the grid block conditions (equation 8.54). For gas injectors with very large drawdowns, the difference between the wellbore and grid cell pressures will result in small differences in gas injection flow rates between the simulators.

If gas or water is injected into a grid block initially containing oil, this relationship will cause the well's injectivity to vary until the grid block is completely flooded out with the injected phase. In reality however, most of the pressure drop occurs over a region close to the well, and when this region has flooded out the injectivity will stay approximately constant. If the grid block size is much larger than this region, the calculated injectivity will be incorrect until the whole grid block has flooded out. For cases like this, an alternative approach is available that allows you to supply a constant value of the injected phase relative permeabilities. In Eclipse 100 the keyword `COMPINJK` can supply a single relative permeability of the injected phase for non-crossflowing injectors. In Eclipse 300 the keywords `COMPKRI` and `COMPKRIL` supply relative permeabilities for all three phases, which can be applied to injectors regardless of their crossflow status; alternatively `COMPMOBI` and `COMPMBIL` can supply a fixed value for the injection mobility.

Eclipse 100

If crossflow occurs through the well bore, or if the well is a multiphase injector, it is possible for the well to inject a mixture of phases into the formation through one or more connections. The injection mobilities must be adapted to represent the fluid mixture present in the well bore. Eclipse 100 determines the proportion of each phase present in the mixture entering the well bore, and ensures that the phases flow in the same proportions through all the injecting connections. The treatment of crossflow is thus based on the assumption that the mixture of phases is uniform throughout the well bore at formation level. To determine the injection mobilities, a set of intermediate quantities is first calculated, which we call the "well volume factors". The well volume factor α_p of a given phase p is defined to be the stock tank volume of the phase that is contained within a unit volume of the well bore at formation level. Its value depends on the well's bottom hole pressure and the mixture of phases entering the well bore. The injection mobilities are set in proportion to the well volume factors. Assuming that the total injectivity of the connection is still proportional to the total fluid mobility within the grid block, the injection mobilities are given by

$$M_{p,j} = \alpha_p \left[\frac{\frac{k_{o,j}}{\mu_{o,j}} + \frac{k_{w,j}}{\mu_{w,j}} + \frac{k_{g,j}}{\mu_{g,j}}}{\alpha_{fo} B_{o,j} + \alpha_w B_{w,j} + \alpha_{fg} B_{g,j}} \right] \quad \text{Eq. 8.56}$$

The expression reduces to the standard injection mobility, equation 8.54, when the well bore contains only one free phase.

If a flow-dependent skin factor is used, the mobility reduction factor D_m is calculated as in equation 8.14. It is applied to every phase flowing in the injecting connection, so that the injection rates remain in proportion to the well volume factors.

If the Russell-Goodrich or pseudo-pressure equations are applied, they affect the mobility of free gas within the grid block containing the connection, unlike the skin factor whose influence is restricted to the immediate vicinity of the well bore. Accordingly the term $k_{g,j}/\mu_{g,j}$ in equation 8.54 or equation 8.56 is replaced by the term $k_{g,j}\lambda_{g,j}B_{g,j}$, where $\lambda_{g,j}$ is evaluated in the form appropriate to each equation, as described above for producing connections.

High mobility injection wells

Eclipse 300

It is known that injector wells can cause convergence problems when the mobilities are high in the connecting grid cells. Specially in thermal simulation this might be the case when steam is injected at high pressure resulting in fracturing near the well. The high mobility injection well use the expression

$$\widehat{q}_{p,j} = T_{wj}M_{pj}\left(\frac{\sum_i M_{pi}(P_i - H_{wi})}{\sum_i M_{pi}} - P_w\right)$$

for the volume injected by a completion j. The mobilities are the total mobilities as given above. See equation 8.1 and above for the standard model.

Using the high mobility injection well option, set by item 16 of the keyword `WELSPECS`, an average completion pressure-term is calculated. This gives a fixed drawdown for all the completions of the well. However the total volume injected at a given reservoir pressure and bottom hole pressure is maintained by the model. It is important to note that this model does not reflect the instantaneous flow that would occur on a short time scale as the model is designed to average out injections oscillating between completions and thus avoiding convergence problems for larger timesteps.

The wellbore head term

The wellbore head term H_{wj} is the pressure difference in the wellbore between the connection j and the well's bottom hole datum depth. Friction effects are usually small within the well bore at formation level, and they are neglected. H_{wj} therefore represents the hydrostatic head through the mixture in the well bore between the connection depth and the bottom hole datum depth.

If we treat the mixture as uniform throughout the well bore at formation level, the hydrostatic head depends on the average mixture density. This can be readily calculated from the well volume factors,

$$\overline{\rho} = \rho_{o,surf}\alpha_o + \rho_{w,surf}\alpha_w + \rho_{g,surf}\alpha_g \qquad \text{Eq. 8.57}$$

where the subscript 'surf' refers to the phase densities at stock tank conditions.

However, the assumption that the mixture density is uniform throughout the well bore at formation level may result in inaccuracy. Consider a well having two connections, in which oil enters through the lower connection and a high GOR. mixture enters through the upper connection. In reality the density of the fluid in the well bore between the two connections would be close to the density of the oil in the lower connection, whereas equation 8.57 would predict a lower density corresponding to a mixture of higher GOR. This could give inaccurate results if the head between the two connections were a significant fraction of the drawdown. To overcome this, Eclipse has an option to calculate the head between each pair of connections separately, according to the local mixture density. The well bore is segmented into sections beginning and ending at adjacent connections. Within each section, the local mixture density is calculated from the accumulated upstream connection flows, assuming there is no slip between the phases,

$$\rho_{\text{local}} = \frac{\rho_{o,\text{surf}} Q_o + \rho_{w,\text{surf}} Q_w + \rho_{g,\text{surf}} Q_g}{Q_{fo} B_o + Q_w B_w + Q_{fg} B_g} \quad \text{Eq. 8.58}$$

Q here represents the accumulated flow through the upstream connections, and the formation volume factors B are evaluated at the average pressure in the wellbore section. The expression on the denominator is the local volumetric flow rate of the mixture in the section. Within the present structure of the program, it is not possible to perform the segmented head calculation implicitly. It is therefore performed explicitly, based on the conditions at the end of the previous timestep. The explicit treatment of this term is not likely to cause instability, and in view of its greater accuracy it has been made the default option in **both simulators**.

The productivity index

Engineers have traditionally used the "productivity index" to define the degree of communication between a well and the reservoir. Unlike the connection factor, it is a quantity that can be obtained directly from field measurements. But whereas the connection factor remains constant over the lifetime of the well (unless the skin factor changes because of silting or stimulation, for example), the productivity index will vary with the fluid mobilities at the well.

The steady-state productivity index is defined as the production rate of a chosen phase divided by the drawdown. The drawdown here is the difference between the well's bottom hole pressure and the pressure in the reservoir at the edge of the well's zone of influence. The extent of the well's influence is known as the "drainage radius", r_d. Thus the productivity index, J, is defined as

$$J = \frac{Q_p}{P_d - P_w} \quad \text{Eq. 8.59}$$

where Q_p is the production rate of the chosen phase, and P_d is the pressure at the drainage radius. Equation 8.59 also defines the corresponding quantity for injection wells, the "injectivity index", if injection rates are treated as negative.

A relationship between J and the connection factors of the well can be derived, assuming steady radial Darcy flow exists with uniform mobility throughout the region bounded by the drainage radius:

$$J = \sum j \left[T_{wj} M_{p,j} \left(\frac{\ln(r_o/r_w) + S}{\ln(r_d/r_w) + S} \right) \right] \quad \text{Eq. 8.60}$$

where Σ_j denotes a summation over all the connections j belonging to the well. The simulator calculates the productivity index of each well from this expression, and can optionally print it out at each well report. If the chosen phase p is gas and the connections have a non-zero D-factor, the non-Darcy skin is added to S in equation 8.60.

The simulator's calculation of the productivity index should not be used for a horizontal well, as the assumptions used in equation 8.60 will not be valid. Equation 8.60 requires that a steady radial flow regime perpendicular to the well bore exists out to the drainage radius. For a horizontal well, this flow regime will rapidly be disrupted by the top and bottom boundaries of the formation, and the ultimate flow regime may be linear or pseudo-radial depending on the geometry of the well and its drainage region. For horizontal wells, therefore, you should calculate the productivity index "manually" from equation 8.59. The reservoir pressure at the drainage radius could be obtained from a representative grid block, or the average pressure of the field or a suitable fluid-in-place region could be used.

The drainage radius r_d is often known only to a poor degree of accuracy. The engineer must supply its value for each well in the input data, but there is a default option which sets its value for each connection equal to the value of the pressure equivalent radius of the connecting grid blocks. In this case equation 8.60 simply reduces to

$$J = \sum_j \left[T_{wj} M_{p,j} \right] \qquad \text{Eq. 8.61}$$

and the PI should be regarded as the PI of the well within its connecting grid blocks rather than within its drainage region.

Additional forms of the well PI can be output to the SUMMARY file using the SUMMARY section keywords WPI1, WPI4, WPI5 and WPI9. These use equation 8.59 where P_d is replaced with a weighted average pressure within a 1-block, 4-block, 5-block and 9-block region around each connecting grid block. The weighting formula is specified with keywords WPAVE and WWPAVE

Eclipse 300

The connection phase mobilities for oil and gas must be modified in the compositional case to take into account phase changes for different components when combined into a single well stream. In this case the mobilities used in equation 8.60 are

$$\widetilde{M}_{p,j} = \frac{O_{p,j}}{\sum_k O_{p,k}} M_{p,j} \qquad \text{Eq. 8.62}$$

where $O_{p,j}$ is the volume flow rate of phase p per mole of hydrocarbon at connection j. Note that this means that the productivity index for oil and gas may differ between Eclipse 100 and Eclipse 300. The mobilities used for the water phase are unaffected.

Eclipse 100

An alternative option is triggered by the engineer setting a negative value for the drainage radius. In this case the well’s production/injection potential is printed in the well reports instead of its productivity index. The potential is defined as the flow rate that the well could sustain in the absence of any rate limits, that is if it were constrained only by the supplied limiting values of the bottom hole pressure and the tubing head pressure, if set. When this option is selected, the well PIs can still be output to the SUMMARY file using the SUMMARY section keyword WPI.

Note: The PI of the preferred phase of a well is written to the SUMMARY file as a quantity without units in FIELD. This is because the SUMMARY file is unable to handle a change in the units if the preferred phase of the well changes between gas (Mscf/day/psi) and liquid (stb/day/psi). The alternative keywords WPIO, WPIG, WPIW and WPIL can be used to output the productivity index for specific phases and are printed with appropriate accompanying units.

Well modeling facilities

Well completions

x	Eclipse 100
x	Eclipse 300

Each well can be completed in several layers of the reservoir. A communication between the well bore and a single grid block is defined as a *connection*. A given well may have any number of connections, as long as it does not exceed the maximum number of connections per well, which is specified in keyword WELLDIMS in the RUNSPEC section. (Any of the WELLDIMS maximum values may be increased on a restart, and in Eclipse 300 it is also possible to increase them during a run using the CHANDIMS keyword.)

The connections belonging to a particular well need not all lie in the same vertical column of grid blocks, so it is possible to model highly deviated wells. Moreover a well can be completed in several grid blocks within the same layer, enabling the simulator to model horizontal wells. This feature is also useful in 3D radial coning studies where the well connects with all the innermost, wedge-shaped, blocks in any given layer.

Inflow performance

The flow rate of a phase (oil, water or gas) through a connection is proportional to the product of three quantities:

- the transmissibility between the grid block and the well bore
- the mobility of the phase in the grid block at the perforations
- the pressure drawdown between the grid block and the well bore.

These quantities are discussed in more detail in "Well inflow performance".

In certain situations the drawdown in one layer may have the opposite sign to the drawdown in the other completed layers. The simulator will allows **crossflow** to take place between the reservoir layers through the well bore. A material balance constraint is observed within the well bore at formation level, so that the mixture of phases flowing out of the well bore corresponds to the average phase or component mixture entering the well bore. A special case in which crossflow may occur is when the well is plugged off above the formation while the completions remain open to the reservoir. In Eclipse 100 the crossflow facility can be turned off if you do not wish this to occur, using a switch in the keyword WELSPECS.

Relative permeabilities in well connections

The relative permeability of a phase in a well connection is a function of the saturations in the grid block containing the connection. By default, the relative permeabilities are calculated using the Saturation Function tables associated with the grid block containing the connection. However, the engineer can supply additional relative permeability tables for calculating the fluid mobilities at the well connections. These tables could take into account the effects of coning and/or partial completion within the grid block, or they could be well **pseudo** tables derived from an earlier fine-grid simulation. The association between the relative permeability tables and the well connections is defined using the keyword COMPDAT.

An alternative means of allowing for coning or partial completion within a grid block is to re-scale the Saturation Function tables to different end-point saturation values. The engineer can optionally supply alternative sets of end-point saturation values for use in particular well connections. These end-point values are used to transform the grid block saturations into the saturations ‘seen by’ the well connections. The alternative end-point saturation values are entered using the keyword COMPRP.

Eclipse 100

In simulation runs which use the Vertical Equilibrium option, the keyword COMPVE should be used instead of COMPRP, to supply depth values for the top and bottom of the perforated interval. The COMPVE keyword may be used to specify partial penetration data in non-VE runs also; Eclipse automatically translates the top and bottom depths of the perforations into COMPRP type scaled end-points.

Special facilities for gas wells

There are two facilities for improving the accuracy of modeling gas inflow. Firstly, the non-Darcy component of the pressure drawdown resulting from turbulence near the well bore is modeled by a flow-dependent contribution to the skin factor. The turbulence skin factor for a well connection is equal to the product of the free gas flow rate in the connection and the connection’s D-factor. Well D-factors are defined using the keyword WDFAC or they may be calculated from a correlation using the keyword WDFACCOR.

The second facility is the option to use a special inflow equation to provide a more accurate model of the flow of gas into the well. Eclipse 100 offers a choice of three models: the Russell-Goodrich equation, the dry gas pseudo-pressure equation and the generalized pseudo-pressure method. These are all methods of taking into account the pressure-dependence of the gas properties between the grid block pressure and the well bore pressure. The generalized pseudo-pressure equation also takes account of the effects of condensate dropout. They are described in more detail in "Special inflow equations". You can select the desired equation by setting a switch in the keyword WELSPECS. The generalized pseudo-pressure equation is also available in Eclipse 300, where it may be activated for specific wells using the WELSPECS keyword, or it may be activated for all production wells using the keyword PSEUPRES.

Scale deposition model

A scale deposition model is available, which accounts for the cumulative effects of scale deposited around the well connections and the resulting degradation of the productivity index.

The rate of scale deposition in each well connection is proportional to the flow rate of water into it and also varies with the amount of impurities dissolved in the water. Formation water and injected sea water contain different impurities and thus the overall rate of scale deposition is dependent on the relative proportion of sea water and formation water.

This scale deposition process is modeled by a table that defines the rate of scale deposition per unit water flow rate as a function of the fraction of sea water present in the water flowing into a connection (see keyword SCDPTAB). The sea water fraction is represented by a user-configured passive water tracer (see keyword SCDPTRAC).

A second scale damage table defines how the current amount of deposited scale alters the PI of the well connections (see keyword SCDATAB).

The calculation for the effect of scale damage on a well is performed explicitly. Thus, at the end of each timestep the mass of scale deposited around each connection is incremented according to the flow of water during the timestep and the current sea water concentration. This is translated into a PI reduction factor for each connection, that is used during the following timestep.

You may supply multiple versions of each of the two table types. The WSCTAB keyword must be supplied to select which tables are to be used by each well for the calculation.

The WSCCLEAN keyword may be used to reduce the amount of scale deposited around well connections after a scale cleanup operation. (The WSCCLENL keyword should be used for wells in local grid refinements.)

Well controls and limits

Individual wells can operate at a target value of any of the following quantities:

- Oil flow rate
- Water flow rate
- Gas flow rate
- Liquid flow rate (oil and water)
- Linearly combined rate (user-specified linear combination of oil, water and gas)
- Reservoir fluid volume (or voidage) rate

Note: Voidage rate is the reservoir volume that the produced or injected fluids would occupy at reservoir conditions. The associated reservoir pressure is usually the average hydrocarbon pressure in the field. However, this may be set to be the average pressure in a nominated fluid in place region using argument 13 in keyword WELSPECS. By obtaining voidage rates at a single pressure for all wells we avoid the effects of higher pressures at injectors and lower pressures at producers; setting equal injection and production voidage rates will then yield approximate pressure maintenance.

- Bottom hole pressure
- Tubing head pressure

Eclipse 300
- Wet gas rate (which is the volume that the hydrocarbon mixture would occupy at standard conditions if it were an ideal gas)

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- Total molar rate

Note: The total molar rate includes the molar rate of water.

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- Calorific value rate.

One of these quantities is selected as the main control target, while limiting values can be supplied for any of the remaining quantities. The limiting flow rates are treated as upper limits, while the pressure limits are treated as lower limits in production wells and upper limits in injectors. The well will automatically change its mode of control whenever the existing control mode would violate one of these limits. The target quantity in the old control mode will then become a limit in the new control mode.

For example, consider a production well operating at a target oil flow rate of 1000 stb/day, with a lower limit of 3000 psia on the bottom hole pressure. The well will produce 1000 stb/day of oil until the bottom hole pressure falls below 3000 psia. The well's control mode will then change automatically to maintain a constant bottom hole pressure of 3000 psia, and the oil production rate will decline. If subsequently the oil production rate were ever to exceed 1000 stb/day (resulting for example from stimulation of the well, or the opening of an extra connection), then the control mode will automatically switch back to the original target oil production rate.

It is possible to set separate customized default values for BHP limits for production and injection wells, which will replace the standard default value whenever the BHP limit is defaulted when setting up the well control data. This is done using the FBHPDEF keyword.

In addition to the control quantities listed above, a maximum allowable drawdown can be specified for each production well, using the keyword WELDRAW. This is converted into a maximum liquid or gas production rate at each timestep (depending on the well's preferred phase), using the current value of the liquid or gas mobility in the well's connections. Thus the liquid or gas rate limit is used to apply constraints on the drawdown.

An alternative method of control can be applied to wells during the history matching process, when their observed oil, water and gas production rates are known. The observed rates are entered with the WCONHIST keyword. A well can be made to produce at a rate which matches one of the phases, or at the equivalent liquid or reservoir volume rate of the observed production. The latter option is useful for making the well produce the correct amount of total fluid from the reservoir before the mobility ratios are fully matched. Thus the rate of pressure decline should be approximately correct. The observed rates and production ratios can be written to the summary file, for graphical comparison with the calculated rates. Observed rates for injection wells may be entered with the WCONINJH keyword

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Note: If a well's control mode is changed in a keyword other than WELSPECS, the well's preferred mode will also be changed to the new control mode. To match the default Eclipse 300 behavior, set item 227 of the OPTIONS keyword to 1.

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Note: If a well's control mode is changed in a keyword other than WELSPECS, the well's preferred mode will not be changed to the new control mode. To match the default Eclipse 100 behavior, set item 318 of the OPTIONS3 keyword to 1.

Economic limits

There is an additional class of limits that can be applied to production wells. The engineer can specify values of the oil or gas production rate, the water cut, the gas-oil ratio and the water-gas ratio which represent the limits of economic operation of the well. The violation of one of these limits will result in the well, or one or more of its connections, being automatically closed. The economic limit options are:

- Lower limits for the oil, gas, liquid and reservoir fluid volume production rates for each well. The well is closed if any of these limits are broken.

  Eclipse 100

  Alternatively, if the well still has one or more connections placed on automatic opening (see "Automatic opening of wells and well connections"), one of these connections will be opened.
- Upper limits for the water cut, the gas-oil ratio, the gas-liquid ratio and the water-gas ratio of each well. There is a choice of actions to be taken if a well violates one of these limits:
  - The worst-offending connection (that is the connection with the highest water cut, gas-oil ratio or water-gas ratio) in the well is closed.
  - The worst-offending connection and all connections below it in the well are closed. Here, "below" is interpreted as "further from the wellhead according to the connection ordering". The connection ordering may be specified in the COMPORD keyword.
  - The well itself is closed.

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  - The well is plugged back, that is the length of perforations in either the top or bottom open connection or the worst-offending connection is reduced by an amount specified in keyword WPLUG.

Keyword WECON is used to set the limiting values and define the choice of workover action.

Note: If a well which is closed still has two or more connections open to the reservoir, it may be possible for crossflow to occur between the completed layers through the well bore. To distinguish a well that can still make its presence felt in this way after being closed, we refer to it as **stopped**. A well that is not allowed to crossflow after being closed is referred to as **shut**. Thus a **stopped** well can still affect the reservoir, whereas a **shut** well is effectively ignored by the simulator. Wells can be opened, stopped or shut manually. The engineer can also stipulate whether a well is to be shut or stopped after it violates an economic limit, using item 9 in the WELSPECS keyword.

Economic limits can also be set for individual connections in production wells. A connection will be closed if it exceeds its limit on water cut, gas-oil ratio or water-gas ratio, or if its oil or gas production rate falls below a specified value. Connection economic limits are set using the keyword CECON.

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A minimum economic rate can also be applied to injection wells, using keyword WECONINJ. This allows injection wells to be automatically closed if they are not injecting at a high enough rate.

Workovers are normally performed on individual well connections, but it is possible to lump sets of connections together into ‘completions’ for simultaneous closure, using the keyword COMPLUMP. When a well with lumped connections is worked over (for example, after exceeding a water cut limit set in WECON), the worst-offending completion is identified (by summing the flow rates of its connections) and all the connections belonging to that completion will be shut together.

The economic limits are checked at the end of each timestep, and the appropriate action is taken if any are violated. The action is taken at the end of the timestep during which the limit was violated. Thus a well may produce ‘uneconomically’ for one timestep before remedial action is taken. However, the engineer can specify a **tolerance fraction** for these limits. If a limit is broken by proportionately more than this tolerance, the timestep will be repeated after the remedial action has been taken. Thus the action will take effect from the beginning of the timestep in which the limit would otherwise have been violated. For example, consider a well which has a maximum economic gas-oil ratio of 10 Mscf/stb, and assume a tolerance fraction of 0.2 has been specified. If the well’s gas-oil ratio increases to 11 Mscf/stb, then remedial action is taken at the end of that timestep, and will be effective for the following timestep. If however the gas-oil ratio reaches above 12 Mscf/stb, the timestep will be repeated after the remedial action has been taken. The tolerance fraction is set with the keyword WLIMTOL. Note that each timestep will be repeated only once. Thus if a well still violates an economic limit at the end of the repeated timestep, that timestep is ‘accepted’. Further remedial action will applied to the well before starting the next timestep.

Wells or connections that are closed after violating an economic limit can be tested periodically to see if they can flow economically again. If they can flow without violating a limit, they will be reopened. This facility can, for example, allow a watered out well to produce again after its cone has subsided sufficiently while it was closed. Instructions for periodic testing of closed wells are entered using the keyword WTEST.

There is an option to make the simulation end automatically if all the production wells in the field, or in a particular group, have been shut or stopped. This saves the simulator from using computer time needlessly when nothing further of interest is going to happen. The option can be switched on using the GECON keyword. In addition, selected wells can be flagged to make the simulation stop if any of them are closed during the run for any reason, using a switch in the WECON keyword. If for example a flagged well is shut because it has exceeded its economic water cut limit, the run will stop and an appropriate message will be printed. The engineer can then examine the state of the reservoir at that time, make any necessary changes to the well completions, and restart the run. This ‘semi-interactive’ mode of working allows the engineer to carry out workover procedures that are more complex than the built-in automatic workover options.

An additional option allows the engineer to nominate a well to be opened when another well is closed automatically for any reason (for example on violating an economic limit). This facility can be used, for example, to open an injector when a producer closes after becoming uneconomic. If the injector has the same completions as the producer, the effect will be equivalent to converting the producer into an injector, although in this case they are treated as two separate wells. The option is activated with item 9 in keyword WECON.

Automatic cutback of flow rate

There is an indirect way of controlling the following production ratios of individual wells subject to gas or water coning:

- Water cut
- Gas-oil ratio
- Gas-liquid ratio
- Water-gas ratio.

Upper limits for one or more of these quantities can be set using the keyword WCUTBACK. Whenever one of these limits is exceeded, the well's production rate is cut back to a specified fraction of its current value. The rate will be cut back successively at each timestep until the limit is no longer exceeded. The rate cutback is performed at the end of the timestep in which the limit is exceeded. However, if the limit is exceeded by proportionately more than the tolerance fraction entered with keyword WLIMTOL (see above), the timestep will be repeated with the well flowing at the reduced rate. The timestep will only be repeated once; if the limit is still exceeded at the end of the repeated timestep, the well's flow rate will be reduced again for the next timestep.

This facility is useful for determining the production rate necessary to keep water or gas coning in a well down to an acceptable level.

The WCUTBACK keyword can also supply a minimum pressure value for the grid blocks in which the well is completed. If the pressure falls below this value, the well's production rate is cut back to the specified fraction of its current value. This facility can be useful for limiting the decline of grid block pressure at the wells.

For each of the five cutback limits, there is a corresponding limit that reverses the cutback process. For example, the gas-oil ratio could have a cutback limit of 5.0 Mscf/stb and a cutback reversal limit of 3.0 Mscf/stb. If the GOR increases above 5.0 Mscf/stb, the production rate will be cut back until the GOR falls below this limit. If subsequently the GOR falls below 3.0 Mscf/stb while the well is operating at the reduced flow rate, the rate will be allowed to increase again, reversing the cutback process. Cutback limits and their corresponding reversal limits are therefore a useful way of making wells operate within a given band of conditions.

The WCUTBACK keyword can also apply to injection wells, in which case the production ratio limits are ignored, the cutback pressure limit is a maximum value, and the cutback reversal pressure limit is a minimum value.

Efficiency factors

Wells can be given 'efficiency factors' to take account of regular downtime (see keyword WEFAC). For example, if a well is 'down' for 10 percent of the time, its efficiency factor will be 0.9.

The well rates and pressures given in the well reports describe the state of the flowing well. The full rates are also used in the well economic limit checks. The efficiency factors are applied when updating the cumulative well and connection flows.

The efficiency factors are also applied when summing the well rates to obtain the group and field flow rates. Thus the group and field flow rates represent the average rates over the timestep, rather than the full rates reached when all the wells are operating.

The underlying formulation used to implement the above is that the well model is solved using inflow equations 8.1 and 8.2 while the reservoir model is solved using well source terms

$$q_{p,\,j} = ET_{wj}M_{p,\,j}\left(P_j - P_w - H_{wj}\right) \quad \text{Eq. 8.63}$$

and

$$\hat{q}_{p,\,j} = ET_{w,\,j}\left(\frac{K_{rp,\,j}}{\mu_{p,\,j}}\right)\left(P_j - P_w - H_{wj}\right) \quad \text{Eq. 8.64}$$

where E is the well efficiency factor and other terms are as defined for equations 8.1 and 8.2. Cumulative production and injection are determined using the same rates as the reservoir solve, so the cumulative production, injection and fluid in place sum to a constant assuming no other source terms such as analytic aquifers or sector model boundary conditions.

The WEFAC keyword is best used when wells are brought down individually on a regular basis for short periods of maintenance. But if all the wells in a group are brought down together for regular maintenance, then the efficiency factor should instead be applied to the group. Group efficiency factors are defined with the keyword GEFAC. The treatment of group downtime is entirely analogous to the treatment of well downtime. The group's flow rates reported in the PRINT and SUMMARY files reflect its full flow rate. Any flow targets and limits for the group will apply to its full flow rate. But the group rates will be multiplied by the group efficiency factor when summing the group's contribution to its parent group in the hierarchy. Thus the flows of any superior groups will reflect this group's time-averaged flow rate.

Automatic opening of wells and well connections

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An automatic opening facility in Eclipse 100 allows wells to be opened in sequence at the earliest opportunity, subject to constraints on the drilling rate and the maximum number of open wells allowed in each group. Wells are placed on automatic opening by setting their OPEN / SHUT status to AUTO, in any of the WCON... keywords or WELOPEN. They will be treated as initially SHUT, and will be opened in sequence in the order in which they are first defined in keyword WELSPECS (although this order is overridden if some of the wells belong to groups that already have their maximum quota of open wells).

Drilling rigs can be allocated to groups with keyword GRUPRIG, and the time taken to drill each well may be specified in keyword WDRILTIM. Together these restrict the rate at which wells can be opened automatically. Wells can effectively be opened midway through a timestep by having their efficiency factors temporarily adjusted to allow for the part of the timestep in which they were still shut. Thus in order to open a series of wells at 30 day intervals, say, it is not necessary to restrict the timesteps to 30 days.

Each group may be given an upper limit to the number of open wells it can contain, using keyword GECON (item 9). When a group has its full complement of open wells, no more of its wells can be opened automatically until one is shut-in. The AUTO facility can therefore be used to keep a group operating with its full complement of wells while some wells are being closed after violating their economic limits.

Well connections can also be placed on automatic opening, by setting their OPEN / SHUT status to AUTO in keywords COMPDAT or WELOPEN. They will remain SHUT until the well is subjected to an automatic workover, as for example is performed when a well or group economic water cut limit is exceeded. A connection on AUTO will be opened each time its well is worked over. One will also be opened each time the well's production rate falls below its minimum economic limit. Within a given well, the connections will be opened in the order in which they are first defined in keyword COMPDAT.

To prevent wells or connections from being automatically opened in areas of the reservoir that have already been depleted, a minimum acceptable oil saturation can be declared using the keyword WELSOMIN. A connection will not be opened automatically in a workover, or opened when its well is opened automatically, if the connecting grid block has an oil saturation below this limit. A flag can also be set, using the keyword WDRILRES, that prevents a well on automatic opening from being opened if it is completed in the same grid block as a well that is already open.

Wells and individual connections can also be opened automatically from a **drilling queue** in response to a group failing to meet its production target. This facility is described in "Group Production Rules".

Group and field control facilities

The multilevel grouping hierarchy

Each well must belong to a particular group, which is named when the well is first declared with keyword WELSPECS. Rate targets and limits, and economic limits, can be applied to each group and to the whole field. This provides a standard three-level (well - group - field) control hierarchy, where controls and limits can be specified independently at each level.

If controls or limits are required on more than three levels, a highly flexible multilevel hierarchy can be built up using the keyword GRUPTREE. This keyword declares additional groups that do not contain wells, but instead act as ‘parents’ to other groups at a lower level. These groups are called ‘node-groups’, to distinguish them from ‘well-groups’ which contain wells.

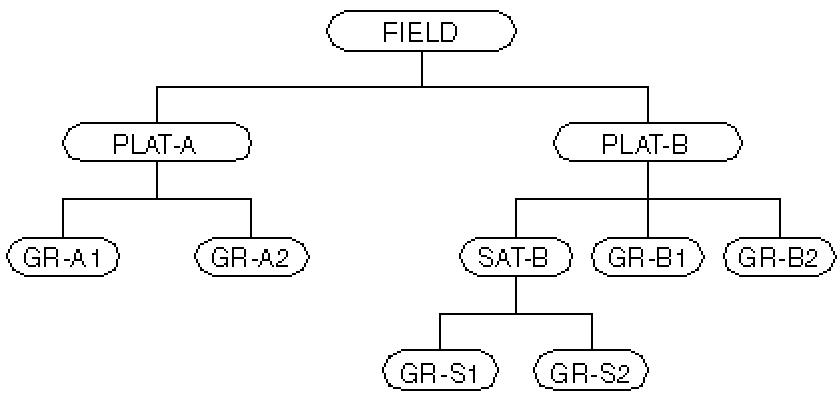

Figure 8.2. A five-level hierarchy

An example of a five-level hierarchy is shown in the figure above. The field occupies the highest level, level 0. PLAT-A and PLAT-B are node-groups at level 1. The groups at level 2 are all well-groups except for SAT-B which is a level 2 node-group. Groups GR-S1 and GR-S2 are level-3 well-groups subordinate to SAT-B. GR-A1 and GR-A2, and their respective wells, are subordinate to PLAT-A. GR-B1 and GR-B2, and their respective wells, are subordinate to PLAT-B. GR-S1 and GR-S2, and their respective wells, are subordinate to both SAT-B and PLAT-B. When the wells are included (on levels 3 and 4), the hierarchy has five levels in total.

Group production control options

Group production rate targets - guide rate control

Selected groups at any level in the hierarchy (including the field) can be made to produce at a target value of any one of the following quantities:

- Oil production rate
- Water production rate

- Gas production rate
- Liquid (oil + water) production rate
- Linearly combined production rate (user-specified linear combination of oil, water and gas)
- Reservoir fluid volume rate

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- Wet gas rate (which is the volume that the hydrocarbon mixture would occupy at standard conditions if it were an ideal gas)
- Reservoir volume production balancing fraction (which limits the group's reservoir fluid volume production rate to the specified fraction of its reservoir fluid volume injection rate).

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- Surface gas volume production balancing fraction (which limits the group's surface gas volume available rate to the specified fraction of its surface gas volume injection rate).

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- Surface water volume production balancing fraction (which limits the group's surface water volume production rate to the specified fraction of its surface water volume injection rate).

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- Calorific value rate.

The control target is set using keyword `GCONPROD`. The group's target rate is apportioned between the individual wells in proportion to each well's specified **guide rate**. A guide rate can optionally be defined for any phase; oil, water, gas, liquid or reservoir fluid, using keyword `WGRUPCON`. If the phase differs from the phase under group control, the guide rate is translated into a guide rate for the controlled phase using the well's production ratio for the previous timestep. Thus, for example, the target oil production rate of a group can be apportioned among its subordinate wells such that each well produces similar amounts of liquid. This can be done by giving each well the same guide rate for the liquid phase.

If a well's guide rate has not been specified, it is set at the beginning of each timestep to a value that depends on the well's **production potential**. The production potential is defined as the instantaneous production rate that the well would initially have in the absence of any constraints on flow rate, at the current grid block conditions (see "Well potentials"). By default, the well's guide rate is set equal to its production potential of the phase under group control. This enables, for example, a group's target oil production rate to be apportioned among the individual wells such that the wells produce in proportion to their potentials. Alternatively, the keyword `GUIDERAT` can be used to specify the coefficients of a general formula for well guide rates as a function of their oil, water and gas production potentials. This can enable, for example, wells with high water cut to be given a progressively smaller guide rate as they water out.

The individual wells can be subject to their own flow and pressure constraints. For example, if a well would violate its bottom hole pressure limit when producing its full share of the group's production target, the well will operate at its bottom hole pressure limit. The production rates of the other wells will increase in proportion to their guide rates (or potentials) to meet the group's target, as long as they do not violate any of their own flow or pressure limits. A well that is producing its full share of the group's production target is said to be under **group control**, while a well constrained by its own flow or pressure limits is said to be under **individual control**. If required, certain wells can be prevented from coming under group control. These wells will continue to operate according to their individual flow or pressure constraints, regardless of any group flow target.

Drilling queue

At some stage in the simulation, it may happen that all the group-controlled wells in the group have hit one of their constraints, and have either dropped out of group control or have been closed. The group can then no longer meet its flow target, and its production rate will decline. This situation can be postponed by placing additional wells in the **drilling queue**, using either keyword `QDRILL` or keyword `WDRILPRI`. When a group can no longer meet its production target, the simulator will scan the drilling queue and open a suitable well. The well must be subordinate to the group that is unable to meet its target, but not subordinate to any lower level group that is operating at its own production target or limit for the same phase. The drilling queue may either be a **sequential** queue, from which wells are opened in the sequence in which they were placed in the queue, or a **prioritized** queue from which wells are opened in decreasing order of their **drilling priority**. Well drilling priorities may be set manually in keyword WDRILPRI or calculated from their current potentials according to a function defined with keyword DRILPRI.

The rate of opening new wells from the drilling queue can be limited by specifying the time taken to drill each well with keyword WDRILTIM and allocating drilling rigs to groups with keyword GRUPRIG.

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Additionally, Eclipse 100 can also place an upper limit on the number of open wells in each group, with keyword GECON. A well from the drilling queue is not opened if it would cause the limit to be exceeded. The restrictions set with the keywords WELSOMIN and WDRILRES described in "Automatic opening of wells and well connections" also apply to wells opened from the drilling queue. In addition to drilling new wells, Eclipse 100 can take other actions to postpone a decline in production rate. These include opening new connections in existing wells, decreasing the wells' THP limits, retubing the wells and applying artificial lift. The actions are specified with the keyword PRORDER. These are described more comprehensively in "Group production rules".

Note: The drilling queue should not be confused with the facility described in "Automatic opening of wells and well connections". They are both subject to the same constraints on drilling rate, rig availability and maximum number of wells per group. But wells on AUTO are opened as soon as these constraints allow, whereas wells in the drilling queue are opened only when a group or the field cannot meet its rate target.

Group production rate limits

The engineer can impose **upper limits** on any of the quantities listed on "Group production rate targets - guide rate control", for selected groups and/or the field, using keyword GCONPROD. These limits can be imposed whether or not the group or field has a production target to maintain. If a group's production rate exceeds one of these limits, there is a choice of actions to follow:

1. The worst-offending connections in the worst-offending wells will be closed successively until the rate limit is honored. The 'worst-offending' well or connection here is the one that has the highest ratio of the production rate of the phase violating the limit to the production rate of the well's preferred phase.
2. As 1 above, except that all connections below the worst-offending one in the well will also be closed.
3. The worst-offending wells will be closed successively until the rate limit is honored.
4. *Eclipse 100* The worst-offending well will be plugged back, that is, a length of perforations in the top or bottom open connection is reduced by an amount specified in keyword.
5. The group will be made to produce the offending phase at its maximum allowed rate, according to the group control facilities described in "Group production rate targets - guide rate control". If the group was previously meeting a target rate for another phase, that target will no longer be met but will act as an upper limit for the production rate of that phase. The violated limit in effect becomes the new target.

Note: When a group which was operating under a rate target for one phase, switches to rate control by another phase in this way, the guide rates for any children of this group are not allowed to be zero.

This is to prevent the following situation. Consider, for example, a group under gas production rate control which also has a limit on the group water production rate to control water breakthrough. When water breakthrough occurs in some of the group's wells and the water rate limit is violated, the group switches to water rate production control. Any wells that do not see any water breakthrough would not have any water production potential, and normally this would mean zero guide rates so these pure gas production wells would shut - which is obviously not what is required for a group which is primarily intended for gas production.

Instead, a small positive guide rate is imposed on such wells force them to stay open and operate at their potential. **Unfortunately, sometimes this means that the original group gas rate target is exceeded.** Because of this, it is strongly recommended that `RATE` is not used to control breakthrough in this way. Two possible alternatives are (i) use an alternative procedure such as `WELL` or `CON` under `GCONPROD` in items 11 onwards, or (ii) use a `GUIDERAT` formula that penalizes wells that see breakthrough.

If a reservoir fluid volume rate limit or a production balancing fraction limit is exceeded, only action 5 is performed.

As an example consider a group of oil wells experiencing water breakthrough, such that the liquid rate is approaching its maximum limit. If action 1, 2, 3 or 4 is selected, the wells or connections with the highest liquid-oil ratio (that is the highest water cut) will be successively closed or plugged back. If the group is operating at a target oil rate, this target will continue to be met until there are no wells remaining under group control. If however action 5 is selected, the group control facility will be applied to make the group produce liquid at the maximum liquid rate. The oil rate will decline steadily as the group's water cut increases.

Guide rate control for node-groups

When a node-group (or the field itself) has a rate target, its target rate will be apportioned among its subordinate wells as described in "Group production rate targets - guide rate control", unless a subordinate group either

- has a target rate of its own,
- has been declared unavailable for control at a higher level,
- has been given a production guide rate.

If a subordinate group has a target rate of its own, its production rates will have already been calculated. (The groups under rate control are processed level by level, from the bottom level up to the field.) For example, in figure 8.2 let PLAT-B have a target rate of 50,000 stb/day for oil and SAT-B have a target rate of 30,000 stb/day for liquid. SAT-B's target rate may be a limit that has become a target after it had been violated. SAT-B's oil rate is subtracted from PLAT-B's target, and the remainder is shared between the wells in GR-B1 and GR-B2.

If SAT-B is required to produce 30,000 stb/day of liquid regardless of PLAT-B's target, it should be declared (in keyword `GCONPROD`) unavailable for control at a higher level. Otherwise its wells could come back under control from PLAT-B if ever SAT-B's share of PLAT-B's oil target fell below its current oil rate when producing 30,000 stb/day of liquid. SAT-B's wells would then produce their own share of PLAT-B's oil target, with a combined liquid rate of less than 30,000 stb/day. Declaring a group unavailable to respond to a higher level control will ensure that it continues to operate at its rate target or maximum production capacity independently of any higher level rate control.

A group can be given a production guide rate (in keyword `GCONPROD`) if it is required to produce a specific proportion of a higher level rate target. For example, if the field has an oil rate target of 100,000 stb/day and platforms PLAT-A and PLAT-B are required to produce equal quantities of liquid, then PLAT-A and PLAT-B should be given equal guide rates for the liquid phase. The liquid guide rates are translated into oil guide rates by multiplying them by each platform's oil-liquid ratio from the previous timestep. Both platforms are then given oil rate targets in proportion to their oil guide rates, and placed under control from the higher (field) level. If PLAT-A is unable to provide its share of the field’s oil target, it will produce what it can according to its own limits or capacity, and the remainder of the field’s target will be allocated to PLAT-B.

Prioritization - an alternative to guide rate control

The prioritization option is an alternative means of applying production rate limits to groups. It can be used instead of the guide rate method described so far. In the prioritization option, wells are turned on in decreasing order of priority, the well with the highest priority flowing first, until a group’s production rate limit is exceeded. The flowing wells operate at their individual targets or limits (WCONPROD). However, the well that exceeds the group’s rate limit is cut back to meet the limit, and the wells with priority lower than this subordinate to that group do not flow. The low priority wells not selected for flow are “priority-closed” until they are eventually selected.

The priority number for each well is calculated from its potentials at regular intervals, according to a general formula with user-defined coefficients. The keyword PRIORITY is used to select the prioritization option and to set the coefficients of the priority equation. The equation allows the priority to be set equal to, for example, the potential oil rate, or the reciprocal of the potential gas rate, or the reciprocal of the water cut. You can set individual well priorities independently using the keyword WELPRI, overriding the general equation.

With the prioritization option, upper limits can be set for the oil, water, gas, liquid and reservoir fluid volume rates of each group. The choice of actions on exceeding a limit is the same as described in "Group production rate limits", except that action 5 causes the rate to be reduced by closing the low priority wells instead of cutting all the wells back to flow in proportion to their guide rates. The group production rate limits and actions must be set using the keyword GCONPRI, and not GCONPROD as in the guide rate method.

Eclipse 100

Optionally, a second priority equation can be defined, which may be used instead of the first equation when prioritizing wells to cut back production to obey specific rate limits. For example, if a group gas rate limit is exceeded you may wish to close wells with a high GOR, while if a water rate limit is exceeded you may wish to close wells with a high watercut. When two priority equations are defined, the calculation procedure differs slightly from the single priority equation case. All the available wells are opened initially, then if a group rate limit is violated the wells with the lowest priority according to the formula chosen for that particular limit are closed in turn until the rate limit is no longer exceeded. The last well to be closed is then reopened and made to produce at a rate calculated to meet the limit. If two or more group rate limits are exceeded which use different priority formulae, the lowest priority well is selected according to the formula appropriate to the limit that is exceeded by the greatest fraction. The ‘active’ formula may of course change as each well is closed.

It is possible to mix the two group control strategies - prioritization and guide rates - in a single run. Groups to be controlled by prioritization should have their rate limits set with keyword GCONPRI, while groups using the guide rate method have their rate limits set with keyword GCONPROD. A group can be changed from one method to the other simply by re-specifying its rate limits with the appropriate keyword. The mixing of the two methods, however, is subject to the following restriction: the guide rate method cannot be used by a group subordinate to a group using the prioritization method. Thus, for example, no groups can use the guide rate method if the field has a prioritization rate limit.

When the two group control strategies are mixed, the well rates are allocated according to the following method. First, the production wells subordinate to prioritization groups are solved, selecting the wells for flowing in decreasing order of their priorities. Next, the producers in the remaining parts of the grouping tree are solved, employing the standard guide rate method to apportion the flow rates between them to meet any GCONPROD group targets. If a prioritization group is subordinate to one with a GCONPROD rate target, the wells in the prioritization group will already have had their flow rates set, and the remaining wells will have their rates set to make up the rest of the GCONPROD group's target. Wells from the drilling queue (keyword QDRILL or WDRILPRI) that are not subordinate to a prioritization group will be opened when necessary to meet the GCONPROD group's rate target. The drilling queue is **not** used for opening wells subordinate to prioritization groups, as the prioritization algorithm itself handles the opening and closing of these wells.

Economic limits and cutback limits

The set of economic limits described earlier for wells can also be applied to the overall production of groups and the field as a whole. If the oil or gas production rate of the group (or field) falls below a minimum value, all its production wells are closed. If the group (or field) water cut, gas-oil ratio or water-gas ratio exceeds a maximum value, then the choice of workover options described earlier for wells will be performed on successive worst- offending wells in the group (or field). The economic limit values are set using keyword GECON.

As with the well economic limits, the remedial actions for the group and field economic limits are performed at the end of the timestep in which they were broken. But if a limit is broken by proportionately more than the **tolerance fraction** specified with keyword WLIMTOL, the timestep will be repeated after the action has been taken.

Eclipse 100

There is also an option to cut back the group production rate by a specified fraction each time the group's water cut, gas-oil ratio, gas-liquid ratio, or water-gas ratio exceeds a specified value. The option is invoked with the keyword GCUTBACK. It works in a similar way to the well rate cut-back option controlled by the keyword WCUTBACK.

Group injection control options

Groups, and the field as a whole, can be given targets and limits which govern the injection rate of each phase. The injection controls can be applied independently of any production controls the group may have. The injection controls are set using the keyword GCONINJE.

A group at any level (including the field) can have its injection rate of water and/or gas (and oil in Eclipse 100) controlled to meet one of the following targets:

- Surface injection rate of a particular phase
- Total reservoir volume injection rate of all phases
- Reinjection fraction of a particular phase
- Total voidage replacement fraction of all injected phases.

Eclipse 300

- Wet gas rate (which is the volume that the injected hydrocarbon mixture would occupy at standard conditions if it were an ideal gas).

The first control mode gives the group a user-specified surface injection rate target. The second control mode gives the group a target reservoir volume injection rate for the phase that is equal to the user-specified total rate minus the reservoir volume injection rate of any other phases in the group. The third control mode gives the group a surface injection rate target equal to its production rate of the phase (or alternatively the production rate of another named group) multiplied by the user-specified reinjection fraction. If the phase is gas, gas consumed by the group and any subordinate groups is subtracted from the production rate, and gas imported to the group and any subordinate groups is added to the production rate, before multiplying by the re-injection fraction. (A gas consumption rate and import rate can be specified for any group using keyword GCONSUMP in Eclipse 100 and keywords GRUPFUEL and GADVANCE in Eclipse 300.) In Eclipse 300 gas reserved for sale (keyword GRUPSALE) is also subtracted from the production rate. The fourth control mode gives the group a target reservoir volume injection rate for the phase that is calculated such that the group's total reservoir volume injection rate equals its voidage production rate (or alternatively the voidage production rate of another named group) multiplied by the user-specified voidage replacement fraction.

If the group has a target set for reservoir volume injection rate or voidage replacement fraction for a particular phase, that phase is designated the 'top-up' phase for the run. It is used to top up the total group injection rate to the required target. Its injection target depends on the injection rates of the other phase(s) within the group, and therefore the 'top-up' phase must have its injection controls handled last of all. There can only be one 'top-up' phase at any given time in the simulation - different groups cannot have different 'top-up' phases. An example application of injection control is for a platform to reinject all its available gas (that is, produced gas minus gas consumed for power), and to inject water so that the water and gas injection together replaces the platform's production voidage. Water is the 'top-up' phase here.

Upper limits can be imposed on the four quantities at any group or the field as a whole. If the injection rate exceeds one of a group's upper limits, the violated limit becomes a target for the group. If the group was already injecting at a target rate, the original target will no longer be met but will act as an upper limit for that quantity. For example, a group can inject water to replace its voidage production, subject to a surface rate limit that represents its injection water handling capacity. Obviously only the 'top-up' phase can have upper limits specified for quantities 2 or 4 since a limit could become a target at some stage in the simulation.

Defining the composition of injected gas

Eclipse 300

When gas is injected in compositional simulations, the nature of this gas must be defined. The injected gas may be provided in the following ways:

- The gas may be designated as a named wellstream, whose composition is defined explicitly in keyword `WELLSTRE`.
- The gas may be provided by the production stream of a named group. Gas used for fuel (`GRUPFUEL`) and sales (`GRUPSALE`) is by default subtracted from a source group's available gas supply for reinjection, but the keyword `WTAKEGAS` may be used to alter the priorities for re-injection, fuel and sales. The injection gas supply from the source group may be augmented by advance gas defined in keyword `GADVANCE`. The advance gas will be used for reinjection prior to the gas produced by this group from the reservoir.
- The gas may be provided by the production stream of a named well. The gas supply from the source well may be augmented by advance gas defined in keyword `WADVANCE`. The advance gas will be used for reinjection prior to the gas produced by this group from the reservoir.
- The injected gas may be a mixture from a number of sources. The mixture may be specified by the fractional contribution from each source (keyword `WINJMIX`). Alternatively the gas from a set of sources may be taken in a specified order (keyword `WINJORD`).

Groups with gas injection targets should have the nature of their injected gas defined in one of the above ways with the keyword `GINJGAS`. If there is not enough available gas from the source to fulfil the group's injection requirement, an additional type of gas import named "make-up gas" may be defined for the injecting group. This comes from a named wellstream, which has an infinite supply and will make up the extra requirement to fulfil the group's injection target.

In general, therefore, the composition of the injected gas can vary with the injection rate and the supply of available gas from the sources. When the gas comes from a limited source (namely a group or well production stream), the injection gas is taken from the following supplies in the stated order:

1. The source's advance gas stream (if defined), up to its specified maximum rate.

2. The source's production stream, minus the specified fuel usage and the gas reserved for sale if the source is a group (unless the priorities are reordered by WTAKEGAS).

3. The injecting group's make-up gas stream.

If the injecting group does not have a make-up gas stream defined, there may possibly be a shortfall between its injection requirement and the injection gas supply. By default, the group will continue to fulfil its injection requirement and the shortfall between supply and injection rate will show up as a negative excess in the gas accounting tables (see below). However, if the WAVAILIM keyword is entered the injection rate will be limited by the available gas supply. A shortfall between supply and injection demand will cause the group to reduce its injection rate to match the available gas supply; the group will then be under **availability** control for gas injection.

When the injected gas composition may vary with the injection rate (when there is make-up or advance gas, or when the injection stream is defined by WINJORD), the injection calculation may be iterated a number of times until the estimated injection rate (from which the composition is determined) and the actual injection rate converges. Similarly, when the WAVAILIM keyword is entered and a limited source of gas is shared between two or more injection groups, the injection calculation may have to be iterated a number of times to ensure that, if any groups inject less than their allocated share of available gas, their unused share of available gas is re-allocated to other groups. The maximum number of these iterations, and their convergence tolerance, is controlled by items 3 and 4 of keyword GCONTOL.

Gas accounting tables

Eclipse 300

The gas rate accounting table is generated in the print file at each report step, and provides information on how produced gas is used. Each column is described below:

1. Fuel volume rate

   The rate at which produced gas is used for fuel. (Controlled by the GRUPFUEL keyword.)

2. Sales volume rate

   The rate at which produced gas is used for sales. (Controlled by the GRUPSALE keyword.)

3. Reinjection volume rate

   The rate at which produced gas is reinjected into the reservoir. (Controlled by the GCONINJE and WCONINJE keywords.) This also includes gas which is injected into the reservoir during production balancing of surface gas (controlled by the GCONPROD keyword).

   If item 108 of the OPTIONS3 is non-zero, then this will **not** include any gas injected during the surface gas production balancing group control mode.

4. Excess volume rate

   The rate difference between the gas produced and the gas used to meet the fuel, sales and reinjection requirements.

   If OPTIONS3 item 108 is non-zero, then gas injected during the surface gas production balancing group control mode will **not** be used in the reporting of the excess (only the reinjected gas during the reinjection group control mode).

5. Sum volume rate

   This is the sum of the first four columns and should equal the gas produced by the field for separators that simply accumulate the gas from each stage. (This is the default separator configuration). If gas from an intermediate stage is used as a feed for a subsequent stage and is also a source of reinjection gas, then the sum of the first four columns will not necessarily equal the produced gas rate. This is discussed further in the "Export tables".

6. Export volume rate

   The sum of the excess gas rate and the sales gas rate, that is the rate produced gas is available for export after the fuel and reinjection requirements have been met.

   If `OPTIONS3` item 108 is non-zero, then gas injected during the surface gas production balancing group control mode will **not** be used in the reporting of the export (only the reinjected gas during the reinjection group control mode).

7. Advanced gas volume rate

   The rate at which advanced gas is used to augment produced gas in a combined reinjection stream. (Controlled by the `GADVANCE` or `WADVANCE` keywords.) The advanced gas is only used for reinjection and not as a source of fuel or sales gas.

   This column is not output if the field is operating with a surface gas production balancing limit/target.

8. Make-up gas volume rate

   The rate at which a make-up gas stream is used to top-up advanced and re-injected produced gas in order to meet an injection target. (Controlled by the `GINJGAS` and `WINJGAS` keywords.)

   This column is not output if the field is operating with a surface gas production balancing limit/target.

9. Reinj + Adv + Make

   The sum of volume rates of reinjected produced gas, advanced gas and make-up gas. Note that this may differ from the total field injection gas rate, which could include wellstreams or mixtures not associated with produced gas.

   This column is not output if the field is operating with a surface gas production balancing limit/target.

Export tables

Eclipse 300

Export tables are written to the print file at each report step only when both of the following conditions are met:

- A separator has been defined in which the gas from an intermediate stage is fed into a subsequent stage or gas plant, instead of the stock tank (see item 7 of the `SEPCOND` keyword).
- Gas from an intermediate stage of a separator is used as the source of re-injection gas (see item 5 of the `GINJGAS` keyword).

If the group has a target set for reservoir volume injection rate or voidage replacement fraction for a particular phase, that phase is designated the ‘top-up’ phase for the run. It is used to top up the total group injection rate to the required target. Its injection target depends on the injection rates of the other phase(s) within the group, and therefore the ‘top-up’ phase must have its injection controls handled last of all. There can only be one ‘top-up’ phase at any given time in the simulation - different groups cannot have different ‘top-up’ phases. An example application of injection control is for a platform to reinject all its available gas (that is, produced gas minus gas consumed for power), and to inject water so that the water and gas injection together replaces the platform’s production voidage. Water is the ‘top-up’ phase here. In these circumstances, reinjection from an intermediate separator stage reduces the feed to a subsequent stage and the separator oil rate is thus reduced. This reduced oil rate is reported in the table under the Oil Volume Rate column. This export rate should be distinguished from the produced oil rate, reported elsewhere in the print file, which is the ‘formation’ oil rate prior to any reinjection work. The difference then between the formation oil and export oil is the oil vapor that is contained in the reinjected gas that would otherwise have been separated in the subsequent separator stage. Note that it is the produced oil rate and not the export oil rate that may be controlled by the GCONPROD keyword. (This is also the case for gas production rates, that is, it is the produced gas rate and not the export gas rate that may be controlled.)

The export gas rate is also included in this table. This is the remaining gas rate after reinjection and fuel usage have been accounted for. For this type of separator configuration the sum of the reinjected, fuel, sales and excess gas rates will not necessarily equal the produced gas rate.

If a Gas Plant table has been specified in the separator train that produces an NGL stream (see GPTABLEN or GPTABLE3), then the Export table also reports the reduced NGL rate.

The export oil, gas and NGL rates and totals may also be written to the SUMMARY file using the SUMMARY section keywords (F,G)E(O,G,N)(R,T).

Note: The export rate calculation should be treated with caution when the intermediate separator stage gas is used as part of a mixture to be injected (see the WINJMIX and WINJORD keywords) as it may be unable to take the re-injection into account.

Apportioning group injection targets by guide rate

A group or field target injection rate is apportioned between the subordinate wells or groups in a precisely analogous manner to a production target, as described in "Group production rate targets - guide rate control". A well-group’s target is shared between its group-controlled injectors in proportion to their guide rates, after subtracting the injection rate of any injectors under independent control. An injector’s guide rate can be set with keyword WGRUPCON, otherwise it will be set equal to the well’s injection potential at the beginning of each timestep. Wells are placed under independent control if they cannot inject their share of the group’s target, or if they have been declared unavailable for group control in keyword WGRUPCON. If a group cannot inject its target rate with its existing wells, the drilling queue is scanned for a suitable injector to open.

Similarly a node-group’s (or the field’s) target rate is apportioned among its subordinate wells unless a subordinate group either

- has a target injection rate of its own for that phase, or
- has been declared unavailable for injection control at a higher level for that phase, or
- has been given an injection guide rate for that phase.

Eclipse 100

Group injection guide rates can be used to control the distribution of the injected phase between various groups in the same way as for production control. But as an alternative to setting the guide rates specifically, they can be calculated at the beginning of each timestep according to a choice of two options. One option sets them equal to the groups’ voidage production rate. The other option (available only to the ‘top-up’ phase) sets them equal to the groups’ net voidage, that is the production voidage minus the reservoir volume injection rate in the other phases. An example application is for a platform to reinject its available gas, distributing it to selected groups in proportion to their voidage production rate. The platform could also inject water to replace the overall production voidage, distributing it to all groups in proportion to their net voidage rate. This would give less water to the groups that have gas injection.

Sales gas production control

Eclipse 100

The associated gas from an oil field, which may at the same time have controls on the oil production rate. The option is selected by using the keyword GCONSALE. The sales gas production rate of a group or the field is defined as its total gas production rate, minus its total gas injection rate, minus the gas consumption rate of the group and any subordinate groups, plus the gas import rate of the group and any subordinate groups. (The group gas consumption and import rates are set with the keyword GCONSUMP.)

The sales gas production rate is controlled by reinjecting the surplus gas not required for sale. Sales gas control can be applied independently of any other production controls on the group or field, provided that there is enough injection capacity to inject the surplus gas. There must be one or more gas injectors subordinate to each group with a sales gas target. The group or field is automatically placed under gas reinjection control, and its target reinjection fraction is determined dynamically at each timestep to inject the surplus gas. Other limits on gas injection for the group or field (for example maximum surface rate) can be simultaneously applied using the keyword GCONINJE. The distribution of the injected gas between any subordinate groups can also be controlled by giving these groups upper rate limits and guide rates for gas injection, with the keyword GCONINJE.

If the group or field is producing more surplus gas than the injectors can handle, the drilling queue will be scanned for a new gas injector to open. Failing this, a choice of actions can be performed at the end of the timestep to reduce the gas production in the next timestep. These include closing or working over the worst-offending well, or controlling the group or field gas production rate. There are two options for controlling the group/field production rate.

- The RATE option calculates the gas production rate necessary to meet the sales gas target after allowing for any consumption and the current rate of reinjection, then places the group/field on gas production rate control. The production rate target will remain at this value unless another limit is violated. Note that if the injection capacity were subsequently to increase, the production rate would not increase to take advantage of the increased injectivity.
- The alternative MAXR option also places the group/field under gas production rate control with a similarly calculated production target. But it also removes the reinjection fraction limit (by setting the reinjection fraction limit to 1) and continues to recalculate the gas production rate target according to what the injection rate was in the previous iteration. This option therefore maximizes the production rate when the system is constrained by its injection capacity, because if the injection capacity were subsequently to increase then it would allow the production rate to increase accordingly while still meeting the sales target.

  There is a flaw in this algorithm if the net sales, import and consumption is zero: stagnation of the injection and production amounts results even if increased injection capacity is available. In this special case, the reinjection fraction limit is set to a larger value (100 by default) so that any increased injection capacity can be used. The value of this increased reinjection fraction limit can be controlled using item 103 of the OPTIONS keyword.

  The sales target may not be met exactly, however, because the production rate calculation uses the injection rate from the previous iteration, and also the well rates will not be updated after the first NUPCOL iterations of each timestep. The MAXR option will overwrite the group's gas production rate limit with a value equal to the sales gas target plus the injection rate plus any gas consumption minus any import rate, while the group is operating on gas production rate control. Any user-defined limit on this quantity will be erased.

The RATE and MAXR options require guide rate group control, and are not allowed if prioritization (keyword GCONPRI) is used to control the group/field's production.

If on the other hand the group or field is not producing enough gas to fulfil its sales requirement, corrective action will be taken at the end of the timestep to increase its gas production rate.

- If it is already under gas production rate control, the target rate will be increased by the amount required to meet the sales target.

- If there is a special gas producer (declared with keyword WGASPROD) open and able to increase its production rate, its gas rate target will be increased by a specified increment.
- Otherwise the drilling queue will be scanned for a suitable production well to open, preferentially opening any special gas producers in the queue.

Eclipse 300

A more limited means of controlling sales gas production is available in Eclipse 300. The rate of produced gas required for sale is specified with keyword GRUPSALE. A fuel usage rate may also be set with keyword GRUPFUEL. The group can then be given a reinjection fraction target of 1.0, and it will reinject its produced gas minus its fuel and sales requirements. However, there is no feedback on the production controls if there is an excess or shortfall of gas. If there is insufficient injection capacity to reinject all the surplus gas, the surplus gas that cannot be injected will show up as an “excess” in the gas accounting tables. Similarly, no remedial action will be taken if there is insufficient produced gas to meet the sales target.

Pressure maintenance

The voidage replacement control for group injection (keyword GCONINJE) provides one method of maintaining the reservoir pressure at its current value. This method, however, achieves only approximate pressure maintenance, and for a number of reasons the average reservoir pressure may drift gradually from its original value. A more flexible alternative method of maintaining the reservoir pressure is provided by the keyword GPMAINT.

Using the GPMAINT keyword you can designate one or more groups to maintain the average pressure of the field or a particular fluid-in-place region at a specified target value. These groups will automatically adjust either their production rate or their injection rate of a nominated phase, in response to the difference between the current pressure and its target value. Thus, for example, a group may adjust its injection rate to bring a region's pressure up to a desired target value and maintain it there. Alternatively, if injection capacity is limited, the group may adjust its production rate to maintain the pressure while the injectors operate at their user-specified targets or limits.

The target production or injection rate of each pressure maintenance group is calculated at every timestep from an expression derived from elementary control theory. The control strategy employed is that of a ‘proportional + integral controller’. There are two associated control parameters which you can use to adjust to achieve an acceptable compromise between the speed of response and the stability of the pressure. The group's flow rate will, honor any additional constraints that you apply.

Calculating flow targets for wells under group control

When any form of group control option is being used (keywords GCONPROD, GCONPRI, GCONINJE and GCONSALE), the flow targets of certain wells are directly influenced by the behavior of other wells in the field. For example, the target rates of producers under group control must allow for the production from wells that are under other modes of control within the group. Also injection wells performing reinjection or voidage replacement have flow targets that depend on the behavior of the production wells.

For the group targets to be met exactly, the flow targets of the wells under group control must be recalculated at every Newton iteration. But when the well flow targets change at each iteration, the convergence rate of the iterations is reduced. This is because the Jacobian matrix calculation does not include the terms representing the mutual dependency of the well rates.

As a compromise, the well flow targets are updated in the first NUPCOL Newton iterations of each timestep, and are kept constant for any subsequent iterations. (NUPCOL’s value is initialized by keyword NUPCOL in the RUNSPEC section, and can be reset at any time in the simulation with keyword NUPCOL or GCONTOL in the SCHEDULE section.) Thus the group/field flow targets will be met exactly if the timestep converges within NUPCOL iterations. Any changes in the solution after the first NUPCOL iterations may cause the group/field flow rates to drift slightly away from their targets. In general, larger values of

NUPCOL will result in the group/field flow targets being met more accurately, but perhaps at the expense of requiring more Newton iterations to achieve convergence.

Eclipse 100

The discrepancy between a group/field flow target and its actual flow rate can be limited to within a specified tolerance fraction by using the keyword GCONTOL. If a flow target is not met to within this tolerance, when there is sufficient capacity to meet the target, the well flow rates are recalculated to meet that target regardless of the value of NUPCOL.

Workover and drilling rig constraints

An option is available to limit the rate at which workovers are performed and new wells are drilled, according to the availability of workover and drilling rigs. Each group may have up to 5 workover rigs and up to 5 drilling rigs assigned to it, using the keyword GRUPRIG. Several groups may share the same rig, if required. The default is that one rig of each type is shared by the whole field.

The availability of a workover rig determines the rate at which automatic workovers can be performed. Automatic workovers are performed after violation of a well or group economic limit, when in Eclipse 100 connections on AUTO opening are being opened, and optionally after a group flow limit is exceeded. The time taken for each well to be worked over is set with keyword WORKLIM, and the appropriate rig is made unavailable for other workovers for this duration. A workover on a particular well is postponed if all its available workover rigs are already occupied until the end of the timestep. When a workover is postponed, the well continues to operate ‘uneconomically’ until its rig is available. A flag can be set with keyword WDRILTIM which causes a well to be temporarily closed while a workover takes place.

Similarly, the availability of a drilling rig determines the rate at which new wells can be drilled automatically (for example, from the drilling queue set up with keyword QDRILL or WDRILPRI, or in Eclipse 100 using the 'AUTO' option in keyword WCONPROD or WCONINJE). The time taken for each well to be drilled can be set individually with keyword WDRILTIM, and the appropriate rig is not able to drill any other wells for this duration. The drilling of a well is postponed if all its available drilling rigs are already occupied until the end of the timestep. It is drilled during the timestep in which a rig becomes available.

Vertical flow performance

Calculations involving tubing head pressure are handled using Vertical Flow Performance (VFP) tables, which are supplied as input data. These tables relate the bottom hole pressure of a well to the tubing head pressure at various sets of flowing conditions.

There are two types of VFP table:

- Injector tables, which describe the vertical flow performance of injection wells, entered with keyword VFPINJ. These are tables of:
  - bottom hole pressure versus injection rate
  - bottom hole pressure versus tubing head pressure
- Producer tables, which describe the vertical flow performance of production wells, entered with keyword VFPPROD. Producer tables may also be used describe how the temperature of the fluid at the tubing head varies with flowing conditions in the well (see also WHTEMP keyword). These are tables of bottom hole pressure (or) tubing head temperature versus:
  - production rate of oil, liquid or gas
  - tubing head pressure

- water-oil ratio, water cut, or water-gas ratio
- gas-oil ratio, gas-liquid ratio, or oil-gas ratio
- artificial lift quantity

Note that there are three different ways of defining the flow rate, water fraction and gas fraction in producer VFP tables. The engineer should select the most appropriate definitions for the type of well being modeled. For example, if it is an oil well which eventually waters out, a VFP table defined in terms of the liquid rate, water cut and gas-liquid ratio would be suitable. For gas wells, the VFP table should be defined in terms of the gas rate, water-gas ratio and oil-gas ratio.

The artificial lift quantity (ALQ) is a fifth variable that can be used to incorporate an additional lookup parameter, such as the level of artificial lift. For example, if gas lift is being applied, the artificial lift quantity could refer to the injection rate of lift gas into the well bore. Or if the lift mechanism is a downhole pump, the artificial lift quantity could refer to the pump rating. In Eclipse 100 artificial lift can be switched on automatically when a well's production rate falls below a specified limit (see keyword `WLIFT`), subject to optional constraints on the total group or field lift capacity (see keyword `GLIFTLIM`).

Eclipse 100

By default, VFP tables are interpolated linearly in all variables. However, there is an option to apply cubic spline interpolation to the ALQ (Artificial Lift Quantity) variable. This option is primarily intended to be useful for the Gas Lift Optimization facility, in which the gradient of the wells' performance with respect to the lift gas injection rate is important for deciding which wells could gain the most benefit from an increased lift gas supply. The option is enabled with the `VFPTABL` keyword.

Note: If the actual simulation conditions fall outside the range given in a VFP table, then linear extrapolation is used. This can produce unrealistic results and should be avoided (by ensuring that each VFP table covers all conditions that will arise during the simulation).

Two or more different wells may use the same VFP table, provided they have similar depths, deviations and internal diameters. Any difference between the bottom hole datum depths of the wells and that of the VFP table will be taken into account using a simple hydrostatic correction to the BHP obtained from the VFP table. In Eclipse 100 individual wells may also have a further user-defined adjustment made to their BHPs obtained from the table, specified with keyword `WVFPDP`. This could be applied, for example, to help match a well's flow rate at a given THP.

When a well is under THP control, its flow rate is determined by the intersection of its inflow performance relationship (IPR) and the VFP curve interpolated at the current value of its water and gas flowing fractions, on a plane of BHP versus flow rate. Often the VFP curve for a producer is U-shaped, and when there are two intersections the one with the higher flow rate is chosen (the other being unstable). If the IPR does not intersect the VFP curve, the well is deemed to have died. Occasionally, wells dying suddenly in this fashion may cause problems in the simulation. The Eclipse 100 keyword `WVFPEXP` controls some possible remedies, either by using explicit values of the water and gas fractions for the interpolation of the tables or by the use of 'stabilized' VFP curves.

Producer tables are input with the keyword `VFPPROD`, and injector tables are input with the keyword `VFPINJ`. A separate interactive program VFP *i* is available to calculate VFP tables, using a choice of multiphase flow correlations to calculate the pressure losses within the tubing string. The program has been specifically tailored for Eclipse, in that it can produce any type of VFP table required by the simulator in a form which can be included directly in the input data. It contains extensive graphical facilities for examining and visually editing VFP tables in cross-section and displaying them in 3D.

Well name and well list template matching

The template matching feature of Eclipse enables multiple wells to be referenced using a single template containing wildcard characters. The wildcard characters supported by Eclipse are the asterisk (*) to match zero or more characters, and the question mark (?) to match a single non-empty character. When these characters are required at the start of a template they should be preceded by a backslash (\) to differentiate between alternative uses of the symbols, for example to indicate a well list (*) and as an alias for the triggering wells with an action block (?). All well name matches are case insensitive.

Consider the specification of the following wells in the `SCHEDULE` section.

```
WELSPECS
 'PR10-OIL'   GR-A1   3   2   1*   OIL /
 'PR11-OIL'   GR-A1   3   5   1*   OIL /
 'GAS-PR1'    GR-A2   3   8   1*   GAS /
 'GAS-PR2'    GR-A2   3   3   1*   GAS /
 'IN1-GAS'    GR-A3   4   4   1*   GAS /
 'IN2-GAS'    GR-A3   4   6   1*   GAS /
 'IN10-WAT'   GR-A4   5   2   1*   WAT /
 'IN11-WAT'   GR-A4   6   5   1*   WAT /
 'X'          GR-A5   7   2   1*   LIQ /
 'Y'          GR-A5   7   5   1*   LIQ /
/
```

New well lists consisting of all the wells containing the sub-string ‘PR’ and another consisting of all the wells containing the sub-string ‘IN’ could then be created in the following manner:

```
WLIST
 '*PR-WELS'   NEW    '\*PR*' /
 '*IN-WELS'   NEW    '\*IN*' /
/
```

Similarly to shut all wells with names composed of seven characters the question mark symbol can be used.

```
WELOPEN
 '\???????'    SHUT /
/
```

In this context the preceding backslash (\) is not strictly required; however, if the keyword appears within an action block it is necessary to differentiate between the wildcard character and the alias for the triggering wells. For example, when the following action is triggered all wells with a water cut greater than 0.25 will be shut, whilst all wells with single character names will have their liquid rate target / limit halved.

```
ACTIONX
 ACT1 100000 /
 WWCT '*' > 0.25 /
/
-- either use the alias ? to select triggering wells
WELOPEN
 '?'   SHUT /
/
-- or match template against all the wells in the field with a single
-- character name
WTMULT
 '\?'   LRAT   0.5 /
/
ENDACTIO
```

To maintain back-compatibility a single asterisk still represents all the wells in the field. Alternatively, the above action could test only a subset of wells, for example those belonging to a well list

```
ACTIONX
 ACT1 100000 /
```

```
 WWCT '*PR-WELS' > 0.25 /
/
```

or matching a given template

```
ACTIONX
 ACT1 100000 /
 WWCT '\*GAS*' > 0.25 /
/
```

However, note that it is not possible to use the template matching mechanism to select a subset of the wells triggering the action.

Wildcard characters can be included in well names and well lists in all appropriate `SCHEDULE` keywords. In addition wildcard characters can be included in well names and well lists for `W*` and `LW*` keywords in the `SUMMARY` section. For example

```
-- well water production rate for wells containing the
-- substring 'PR' and the wells X & Y
WWPR
 '\*PR*' 'X' 'Y' /
-- well mode of control for wells belong to well lists
-- containing the substring 'WELS'
WMCTL
 '**WELS*' /
```

Note: For all keywords, well name and well list templates containing wildcard characters must be enclosed within quotes.

The `WTEMPQ` keyword can be used to further explore the well template matching mechanism. The `WTEMPQ` keyword writes a list of all the currently defined wells matching the supplied well name or well list template to the PRT file.

Eclipse 100 The pre-2006.1 well name and well list behavior can be restored through item 93 of the `OPTIONS` keyword.

Eclipse 300 The pre-2006.1 well name and well list behavior can be restored through item 94 of the `OPTIONS3` keyword.

Multisegment wells

x	Eclipse 100
x	Eclipse 300

The multisegment well model is a special extension, which is available in both Eclipse 100 and Eclipse 300. It provides a detailed description of fluid flow in the well bore. The facility is specifically designed for horizontal and multilateral wells, although it can of course be used to provide a more detailed analysis of fluid flow in standard vertical and deviated wells. Like the standard well model, the equations are solved fully implicitly and simultaneously with the reservoir equations, to provide stability and to ensure that operating targets are met exactly. A description of the model, with two example applications, is published in *Holmes et al (1998)*.

The detailed description of the fluid flowing conditions within the well is obtained by dividing the well bore (and any lateral branches) into a number of one-dimensional segments. Each segment has its own set of independent variables to describe the local fluid conditions. In Eclipse 100 there are four variables per segment: the fluid pressure, the total flow rate and the flowing fractions of water and gas. In Eclipse 300 there are N_{comps} +3 variables per segment (where N_{comps} is the number of hydrocarbon components), the fluid pressure, the total molar flow rate and the flowing molar densities of water and each hydrocarbon component. Each segment may have completions in one or more reservoir grid blocks, or none at all if there are no perforations in that location. The variables within each segment are evaluated by solving material balance equations for each phase or component and a pressure drop equation that takes into account the local hydrostatic, friction and acceleration pressure gradients. The pressure drop may be calculated from a homogeneous flow model where all the phases flow with the same velocity, or a 'drift flux' model that allows slip between the phases. Alternatively, the pressure drop may optionally be derived from precalculated VFP tables, which can potentially offer greater accuracy and provide the ability to model chokes.

Benefits

The multisegment well model has all the capabilities of the Wellbore Friction option, and provides several additional features and greater flexibility. The benefits of using this well model include:

- Improved handling of multilateral topology. Data input has been designed for ease of use with multilateral wells. There are two separate keywords to define the segment structure and the perforated lengths. Individual branches are identified by their branch number. Branches may, if required, have sub-branches.
- Improved modeling of multiphase flow. Use of the drift flux model or precalculated pressure drop tables can produce more accurate pressure gradients than the homogeneous flow treatment used in the other well models. The pressure gradient can vary from segment to segment throughout the well; it is calculated fully implicitly in each segment using the local flowing conditions.
- Flexibility. The variable number of grid block completions per segment allows the choice between using many segments for greater accuracy or fewer segments for faster computation.
- The multisegment well model contains several facilities for modeling advanced wells. Specific segments can be configured to model flow control devices such as chokes, by providing a pressure drop table that describes their pressure loss characteristics as a function of flow. Various 'built in' flow control device models are also provided:
    - Variable pressure loss multipliers can be used to represent 'smart' devices which react to isolate high GOR or WOR regions.
    - Flow limiting valves can also be used to represent 'smart' devices which react to limit the flow of oil, water or gas through a segment.

- Other advanced well components such as sub-critical valves, ‘labyrinth’ devices and downhole separators can also be modeled.

- Crossflow can be modeled more realistically, as the fluid mixture can vary throughout the well. Complex crossflow regimes may potentially occur in multilateral wells, including branch-to-branch crossflow and crossflow within individual branches. In the other well models, if crossflow occurred, the mixture flowing back into the reservoir would reflect the average contents of the well bore, regardless of which parts of the well were crossflowing.
- Wellbore storage effects can be modeled more accurately by dividing the well bore into several segments. The drift flux model allows phases to flow in opposite directions at low flow rates, so that phase redistribution within the well bore can occur during shut-in well tests.

The multisegment well model

Segment structure

A multisegment well can be considered as a collection of segments arranged in a gathering tree topology, similar to the node-branch structure of a network in the Network option. A single-bore well will, of course, just consist of a series of segments arranged in sequence along the wellbore. A multilateral well has a series of segments along its main stem, and each lateral branch consists of a series of one or more segments that connects at one end to a segment on the main stem (figure 8.3). It is possible, if required, for lateral branches to have sub-branches; this may be useful when modeling certain inflow control devices as part of the network of segments.

The segment network for each well may thus have any number of ‘generations’ of branches and sub-branches, but it must conform to gathering-tree topology. Segments within the gathering tree topology can be connected in order to form loops - this facility is described in "Looped flow paths".

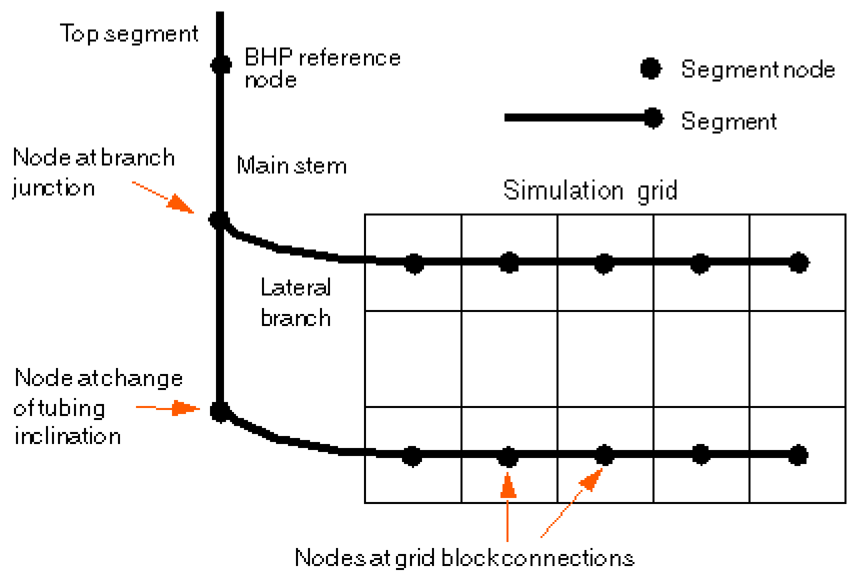

Figure 8.3. A multilateral, multisegment well

Each segment consists of a **node** and a **flow path** to its parent segment’s node. (The term ‘flow path’ is used here rather than ‘branch’, as the latter term is reserved for lateral branches in multilateral wells, each of which may contain several segments.) A segment’s node is positioned at the end that is furthest away from the wellhead (figure 8.4). Each node lies at a specified depth, and has a nodal pressure which is determined by the well model calculation. Each segment also has a specified length, diameter, roughness, area and volume. The volume is used for wellbore storage calculations, while the other attributes are properties of its flow path and are used in the friction and acceleration pressure loss calculations. Also associated with each segment's flow path are the flow rates of oil, water and gas, which are determined by the well model calculation.

Note: The top segment is a special case. It does not have a parent segment so no pressure drop is calculated along this segment's flow path. Hence there is no need to specify diameter, roughness and area for this segment. The equation for the pressure drop along the flow path is replaced by a boundary condition appropriate to the well's mode of control. See "The top segment" for details.

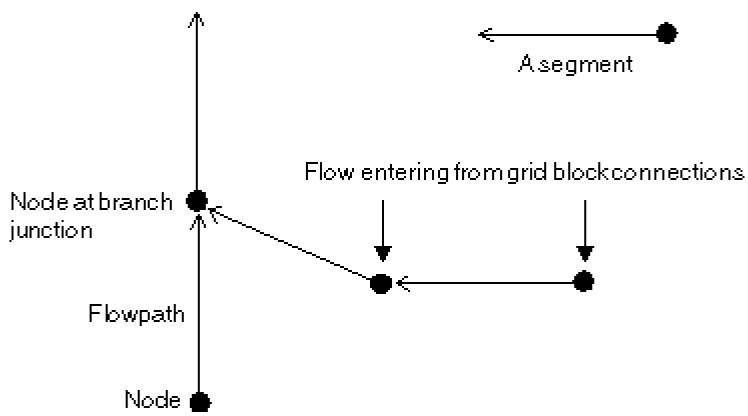

Figure 8.4. Well segments

A segment node must be positioned at each branch junction. Flow from the formation through grid-block-to-well connections also enters the well at segment nodes. A segment (or to be more precise, its node) can accept flow from any number of grid block connections (figure 8.5). If you wish to have a separate segment node for each connection (for greatest accuracy) then it is best to position the nodes to lie at the center of their corresponding grid block connections. But it is also possible to reduce the number of segments by allocating two or more grid block connections to each segment. In this case it best to position each segment node to lie at the center of the perforated interval that is allocated to the segment. However, it is not possible to allocate a single grid block connection to more than one segment.

Eclipse allocates each grid block connection to the segment whose node lies nearest to it. If necessary, you can override this by explicitly specifying the segment to which the connection should be allocated in the `COMPSEGS` (or `COMPSEGL`) keyword.

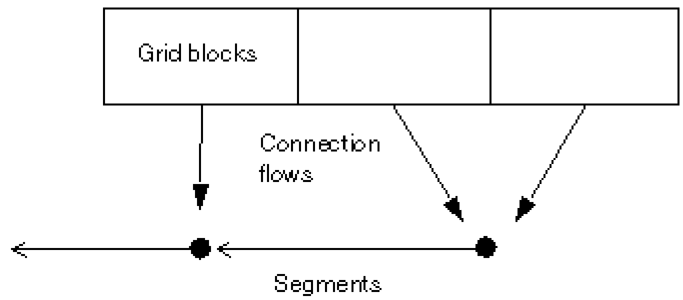

Figure 8.5. Allocating connection flows to segments

It is also possible to position segment nodes at intermediate points along the wellbore, for example where the tubing properties or inclination angle change. Additional segments can be defined to represent chokes or inflow control devices. In common with other aspects of numerical simulation, the optimum number of segments required to model a particular well depends on your preferred compromise between speed and accuracy.

Inflow performance

The flow of fluid between a grid block and its associated segment's node is given by the inflow performance relationship

$$q_{pj} = T_{wj} M_{pj} \left(P_j + H_{cj} - P_n - H_{nc} \right) \qquad \text{Eq. 8.65}$$

where

q_{pj} is the volumetric flow rate of phase p in connection j at stock tank conditions in Eclipse 100 or reservoir conditions in Eclipse 300. The flow is taken as positive from the formation into the well, and negative from the well into the formation.

T_{wj} is the connection transmissibility factor.

M_{pj} is the phase mobility at the connection. If the node is injecting fluid into the formation, the ratio of the phase mobilities reflects the mixture of phases in the segment

P_j is the pressure in the grid block containing the connection.

H_{cj} is the hydrostatic pressure head between the connection's depth and the center depth of the grid block.

P_n is the pressure at the associated segment's node n.

H_{nc} is the hydrostatic pressure head between the segment node n and the connection's depth.

Note the two components of the hydrostatic head, H_{cj} and H_{nc}. In multisegment wells the connection depths are not forced to be equal to the grid block center depths as they are in wells (Figure 8.6). They may be set independently in keyword `COMPSEGS` (or `COMPSEGL`), or defaulted to allow Eclipse to calculate them from the segment depths and their position within the segment. The hydrostatic head H_{nc} between a connection and its corresponding segment node is calculated implicitly in the usual way from the density of the mixture in the segment. The hydrostatic head H_{cj} between the connection and the grid block center is calculated from an average of the fluid densities in the grid block.

Eclipse 100

The average is weighted according to the fluid relative permeabilities, to give a smooth variation as the fluid in the block is displaced by another phase. This average density is calculated explicitly at the beginning of each timestep. If the use of explicit quantities causes oscillations from one timestep to the next, they can be damped by setting item 63 of the `OPTIONS` keyword, which averages the head term with its value at the previous timestep.

Eclipse 300

The average is weighted according to the implicit fluid saturations, to give a smooth variation as the fluid in the block is displaced by another phase.

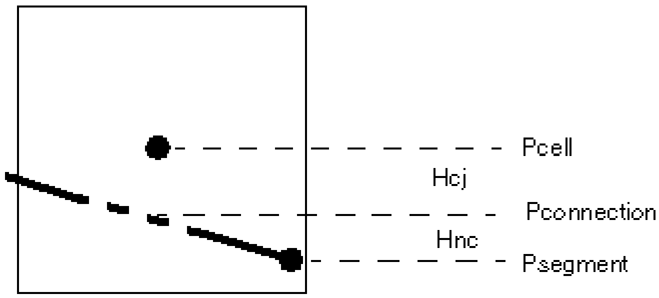

Figure 8.6. Hydrostatic head components

The pressure loss calculation

The following options are available for calculating the pressure drop

Using a built-in homogeneous flow model

This assumes there is no slip between the phases. The hydrostatic term depends on the density of the fluid mixture within the segment, which is simply a flow-weighted average of the phase densities. The frictional pressure drop calculation is the same as the method used by the Wellbore Friction option (see "Calculating the frictional pressure loss"). The acceleration pressure drop across the segment is the difference between the velocity heads of the inflowing and outflowing mixture (see "Acceleration pressure loss").

Interpolating the pressure drop from pre-calculated VFP tables.

VFP tables can be constructed (using, for example, the program VFP *i*) to describe the pressure drop along a certain length of tubing at the appropriate angle of inclination. The pressure drop along a segment is interpolated from the respective VFP table and can be scaled according to the length or depth span of the segment. Optionally, the table can be constructed to contain just the friction pressure drop, and Eclipse computes the hydrostatic pressure drop from the properties of the fluid within the segment (an option useful for undulating sections of tubing). VFP tables can also be used to describe pressure losses across chokes (the program VFP *i* contains the Sachdeva choke model).

Using a built-in 'drift flux' slip model.

This is a simple correlation that is continuous across the range of flowing conditions. It includes the effects of slip as two components: a local vertical slip velocity superimposed on a velocity ratio resulting from the profiles of holdup fraction and flow velocity across the tubing cross-section. The local vertical slip velocity will produce countercurrent flow (phases flowing in opposite directions) when the mixture velocity is small enough.

Using a built-in model for a specific flow control device.

There is a built-in model to represent sub-critical flow through a valve with a specified throat cross-section area. See keyword `WSEGVALV` for further details.

There is a built-in 'flow limiting valve' model, which is a hypothetical device that reacts dynamically to limit the flow rate of oil, water or gas (at surface conditions) through a segment to a specified maximum value, by sharply increasing the frictional pressure drop across the segment if the limit is exceeded. See keyword WSEGFLIM for further details.

There is also a built-in model that corresponds to a ‘labyrinth’ flow control device, which may be used to control the inflow profile along a horizontal well or branch by imposing an additional pressure drop between the sand face and the tubing. See keyword WSEGLABY for further details.

Another built-in model calculates the pressure drop across an inflow control device that is placed around a section of the tubing and diverts the inflowing fluid from the adjacent part of the formation through a sand screen and then into a spiral constriction before it enters the tubing. See keyword WSEGSICD for further details.

There is also a built-in model that corresponds to an ‘autonomous’ inflow control device. This works in a similar way to the ‘spiral’ inflow control device. See keyword WSEGAICD for further details.

The homogeneous flow model is the default method; it is simple to compute, and gives results similar to the Wellbore Friction option (when there is no crossflow). VFP tables provide the most flexible, and potentially the most accurate, method of calculating pressure losses. They can be constructed (for example with VFP *i*) using an appropriate choice of multiphase flow correlation. The drift flux model is less accurate than using properly calculated VFP tables, because of the need for simplicity and continuity in the built-in correlation. The main reason for its presence is to provide a means of modeling phase separation and countercurrent flow at low flow velocities, such as in shut-in well tests. Various parameters in the drift flux model that govern the slip velocity can be adjusted with the keyword WSEGFMOD.

The frictional pressure loss across individual segments can be adjusted by specifying a scaling factor with the keyword WSEGMULT. Optionally, the scaling factor can be defined to vary with the water-oil ratio and gas-oil ratio of the mixture flowing through the segment. A variable multiplier may be applied, for example, to model the operation of an adjustable flow control device whose purpose is to choke back production from high WOR or GOR regions.

The top segment

The top segment of the well is special. Its node corresponds to the well’s bottom hole reference depth, so that the top segment’s pressure is the same as the well’s BHP. Since there is no parent segment above it, no pressure losses are calculated across the top segment. Its pressure loss equation is replaced by a boundary condition appropriate to the well’s mode of control. The top segment may, however, have a volume, which is used in wellbore storage calculations.

It is natural to consider whether there is any advantage in modeling the whole wellbore in segments up to the tubing head, in which case the well’s reported BHP is really its THP. In general, this is not recommended. It is best to position the top segment just above the perforations and any lateral branch junctions, and to model the pressure loss between this point and the tubing head in the standard way with a VFP table. One reason is that, if the well cannot operate at a fixed THP because its IPR lies below the bottom of the ‘U’ shaped VFP curve, the standard logic in Eclipse recognizes this situation and shuts the well in cleanly; whereas if the whole well were modeled as a series of segments up to the tubing head, the Multisegment Well Model may simply not converge.

A situation where it may be useful to model the whole wellbore as a series of segments is when a well test is being simulated and you need to model wellbore storage effects in detail. Here the drift flux model could be applied to model the process of phase separation in the wellbore during a shut-in test.

External sources and sinks

It is possible to import water or gas into a segment from a source that is external to the reservoir grid. The import rate may be defined either as a constant value or as a function of the segment’s pressure (see keyword WSEGEXSS). The imported fluid is added to fluid flowing through the segment. There are a number of possible applications for this facility. Gas import may be used to model the effects of lift gas injection, to examine the kickoff process for example. Water import may be used to take account of the water used to power a downhole hydraulic pump if this is injected into the wellbore.

Eclipse 300

There is also an option to remove fluid from a segment into an external sink, at a rate that is a function of the segment's pressure. This facility is intended for use with the Eclipse 300 Thermal option to model recirculating wells.

Secondary wells

Eclipse 300

As an extension to the facilities afforded by the keyword WSEGEXSS, further controls may be exerted over well segments by defining one or more segments to be secondary wells. This is accomplished using the keyword WSEGWELL which associates a specified well segment with a secondary well name. Once this association has been established, the secondary well keywords SCONPROD, SCONINJE and STEST may be employed to control the production from and injection into the segments. The keywords WSEGEXSS and WSEGWELL are mutually exclusive for a given well segment. With the exception of the top segment, any segment in the well topology may be associated with a secondary well name.

There are a number of possible applications for this facility. For example, a dual-tubing well may be modeled in terms of a multisegment well where the conventional (primary) well can be used to represent one tubing and a well segment can be used to represent the other tubing by defining this as a secondary well. There are, however, a number of factors to consider when constructing this form of model. For example, if the primary well shuts, the secondary well will also shut. Moreover, the multisegment well topology can become complex if both secondary producers and injectors are modeled via secondary wells. In addition, the controls available with a secondary well are more limited than those available with a conventional well.

The fluid production and injection rates for secondary wells may be specified in terms of a target rate plus one or more limits. The limits include the minimum operating pressure for secondary producers (item 10 of the SCONPROD keyword) and the maximum operating pressure for secondary injectors (item 8 of the SCONINJE keyword). In some cases, depending upon the multisegment well topology, it may be conveniently an additional pressure constraint to an alternative segment to that nominated as the secondary well. For example, for a producer, it may be more appropriate to specify a minimum operating pressure for the secondary tubing segment whose depth is similar to the reference depth for the primary well BHP than for the secondary well segment itself with a surface outlet.

The alternative pressure constraint is optional and is specified in terms of an alternative segment number and associated pressure limit (items 12 and 13 respectively of the SCONPROD keyword for producers and items 13 and 14 respectively of the SCONINJE keyword for injectors). For secondary producers, the calculation of the molar rate required to provide the specified target fluid rate utilizes the minimum operating pressure of the secondary well segment (item 10 of the SCONPROD keyword). If the pressure in the secondary segment reduces to within a few percent of this minimum operating pressure, in some circumstances fluctuations in the secondary production rate can occur. In such cases, items 12 and 13 of the SCONPROD keyword may be used to define a cutoff pressure for the secondary well segment by specifying item 12 as the secondary well segment number and item 13 as the cutoff pressure. In this case a suggested value for item 13 is 10% higher than the value for item 10.

Looped flow paths

The gathering tree topology required by multisegment wells normally prevents the formation of looped flow paths within the well itself. However, it is possible to add looped flow paths as follows:

- The basic topology of the well must be specified in the normal manner using the WELSEGS keyword. One or more pairs of nodes of different segments must be made physically coincident. (In practice, this just means that the segment nodes should be specified as being at the same true vertical depth in the WELSEGS keyword.) Note that this in itself is not sufficient to form a looped flow path and these node pairs will still act as though they are unconnected in any way.
- Each such pair of segment nodes must be “linked” using the WSEGLINK keyword. This is the key step that instructs each pair of segment nodes to communicate and form looped flow paths. At least one of these segments must be at the end of a branch.

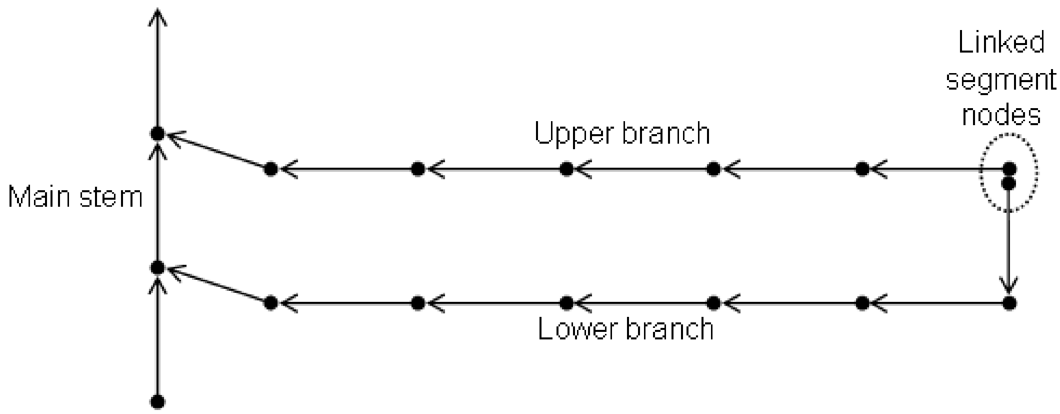

Figure 8.7. Simple looped multisegment well

A pair of linked segment nodes is shown in the previous figure. This link forms a looped flow path between the upper and lower branch in this multisegment well. Flow from the lower branch can pass into the main stem of the well directly or via the upper branch, and flow from the upper branch can pass into the main stem directly or via the lower branch. The pressure variations in the upper and lower branches will determine the direction of the flow.

The pressure drop equations for the segments whose nodes are linked are unaffected, but the material balance equations for one of the segments are replaced instead by equations that enforce equality of the solution (pressure, flowing fraction of water and flowing fraction of gas) at the linked segment nodes.

Handling of linked segment properties

In the WSEGLINK keyword, the first segment of each pair specified is known as the **primary linked segment**, while the second segment of the pair is known as the **secondary linked segment**. If the first segment of the pair is at the end of a branch, but the second is not, then the first segment will be treated as the secondary link segment instead. It is the secondary linked segment that has its material balance equations replaced by the solution equality equations.

- If connection flows or external source /sink flows are specified for both the linked segments (using either the COMPSEGS or COMPSEGL keywords), then these flows will all be assigned to the primary linked segment (as it is the only one of the pair that has the material balance equations which incorporate these terms).
- Any volume associated with the secondary linked segment will be allocated as additional volume in the primary linked segment.
- part from the replacement of the material balance equations by linked segment node solution equality equations in the secondary linked segment, each of these segments is exactly the same as any other

segment used to define a multisegment well. Hence the pressure drop equation in either of the linked segments can be modified in any of the ways described in "The pressure loss calculation".

Example application: Modeling an unpacked annulus

As an example of an application of looped flow paths within multisegment wells, consider the modeling of flow in an unpacked annulus.

One branch of the well can be used to model the tubing and may contain multiple segments, each in direct communication with its immediate neighbors. This tubing branch is shown as a dashed line in figure 8.8.

Another branch of the well can be used to model the annulus; the segments on this branch are also in direct communication with their immediate neighbors. This annulus branch is shown as a dotted line in figure 8.8. Flow only enters the well via connections at annulus segments.

Flow can move freely along the annulus and pass into the tubing via control valves at intervals along the well. Without the ability to model looped flow paths, the flow from a certain annulus segment would have been constrained so that it would have to flow into the tubing using a specific valve. However, due to the presence of looped flow paths, flow in an annulus segment is free to pass through any of the valves into the tubing.

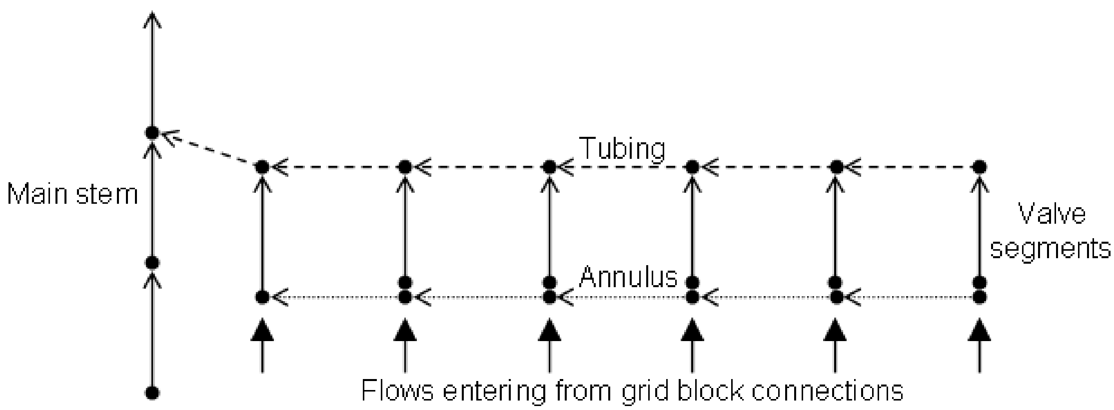

Figure 8.8. Schematic of a multisegment well with an unpacked annulus

Impact of looped flow paths on the well linear equations

From *Holmes et al (1998)*, in the absence of such linked segment nodes, the linear system arising from the well equations can be solved very efficiently due to the special structure of the well matrix. However, as soon as loops are added, this special structure is lost and an alternative method for solving the well system must be used. Eclipse uses an iterative method called GMRES (*Saad, 1996*) to solve the well linear system arising from the presence of loops, with an incomplete LU factorization pre-conditioner used to accelerate the convergence. If required, the parameters that control the GMRES solver and the pre-conditioner can be modified using the `WSEGSOLV` keyword.

Eclipse 100

If the pre-conditioned GMRES solver fails to converge or encounters serious problems then the system will be solved again using a pivoted Gaussian elimination, hereafter referred to as a “direct” solution. This is an expensive method both in memory and CPU cycles, but is likely to succeed should GMRES fail. The simulator will issue warning messages saying that a direct solution had been used; you should alter settings to eliminate these messages if possible. The Gaussian elimination `WSEGSOLV` can be suppressed by setting item 182 in the `OPTIONS` keyword; in this case the unconverged GMRES result will be accepted.

Restrictions

- One of the segments linked using the WSEGLINK keyword must be at the end of a branch.
- It is not permissible for the secondary linked segment to have zero effective absolute roughness. That is, item 8 in the WELSEGS record defining the secondary linked segment properties must be non-zero.
- A segment cannot be both a secondary link segment and a labyrinth inflow control device with non-positive configuration number (specified using the WSEGLABY keyword). If the need for this particular situation arises, the specification of the primary and secondary link segments can be reversed in the WSEGLINK keyword.
- The top segment of the well cannot be part of a loop so it cannot appear in a row of the WSEGLINK keyword.

Pressure drops from VFP tables

VFP tables offer the most flexible, and potentially the most accurate, means of determining the pressure drop across each segment. Interpolating the pressure drop from a table is also considerably faster than calculating it from a multiphase flow correlation. It is not necessary to supply a separate table for each segment. Segments of similar diameter, inclination and roughness can share the same table; Eclipse scales the pressure drop according to the length or depth span of each segment.

VFP tables should be constructed (by, for example, the program VFP *i*) for a representative length of tubing at the appropriate angle of inclination, using a suitable multi-phase flow correlation. The tables should be in the form of production well VFP tables, and are entered in the SCHEDULE section with the keyword VFPPROD. Standard VFP tables give the BHP as a function of the flow rate, the THP, the water and gas fractions, and optionally the ALQ. When applied to a well segment, the BHP should be interpreted as the segment's nodal pressure (which is its inlet pressure), and the THP should be interpreted as the pressure at the node of its neighboring segment towards the wellhead (that is the segment's outlet pressure). This is similar to the interpretation used for VFP tables constructed to model branch pressure drops in the Network option.

Pressure drop scaling

The second item in the first record of a VFP table (see keyword VFPPROD) normally contains the bottom hole datum depth of the table. However, when the table is used to supply segment pressure drops, it should contain either the length or the depth span (bottom depth - top depth) of the representative length of tubing for which the table was calculated. When Eclipse interpolates a segment's pressure drop from the table, it will scale the pressure drop in proportion to the segment's length or depth span.

Instructions to scale the pressure drop according to the segment's length or depth span are entered in the seventh item of keyword WSEGTABL. The choice should be consistent with the quantity supplied in the second item of the first record of the VFP table. The choice of scaling quantity will only make a difference if the same table is to be used for segments having a range of inclinations. The friction pressure drop is proportional to the length, while the hydrostatic pressure drop is proportional to the depth span; the choice should be based on whichever is the dominant component of the overall pressure drop. Obviously, horizontal segments should be scaled according to length.

There is also the option not to scale the interpolated pressure drop at all, irrespective of the length or depth span of the segment. This choice would be appropriate when the segment represents a choke, since the pressure drop across a choke does not vary in proportion to the overall length of the device.

As well as the automatic scaling according to the segment's length or depth span, a user-defined scaling factor can be applied to the interpolated pressure drop. This is defined in keyword WSEGMULT. A typical use of this would be to inhibit manually the flow through a segment representing a flow control device by increasing the pressure drop across it. Optionally, the scaling factor can be defined to vary with the water-oil ratio and gas-oil ratio of the mixture flowing through the segment, for example to model the operation of an adjustable flow control device whose purpose is to choke back production from high WOR or GOR regions. Alternatively the variation with WOR and GOR could be built directly into the VFP table.

Pressure drop components

VFP tables normally describe the combined effect of friction and hydrostatic pressure losses along the representative length of tubing. The tables do not supply the information necessary to resolve the combined pressure drop into these separate components. Accordingly, the combined pressure drop obtained from a VFP table is reported as the friction pressure drop; the hydrostatic pressure loss is reported as zero.

The tables also usually include an acceleration pressure loss due to expansion of the gas phase in its passage down the pressure gradient. However, they cannot take into account any acceleration pressure loss caused by additional fluid entering the segment (for example through the perforations). Eclipse calculates the total acceleration pressure loss and add it to the overall pressure loss interpolated from the table if this is requested in item 6 in Record 1 of keyword WELSEGS. Ideally, in this case, the VFP tables ought to be constructed without including any acceleration pressure loss, or the component due to gas expansion is accounted for twice. The VFP *i* program has a switch in the Calculation Control panel to turn off the acceleration pressure loss calculation. The acceleration component, however, is generally small in comparison with the overall pressure drop, and in the majority of cases can be neglected.

There is an option to supply VFP tables containing the friction pressure drop only, and have Eclipse calculate the hydrostatic pressure drop across the segment and add it to the interpolated pressure drop. Friction-only pressure drop tables can be constructed, of course, by defining the representative length of tubing used for calculating the table to be horizontal. This option is useful to apply to gently undulating sections of tubing, to overcome the need for separate tables for each inclination angle. The hydrostatic pressure drop calculated by Eclipse is based on a homogeneous flow model, so slip effects are neglected.

Reverse flow

A complication arises when fluid is flowing through the segment in the opposite direction expected for a production well, that is, away from the well head. The sign of the flow is then negative. In production wells this situation may arise only when crossflow occurs, and the flow velocities are not likely to be large. If reverse flow is likely to occur, it is possible to extend the VFP table into the negative flow region. VFP *i* can produce tables whose first few flow values are negative, if requested. The table must not, however, have a point at zero flow, since the water and gas fraction values are singular there.

There are three options for handling cases where the flow is less than (or more negative than) the first flow value in the VFP table. The option for each segment is set in the sixth item in keyword WSEGTABL.

- Reverse the flow, look up the VFP table, then reverse the pressure drop.

  This option is recommended for use in horizontal segments or segments representing chokes, or with ‘friction only’ VFP tables (see "Pressure drop components"), in which the table’s pressure drop has no hydrostatic component. In these cases the pressure drop can simply be reversed when the flow reverses. The first flow value in the VFP table should be positive. For negative flows, the ‘mirror image’ of the table is used. For flows in between + and - the first value, the pressure loss is linearly interpolated between these two points.

- Fix the lookup value of the flow rate at the first flow point in the table.

  This option is recommended for use in vertical or inclined segments in which the friction pressure loss in reverse flow is expected to be small compared with the hydrostatic pressure loss. The ‘reverse flow’ option above is not suitable in these cases as the hydrostatic pressure loss does not reverse when the flow reverses. If the first flow value in the VFP table is positive, it should be low enough to give a negligible friction pressure drop, so that the calculated friction pressure drop will remain negligible during reverse flow.

- Extrapolate the pressure drop linearly with the flow.

  This should be used with caution, as substantial extrapolation (especially into the negative flow region) may result in stability problems.

Chokes

VFP tables can also be constructed to model the pressure loss across flow control devices such as chokes. For example, VFP *i* can produce a table based on the Sachdeva choke model, incorporating both critical and sub-critical flow regimes. The effects of varying the choke diameter can also be incorporated into the table, by using the ALQ variable. VFP *i* can create a table of pressure losses across several different choke diameters, with the diameters represented by the ALQ variable in the appropriate units. The selected ALQ definition for this purpose should be 'BEAN'. It is then possible to model the effects of varying the choke diameter at any time in the simulation by changing the look-up ALQ value of the VFP table (item 8 in keyword WSEGTABL).

Segments representing chokes should **not** have their pressure losses scaled according to length or depth span (WSEGTABL item 7), and the most appropriate option for dealing with reverse flow is to reverse the pressure loss (WSEGTABL item 6).

Injection wells

Injection wells would normally have single-phase water or gas flowing through their segments, so there should be less need to use VFP tables to model the pressure losses. However, multiphase flow can occur if there is crossflow with reservoir fluid entering at one or more perforated intervals. If you wish to use VFP tables to model the pressure loss across segments in an injection well, the possibility of multiphase flow requires the VFP tables to be of the **production** type (keyword VFPPROD). They should therefore be prepared as for a production well, but with negative flow values since the predominant direction of flow is away from the well head.

The drift flux slip model

A slip model enables the phases to flow with different velocities. In general, in all but downward flow, the gas phase will flow with a greater velocity than the liquid phase. Slip thus increases the liquid holdup fraction for a given flowing gas-liquid ratio, thereby increasing the hydrostatic head component.

The program VFP *i* contains several one-dimensional slip correlations which it uses to compute pressure losses in the tubing. These, however, are not suitable for implementing directly in Eclipse. A model suitable for use in multisegment wells ought to have the following attributes:

Simplicity

The calculation has to be done many times in each segment at every timestep and so this requires a model that offers rapid and efficient calculation.

Continuity

The model must cover the complete range of flowing conditions without any first order discontinuities. Any such discontinuities would prevent the iterations of the well solution from converging. This requirement rules out methods involving flow regime maps, because the calculated pressure gradient changes discontinuously from one flow regime to another, unless some form of smoothing is applied.

Differentiability

The fully implicit solution of the multisegment Well Model requires the calculation of derivatives of the phase flow rates and the pressure drop for the Jacobian matrix. Ideally these derivatives should be calculated analytically, as calculating numerical derivatives with respect to each main variable would be costly in CPU time.

In addition to these requirements, it would be useful for the model to work in the countercurrent flow regime, allowing the heavy and light phases to flow in opposite directions when the overall flow velocity is small. This would enable Eclipse to model the separation of phases within the well bore that occurs when a well is shut-in at the surface, and also the accumulation of water in undulating sections of a horizontal well.

Model formulation for gas-liquid slip

Drift flux models express the gas-liquid slip as a combination of two mechanisms. The first mechanism results from the non-uniform distribution of gas across the cross-section of the pipe, and the velocity profile across the pipe. The concentration of gas in the gas-liquid mixture tends to be greater nearer the center of the pipe and smaller near the pipe wall. The local flow velocity of the mixture is also greatest at the center of the pipe. Thus, when integrated across the area of the pipe, the average velocity of the gas tends to be greater than that of the liquid. The second mechanism results from the tendency of gas to rise vertically through the liquid due to buoyancy.

A formulation that combines the two mechanisms is

$$v_g = C_0 j + v_d \qquad \text{Eq. 8.66}$$

where

v_g is the flow velocity of the gas phase, averaged across the pipe area

C_0 is the profile parameter (or distribution coefficient) resulting from the velocity and gas concentration profiles

j is the volumetric flux of the mixture

v_d is the drift velocity of the gas.

The volumetric flux of the mixture is same as the average mixture velocity, and is the sum of the gas and liquid superficial velocities

$$j = v_{sg} + v_{sl} = \alpha_g v_g + \left(1-\alpha_g\right) v_l \qquad \text{Eq. 8.67}$$

where α_g is the gas volume fraction, averaged across the pipe area.

The average liquid flow velocity is thus

$$v_l = \frac{1-\alpha_g C_0}{1-\alpha_g} j - \frac{\alpha_g}{1-\alpha_g} v_d \qquad \text{Eq. 8.68}$$

Several of the one-dimensional slip correlations take the general form of equation 8.66. The task here is to develop relationships for C_0 and v_d that are simple, continuous and differentiable, and which extend into the countercurrent flow regime. The derivation of these relationships is described in *Shi et al (2005b)*. The relationships contain some adjustable parameters that can be tuned to fit observations. In *Shi et al (2005b)* the parameter values are optimized to fit a series of experiments performed in 15 cm (6 in) diameter tubing. In *Shi et al (2005a)* the optimization process was improved and the model was extended to allow for the observation that the presence of free gas can disrupt the slip between oil and water, depending on the pipe inclination. The default values of the parameters in the model in Eclipse are those of the original implementation described in the subsequent sections of this chapter, but you may request Eclipse to use the optimized parameters from *Shi et al (2005b)* or *Shi et al (2005a)* with the keyword `WSEGDFMD`.

If you choose to apply the drift flux model to any segment in a well, Eclipse employs a different set of main variables for all the well's segments. In particular, the first variable is the volumetric flux j instead of the total flow rate.

The profile parameter

Zuber and Findlay (1965) have shown that the value of C_0 can range from 1.0 to 1.5. A number of 1-D flow correlations take a value of 1.2 to apply in the bubble and slug flow regimes: for example, the models of Aziz, Govier and Fogarasi, *Ansari et al (1994)*, and *Hasan and Kabir (1988)*. Accordingly the simulator sets C_0 to a constant value of 1.2 at low values of α_g and j.

At high velocities, however, the profiles flatten out and C_0 approaches 1.0. Moreover, C_0 must approach 1.0 as α_g approaches 1.0. In fact the product $\alpha_g C_0$ must never exceed 1.0.

Because of this, the simulator used the following relation for C_0

$$C_0 = \frac{A}{\left[1 + (A-1)\beta^{*2}\right]} \qquad \text{Eq. 8.69}$$

The coefficient A is defaulted to 1.2, but can be set by you to any value not less than 1.0. The purpose of the term β^* is to make C_0 reduce to 1.0 at high values of α_g and j. It is evaluated from the formula

$$\beta^* = \frac{(\beta - B)}{(1 - B)} \quad \text{subject to} \quad 0 \le \beta^* \le 1 \qquad \text{Eq. 8.70}$$

β is a parameter that approaches 1.0 as α_g approaches 1.0, and also as the mixture velocity approaches a particular value. The velocity of the onset of the annular flow regime is taken to be the velocity at which profile slip vanishes. The transition to annular flow occurs when the gas superficial velocity v_{sg} reaches the 'flooding' value that is sufficient to prevent the liquid from falling back against the gas flow, v_{sgf}. An expression for v_{sgf} is given in the next section, with equation 8.76. This gives the following expression for β

$$\beta = \max\left(\alpha_g \, , F_v \frac{\alpha_g |\ j\ |}{v_{sgf}}\right) \qquad \text{Eq. 8.71}$$

The parameters B and F_v are user-definable constants. B represents the value of the gas volume fraction or the flooding velocity fraction at which the profile parameter C_0 will start to reduce from the value A. It defaults to 0.3. F_v is a multiplier on the flooding velocity fraction. It defaults to 1.0, but the profile flattening can be made more or less sensitive to the velocity by respectively increasing or decreasing its value.

The default values for the parameters A and B can be changed by requesting either of the optimized parameter sets from *Shi et al (2005b)* or *Shi et al (2005a)* with the keyword `WSEGDFMD`. Note that in the `SHI-04` model there is no profile slip, A = 1.0, making the value of B irrelevant.

There is a limit on the maximum value that the parameter B may have. Intuitively you would expect the gas superficial velocity $v_{sg} = \alpha_g v_g$ to increase with both α_g and j. Thus:

$$\frac{\partial}{\partial \alpha_g}(\alpha_g C_0) > 0 \text{ and } \frac{\partial}{\partial j}(jC_0) > 0 \qquad \text{Eq. 8.72}$$

Both these criteria will be satisfied if

$$B < \frac{2-A}{A} \qquad \text{Eq. 8.73}$$

With the default value of $A = 1.2$ the constraint is $B < 0.6667$.

The drift velocity

The derivation of the expression for the drift velocity v_d is described in *Shi et al (2005b)*. The expression is derived by combining data on the limits of countercurrent flow made under a variety of flowing conditions and interpolating between them to avoid discontinuities. The method honors observations of gas-liquid relative velocities at low and high gas volume fractions, and joins them with a ‘flooding curve’.

The rise velocity of gas through a stationary liquid is related to v_d by

$$v_g(v_l = 0) = \frac{v_d}{1-\alpha_g C_0} \qquad \text{Eq. 8.74}$$

As α_g approaches zero, the gas velocity should approach the rise velocity of a single bubble, for which Harmathy recommended the value of $1.53v_c$ where v_c is the characteristic velocity

$$v_c = \left[\frac{\sigma_{gl} g(\rho_l - \rho_g)}{\rho_l^2}\right]^{1/4} \qquad \text{Eq. 8.75}$$

where

σ_{gl} is the gas-liquid interfacial tension,

g is the acceleration due to gravity,

ρ_l and ρ_g are the liquid and gas phase densities.

At high values of α_g, the gas velocity should approach the ‘flooding velocity’, which is just sufficient to support a thin annular film of liquid and prevent it from falling back against the gas flow. Wallis and Makkenchery obtained the relation

$$v_g(v_l = 0) = K_u\left(\frac{\rho_l}{\rho_g}\right)^{1/2} v_c \qquad \text{Eq. 8.76}$$

where K_u is the ‘critical Kutateladze number’, which is related to the dimensionless pipe diameter

$$D^* = \left[\frac{g(\rho_l - \rho_g)}{\sigma_{gl}}\right]^{1/2} D \qquad \text{Eq. 8.77}$$

by the function in table 8.1.

D^*	K_u
≤ 2	0
4	1.0

D^*	K_u
10	2.1
14	2.5
20	2.8
28	3.0
≥ 50	3.2

Table 8.1: The relationship between the dimensionless pipe diameter and the critical Kutateladze number

To interpolate between these two extremes the simulator used the flooding curve described by Wallis to define the limit of the countercurrent flow regime. The curve must be ramped down at low gas fractions in order to match Harmathy's bubble rise velocity. A linear ramp is applied between two values of the gas volume fraction, a_1 and a_2. The resulting relation for the drift velocity is

$$v_d = \frac{\left(1-\alpha_g C_0\right)\ C_0 K\left(\alpha_g\right) v_c}{\sqrt{\frac{\rho_g}{\rho_l}}\alpha_g C_0 + 1-\alpha_g C_0} \qquad \text{Eq. 8.78}$$

where

$K\left(\alpha_g\right) = 1.53/C_0$ when $\alpha_g \leq a_1$

$K\left(\alpha_g\right) = K_u$ when $\alpha_g \geq a_2$

and a linear interpolation of these values when $a_1 < \alpha_g < a_2$. The default values of a_1 and a_2 are set to 0.2 and 0.4 respectively, to match data used by Zuber and Findlay to demonstrate the transition from the bubble flow regime. Their default values can be changed by requesting either of the optimized parameter sets from *Shi et al (2005b)* or *Shi et al (2005a)* with the keyword `WSEGDFMD`.

The formulation described above is intended to model the behavior of gas-liquid mixtures in a vertical pipe. To scale the drift velocity for inclined flow it is multiplied by an inclination factor

$$v_d(\theta) = m(\theta) v_d \,|_{\theta = 0} \qquad \text{Eq. 8.79}$$

where θ is the angle of inclination from the vertical and $v_d|_{\theta = 0}$ is given by equation 8.78. By default Eclipse uses the inclination factor given by *Hasan and Kabir (1999)*

$$m(\theta) = (\cos\theta)^{0.5}(1+\sin\theta)^2 \qquad \text{Eq. 8.80}$$

Hasan and Kabir present the formula as valid for oil/water flow for $\theta < 70°$, but the simulator applies this for all inclination angles. For horizontal segments the drift velocity becomes zero.

Item 2 of keyword `WSEGDFMD` allows you to override this default formula for the inclination factor. You may input your own table of m vs. θ with keyword `WSEGDFIN`. Note that if you supply your own table it is not written to the `RESTART` file, so it must be present in all restarted data sets. Alternatively, you can ask Eclipse to use a built-in table that reproduces the `SHI-03` model described in *Shi et al (2005b)*. This is the recommended option if you select the `SHI-03` model's parameter set in item 1 of keyword `WSEGDFMD`. Note that the built-in table has $m > 1$ for vertical pipes, since it was necessary to rescale v_d at all values of the inclination angle, including $\theta = 0$, to fit the experimental observations. If you opt for the `SHI-04` model described in *Shi et al (2005a)* you are recommended to select the corresponding relation for the inclination factor, which is

$$m(\theta) = 1.85(\cos\theta)^{0.21}(1+\sin\theta)^{0.95} \qquad \text{Eq. 8.81}$$

Note that this also has $m > 1$ for vertical pipes.

Three-phase flow and oil-water slip

The drift flux model described above was formulated for two-phase gas-liquid mixtures. Three-phase mixtures flowing in the wellbore, and oil-water mixtures, use a different model.

Three-phase mixtures, first use the formulation above to model the total slip between the gas phase and the combined liquid phase. The properties of the combined liquid phase (density, viscosity) are calculated as an average of the oil and water properties, weighted according to the volume fraction α of each phase. This treatment neglects any tendency of the oil-water mixture to form emulsions, but it is in common practice as a way of applying two-phase flow correlations to oil-water-gas mixtures in a well. The gas-liquid interfacial tension is calculated as

$$\sigma_{gl} = \frac{\alpha_o \sigma_{go} + \alpha_w \sigma_{wg}}{\alpha_o + \alpha_w} \quad \text{Eq. 8.82}$$

σ_{go} and σ_{wg} are the gas-oil and water-gas interfacial tensions. They can either be supplied as tabulated functions of pressure with the keywords STOG and STWG respectively (in Eclipse 100), or calculated internally from built-in correlations obtained from *Beggs (1991)*. In Eclipse 100 the reservoir fluid temperature is unknown and is arbitrarily set to 160 °F (71.1 °C) for these correlations. In Eclipse 300 the temperature is set to the temperature of the EOS region corresponding to the well. Then, having obtained the velocity of the combined liquid phase, an oil-water slip model is used to resolve the velocities of the oil and water phases.

By default, Eclipse uses the formulation of *Hasan and Kabir (1999)* to model the slip between oil and water in the liquid phase. This takes the form of

$$v_o = C_0' v_l + v_d' \quad \text{Eq. 8.83}$$

This model suggests a value of 1.2 for the profile parameter, which becomes unity when oil is the continuous phase at $\alpha_o > 0.7$. The simulator implements this as a continuous function of the holdup fraction of oil in the liquid phase, $\alpha_{ol} = \alpha_o / (\alpha_o + \alpha_w)$

$$\begin{aligned} C_0' &= A \quad \text{when } (\alpha_{ol} \le B1) \\ C_0' &= 1 \quad \text{when } (\alpha_{ol} \ge B2) \\ C_0' &= A - (A-1)\left(\frac{\alpha_{ol} - B1}{B2 - B1}\right) \quad \text{when } (B1 < \alpha_{ol} < B2) \end{aligned} \quad \text{Eq. 8.84}$$

The values of the coefficients default to $A = 1.2$, $B1 = 0.4$ and $B2 = 0.7$. A condition equivalent to equation 8.73, that the oil superficial velocity should increase with the oil holdup fraction, requires

$$B1 < (2 - A)B2 \quad \text{Eq. 8.85}$$

The oil-liquid drift velocity is calculated as

$$v_d' = 1.53 v_c' (1 - \alpha_{ol})^2 \quad \text{Eq. 8.86}$$

where

$$v_c' = \left[\frac{\sigma_{ow} g (\rho_w - \rho_o)}{\rho_w^2}\right]^{1/4} \quad \text{Eq. 8.87}$$

σ_{ow} is the oil-water interfacial tension. This can either be supplied as a tabulated function of pressure with the keyword STOW (in Eclipse 100) or calculated internally as the difference between the values of σ_{go} and

σ_{wg} obtained from the above correlations. Hasan and Kabir state that the drift velocity should become zero when oil is the continuous phase, but the simulator keeps the form of equation 8.84 because when $\alpha_{ol} = 0.7$ the drift velocity will have diminished by an order of magnitude from its value at $\alpha_{ol} = 0$.

If you select the `SHI-03` or `SHI-04` model in item 1 of keyword `WSEGDFMD`, Eclipse will use a different formulation for oil-water slip based on the results described in *Shi et al (2005b)* or *Shi et al (2005a)* respectively. In these experiments no profile slip was observed between the oil and water phases, and accordingly the parameter A in equation 8.85 is set equal to 1.0. The parameters $B1$ and $B2$ are redundant since $C_0{}'$ is always 1.0. In place of equation 8.79 the oil-liquid drift velocity is calculated as

$$v_d{}' = 1.53 v_c{}'(1 - \alpha_{ol})^{n'} \quad \text{Eq. 8.88}$$

where n' is set equal to 0.95 or 1.0 respectively.

The oil-water drift velocity in inclined segments is scaled in a similar way as described in the previous section for the gas-liquid drift velocity, using an inclination factor $m(\theta)$ as shown in equation 8.80. By default Eclipse uses Hasan and Kabir's relation (*Hasan and Kabir, 1999*) as shown in equation 8.79. Alternatively you can request Eclipse in keyword `WSEGDFMD` to use a table of m vs. θ instead. You may then input your own inclination factor table with keyword `WSEGDFIN` or let Eclipse use its built-in table, which is optimized to fit the experimental results described in *Shi et al (2005b)*. This is the recommended option if you select the optimized set of model parameters in keyword `WSEGDFMD`. Note that the built-in table has different inclination factors for gas-liquid and oil-water drift velocities. If you opt for the `SHI-04` model described in *Shi et al (2005a)* you are recommended to select the corresponding relation for the inclination factor, which is

$$m(\theta) = 1.07\cos\theta + 3.23\sin 2\theta - 2.32\sin 3\theta \quad \text{Eq. 8.89}$$

This relation is valid up to 88 º from the vertical. Between 88º and 90º (horizontal) Eclipse will extrapolate $m(\theta)$ linearly to zero.

The effect of gas on oil-water slip

The presence of free gas entrained in the liquid can disrupt the slip between the oil and water phases, as *Shi et al (2005a)* concludes. It proposes that the inclination factor $m(\theta)$ for oil-water slip should be multiplied by a factor $m_g(\theta, \alpha_g)$ that describes this effect as a function of the pipe inclination and the gas volume fraction:

$$\begin{aligned} m_g(\theta, \alpha_g) &= 1 - \frac{\alpha_g}{a_3(\theta)} \quad \text{when} \alpha_g < a_3 \\ m_g(\theta, \alpha_g) &= 0 \qquad \text{when} \alpha_g > a_3 \end{aligned} \quad \text{Eq. 8.90}$$

The oil-water drift velocity decreases to zero as the gas volume fraction increases from zero to the value a_3 and remains at zero at all higher values of α_g. The cutoff value a_3 is a function of the pipe inclination. At angles between 0 and 70 degrees from the vertical only a small amount of gas is required to eliminate the drift between oil and water, while for near horizontal flows gas has a much smaller effect (due to segregation from the liquid). The following equation approximates the observations of *Shi et al (2005a)*

$$a_3(\theta) = 0.017 e^{\theta_r^{3.28}} \quad \text{Eq. 8.91}$$

where θ_r is the angle from the vertical in radians ($= \pi\theta°/180$).

You can include the gas effect in the calculation of the drift velocity by selecting the `SHI-04` option in item 3 of keyword `WSEGDFMD`. You are recommended to select this option whenever you select the `SHI-04` model in item 1. However, the `SHI-04` gas effect option may be used with any of the drift flux models available in item 1, but note that it will not reduce the profile slip which is present in the ORIGINAL model where $A = 1.2$.

Friction

The calculation of the friction pressure loss is based on the formulation used in the correlation of *Hagedorn and Brown (1965)*

$$\delta P_f = \frac{C_f f L w^2}{A^2 D \rho} \qquad \text{Eq. 8.92}$$

where

f is the Fanning friction factor

L is the length of the segment

w is the mass flow rate of the fluid mixture through the segment

A is the segment's area of cross-section for flow

D is the segment's diameter

ρ is the in-situ density of the fluid mixture

C_f is a units conversion constant

2.679E-15 (METRIC), 5.784E-14 (FIELD), 1.523E-13 (LAB), 2.644E-15 (PVT-M).

The Fanning friction factor is calculated using the same method as in the Wellbore Friction Option (see "Calculating the frictional pressure loss"), but with the Reynolds number formulated as

$$R_e = \frac{C_r D w}{A \mu} \qquad \text{Eq. 8.93}$$

C_r is a units conversion constant

0.01158 (METRIC and PVT-M), 0.01722 (FIELD), 0.02778 (LAB).

The viscosity μ is calculated as an average of the phase viscosities weighted according to their volume fraction.

The above formulation becomes equivalent to the homogeneous model used in the Wellbore Friction option when there is no slip between the phases.

Acceleration pressure loss

The acceleration pressure loss across a segment is the difference between the velocity head of the mixture flowing across the segment's outlet junction and the velocity heads of the mixture flowing through all its inlet junctions (its main inlet and any side branches connecting with the segment).

$$\delta P_a = H_{v\mathrm{out}} - \sum_{inlets} H_{vin} \qquad \text{Eq. 8.94}$$

The velocity head of the mixture flowing through a junction is

$$H_v = \frac{0.5 C_f w^2}{A^2 \rho} \quad \text{Eq. 8.95}$$

For the outlet junction flow, A is the cross-section area of the segment. For inlet junction flows, A is the maximum of the cross-section areas of the segment and the inlet segment. Thus the acceleration pressure loss will give the Bernoulli effect at restrictions but no pressure recovery at expansions. Note also that, by including the full velocity head of any inflow from side branches, the assumption is that the inflow is directed into the segment with no momentum loss.

Inflow of reservoir fluid into the segment through grid block connections is **not** included in the inflow velocity head. The acceleration pressure loss thus takes account of the need to accelerate this fluid from zero velocity to the segment's outflow velocity.

Downhole separators (Eclipse 100)

Downhole separators are available in the Eclipse 100 version of the multisegment well model.

A downhole separator can separate water or free gas from the flowing mixture within the well, and send it along a lateral branch to reinject into the formation.

Defining a downhole separator

A segment is designated as a downhole separator with the keyword `WSEGSEP`. The segment must have three segments connecting with it:

- An inlet segment, from which the wellstream enters the separator.
- A water or gas offtake segment, which should be the beginning of the lateral branch that injects the separated fluid back into the formation.
- An outlet segment, through which the remaining fluid flows to continue its journey towards the wellhead; this should normally be the separator segment's outlet segment on the same branch.

A downhole pump is required to maintain the flow through the separator and reinject the offtake fluid into the formation. Pumps may be placed in any segment except the separator itself, but the best location for the pump to control the separation process is the separator offtake segment, or another segment in the offtake branch (a 'pullthrough pump'). There is a choice of two methods for modeling the pump:

- For a pullthrough pump for a downhole water separator, there is a simple device model with an additional term to limit the carryover of oil into the water offtake branch (keyword `WSEGPULL`).
- Pumps can be modeled by VFP tables (keyword `WSEGTABL`). Note that a pump in the offtake branch should be modeled by a VFP table with negative flow values, since the flow is in the direction away from the wellhead.

Figure 8.9 shows schematically a multisegment well with a downhole separator extracting water to reinject into the formation. The segments show the direction of positive flow (towards the wellhead) to illustrate that the offtake branch has negative flow.

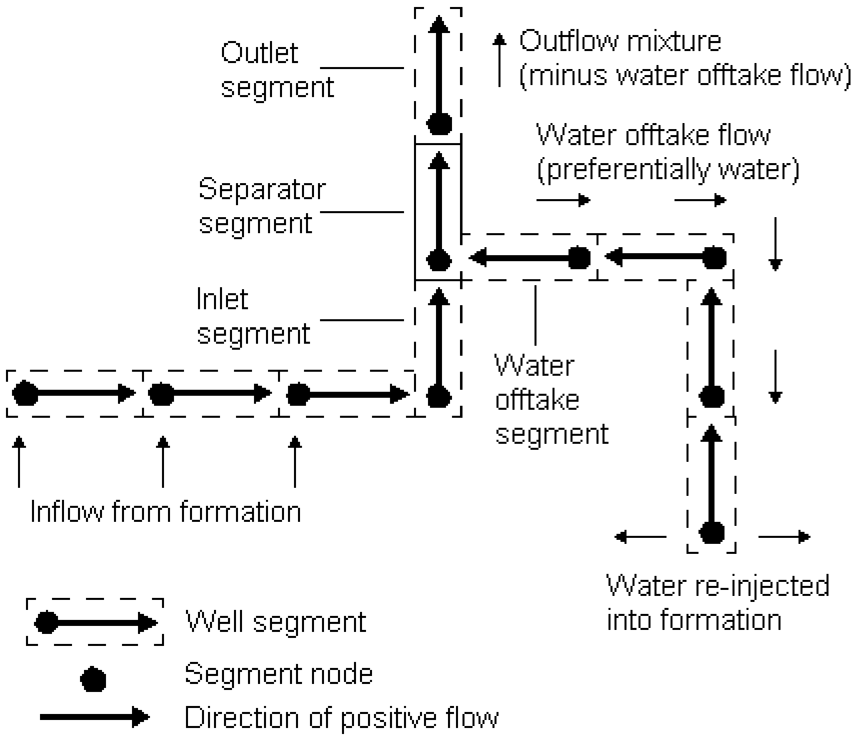

Figure 8.9. Segment diagram for a well with a downhole water separator

The separator model

The flow rate of the fluid along the offtake branch depends on the pressure in the branch (which may be controlled by a pump), the pressure in the formation, the pressure losses along the branch and its injectivity index. The action of the separator is to modify the content of the mixture flowing into the offtake branch through the separator offtake, increasing the fraction of the offtake's preferential phase. This is done by transforming the free oil, water and free gas holdup fractions in the offtake segment.

The holdup fractions α^{of} of the mixture entering the offtake are calculated as a function of the holdup fractions α^{sep} of the remaining fluid within the separator segment and a separation efficiency E. For a water separator offtake the transformation is

$$\begin{aligned} \alpha_w^{of} &= E + (1 - E)\alpha_w^{sep} \\ \alpha_g^{of} &= (1-E)\alpha_g^{sep} \\ \alpha_o^{of} &= (1-E)\alpha_o^{sep} \end{aligned} \qquad \text{Eq. 8.96}$$

Obviously, water cannot be taken off unless it is present within the separator. Thus E must approach zero as α_w^{sep} approaches zero. This functionality is provided by keeping E as a user-defined constant when α_w^{sep} is above a certain value α_{lim} and letting E linearly approach zero when α_w^{sep} falls below the value α_{lim},

$$E = E_{max} \quad \text{when } \alpha_w^{sep} > \alpha_{lim}$$
$$E = \frac{\alpha_w^{sep}}{\alpha_{lim}} E_{max} \text{ when } \alpha_w^{sep} < \alpha_{lim} \qquad \text{Eq. 8.97}$$

E_{max} and α_{lim} are user-defined constants for the offtake, specified in keyword WSEGSEP.

The local volumetric flow rate of each free phase through the offtake is proportional to its holdup fraction α^{of}. Free oil (and free gas) may contain dissolved gas (and vaporized oil) according to the R_s (and R_v) of the phase within the separator segment. The flows at surface conditions are thus

$$Q_o^{of} = Aj\left(\alpha_o^{of} b_o^{sep} + R_v^{sep} \alpha_g^{of} b_g^{sep}\right)$$
$$Q_w^{of} = Aj\alpha_w^{of} b_w^{sep} \qquad \text{Eq. 8.98}$$
$$Q_g^{of} = Aj\left(\alpha_g^{of} b_g^{sep} + R_s^{sep} \alpha_o^{of} b_o^{sep}\right)$$

where

A is the area of the offtake,

j is the volumetric flux of the mixture (that is its flow velocity)

b is the reciprocal of the phase formation volume factors.

The separator segment uses the homogeneous flow model, so all phases flow with the same velocity.

The flows through the outlet junction (towards the wellhead) are similar functions of α^{sep}, the holdup fractions of the fluid remaining within the separator.

Gas separators act in a similar manner to remove free gas (at the separator segment's pressure) and send it preferentially into the offtake segment. They have an equivalent transformation of holdup fractions between the separator and the offtake

$$\alpha_g^{of} = E + (1-E)\alpha_g^{sep}$$
$$\alpha_w^{of} = (1-E)\alpha_w^{sep} \qquad \text{Eq. 8.99}$$
$$\alpha_o^{of} = (1-E)\alpha_o^{sep}$$

where

$$E = E_{max} \quad \text{when } \alpha_g^{sep} > \alpha_{lim}$$
$$E = \frac{\alpha_g^{sep}}{\alpha_{lim}} E_{max} \text{ when } \alpha_g^{sep} < \alpha_{lim} \qquad \text{Eq. 8.100}$$

Behavior of the model

An oil/water separation system powered by electric submersible pumps is described in *Bowers et al (2000)*. A pump is necessary to boost the pressure to reinject the water into the formation. Typically, the pump is placed either at the separator inlet to boost the feed pressure (a 'push-through pump') or at the water offtake to boost the injection pressure (a 'pullthrough pump'). A pullthrough pump allows a more selective control of the offtake flow rate and the overall separation efficiency of the system. Optionally, a second pump may be placed at the separator outlet to lift the produced oil to the surface.

The overall separation efficiency of the system is governed by the 'flow split', the fraction of the inflow that exits through the oil outlet. Let us consider the behavior of the model for a water separator (with $E_{max} = 1.0$) as the flow split is reduced by steadily increasing the flow rate through the water offtake. At low offtake rates, when the flow through the offtake is less than the inflow rate of water to the separator, the offtake flow is purely water. The remainder of the water exits with the oil through the oil outlet. As the offtake flow increases towards the value of the water inflow rate, the water fraction flowing through the oil outlet decreases to a residual value that depends on α_{lim}. As the offtake flow increases further to exceed the water inflow rate, the water fraction in the oil outlet remains at a residual value but there is an increasing carryover rate of oil through the water outlet. The residual water fraction in the oil outlet stream can be made arbitrarily small by decreasing the input value of α_{lim}, but in general the smaller this value is, the more difficult it is for the well solution to converge.

For optimum separation, the flow rate through the water offtake should be kept equal to the inflow rate of water into the separator. But the flow rate in the offtake branch depends on the pressure in the separator, the pressure losses along the branch, its injectivity index and the pressure in the formation. However, the offtake flow can be controlled with a pullthrough pump. If the pump is too weak, the flow rate of water along the offtake branch may not be able to keep up with the flow of water into the separator from the formation. In this case the excess water remains in the production stream that goes to the surface. On the other hand, if the pump is too strong the flow of fluid along the offtake branch may exceed the flow of water into the separator. In this case the flow in the outlet branch also includes some of the oil from the formation, which is reinjected along with the water. The choice of pump power depends on whether the primary objective is to reinject clean water (at the expense of leaving some water in the production stream to the surface), or to remove all the water from the production stream (at the expense of reinjecting some of the produced oil).

Eclipse offers a number of ways of automatically cutting back the pressure differential from an overpowered pullthrough pump to maintain optimum separation:

- Use the built-in device model for a pullthrough pump for a downhole water separator (keyword WSEGPULL). This consists of a simple formula for the pump pressure differential as a function of flow rate, with an additional term to reduce the pressure differential if there is excessive carryover of oil. The maximum power of the pump is user-defined, and the actual applied power (determined from the reduced pressure differential) can be written to the SUMMARY file with the SUMMARY section keyword SPPOW.
- Represent the pump pressure versus flow characteristics by a VFP table (using keyword WSEGTABL). Note that the VFP table must have negative flow values, since the flow in the offtake branch is in the direction away from the wellhead. Further down the offtake branch, include a flow limiting device (keyword WSEGFLIM) with a small negative oil rate limit to restrict the rate of oil carryover.
- Represent the pump pressure versus flow characteristics by a VFP table (using keyword WSEGTABL) as above, but instead of adding a flow limiting device, modify the table to reduce the pump pressure differential when the water cut drops below 1.0. An example of this type of pump table is shown below. The outlet pressure versus flow curve for a WCT (watercut) = 1.0 represents the pump's full power, delivering a maximum 500 psi pressure boost which drops off as the negative flow increases. But the pump pressure also drops off rapidly with decreasing WCT. As the WCT decreases from 1.0 to 0.9 the pressure boost reduces by a factor of 10. The curve for WCT = 0.0 shows no pressure boost at all, except at very low flow rates. In fact the maximum pressure boost is applied at low negative flow rates and all positive flow rates, for all values of WCT, to prevent reverse flow occurring in the offtake branch. Note also that the illustrated pump curve is insensitive to the gas fraction and the inlet pressure, as there is only one value for each of these quantities.

```
VFPPROD
 1  20.0  LIQ  WCT  GLR  THP  ' '  FIELD  BHP /
```

```
-8000  -6000  -2000  -10  -1  10 /  flow values

3000 /  THP (pump inlet pressure) values

0.0  0.9  1.0 /  WCT values

1.0 /  GOR values

0 /  ALQ values

1 1 1 1  3000  3000  3000  3000  3500  3500 /  WCT = 0.0
1 2 1 1  3000  3030  3040  3050  3500  3500 /  WCT = 0.9
1 3 1 1  3000  3300  3400  3500  3500  3500 /  WCT = 1.0
/
```

Whatever model you select for the pump, the non linearities in the separator and pump models can make it difficult for the well solution to converge. You are advised to use the WSEGITER keyword to increase the maximum number of well iterations, and if necessary, also to invoke the option to have multiple iteration cycles with variable pressure loss weighting (see "Performance").

Thermal multisegment well model

For the Eclipse 300 Thermal option. In thermal simulations an extra variable is used, representing the energy density in the segment. Energy is stored in both the fluid and the pipe wall, which are assumed to be at the same temperature. A description of the model and some example applications are published in *Stone et al (2002)*.

Energy inflow

The flow of energy between a grid block and its associated segment's node is given by the inflow performance relationship

$$q_{ej} = T_{wj} \cdot \sum_{p} M_{epj} \cdot \left(P_j + H_{cj} - P_n - H_{nc}\right) \qquad \text{Eq. 8.101}$$

where

M_{epj} is the energy mobility in each phase at the connection. If the node is injecting fluid into the formation, the ratio of the phase mobilities reflects the mixture of phases in the segment.

The other terms in this equation are the same as used in the flow equation, equation 8.101.

Conductive heat transfer

Conductive heat transfer can occur along the well, across the well (between the well tubing and the well casing), and between the well and the formation, which may be inside or outside the simulation grid.

Heat transfer across a small volume is given by the formula

$$Q = -K \cdot A \cdot \frac{\partial T}{\partial X} \qquad \text{Eq. 8.102}$$

where:

K is the thermal conductivity and

A is the cross-sectional area.

This formula is used to calculate heat conduction along the well.

For a length of cylindrical annulus, equation 8.102 can be integrated over the radius to give

$$Q = \frac{L}{R} \cdot \left(T_i - T_o\right) \qquad \text{Eq. 8.103}$$

where the specific thermal resistance of the annulus is:

$$R = \frac{1}{2\pi} \cdot \frac{1}{K} \cdot \ln\frac{r_o}{r_i} \qquad \text{Eq. 8.104}$$

and:

L is the pipe length

T_i is the temperature at the inner radius

r_i and T_o is the temperature at the outer radius r_o.

The resistance of a thin annulus is calculated by substituting $r_o = r_i + \Delta r$ into equation 8.104 to give

$$R = \frac{1}{2\pi} \cdot \frac{\Delta r}{K} \cdot \frac{1}{r_i} \qquad \text{Eq. 8.105}$$

Heat transfer along the well

Equation 8.102 is used to model heat transfer along the well. You can specify a thermal conductivity and a cross-sectional area using the `WELSEGS` keyword. Values are usually chosen to represent the well walls, since conduction in the walls is usually much larger than conduction in the fluid.

Heat transfer between the well and the reservoir or to an external temperature

Equation 8.103 is used to calculate heat conduction from the well to the formation. For heat losses from a segment to a reservoir grid, heat transfer occurs from a bulk flow in the segment across a pipe wall, insulation, annulus, casing wall, cement and altered formation to the unaltered formation. Heat loss from a segment to rocks outside the simulation grid can be modeled in exactly the same way if you specify the external temperature.

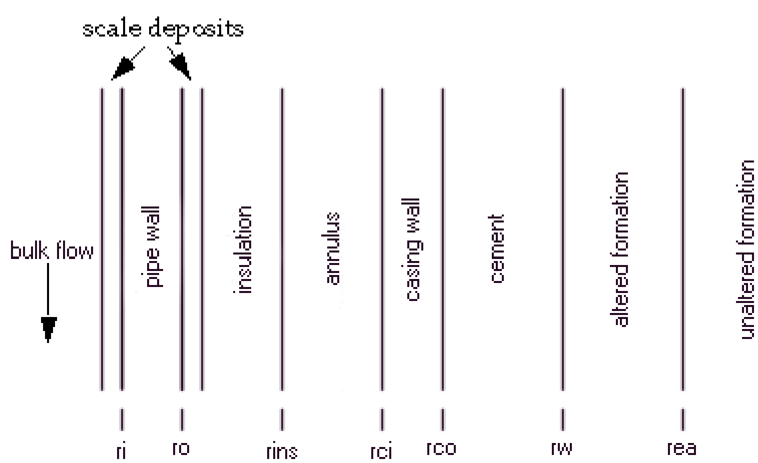

Figure 8.10. Heat conduction between a well and the formation

The specific thermal resistance for these cases is defined by summing terms similar to equations 8.104 and 8.105. See *Prats (1982)*.

$$R_h = \frac{1}{2\pi}\left[\frac{1}{h_f \cdot r_i} + \frac{1}{h_{P_i} \cdot r_i} + \frac{\ln(r_o/r_i)}{\lambda_P} + \ldots\right.$$

$$\ldots + \frac{1}{h_{P_o} \cdot r_o} + \frac{\ln(r_{ins}/r_o)}{\lambda_{ins}} + \frac{1}{h_{rcan} \cdot r_{ins}} + \ldots$$

$$\left.\ldots + \frac{\ln(r_{co}/r_{ci})}{\lambda_P} + \frac{\ln(r_w/r_{co})}{\lambda_{cem}} + \frac{\ln(r_{ea}/r_w)}{\lambda_{ea}} + \frac{f(t_D)}{\lambda_E}\right]$$

where

h_f is the film heat transfer coefficient between the fluid inside the pipe and the pipe wall

h_{P_i} is the heat transfer coefficient across any deposits of scale or dirt at the inside wall

λ_P is the thermal conductivity of the pipe wall

h_{P_o} is the heat transfer coefficient across the contact between pipe and insulation

λ_{ins} is the thermal conductivity of the insulation

h_{rcan} is the heat transfer coefficient due to radiation and convection in the annulus

λ_P is the thermal conductivity of the casing wall

λ_{cem} is the thermal conductivity of the cement

λ_{ea} is the thermal conductivity of the altered earth

$f(t_D)$ is the time function that reflects the thermal resistance of the earth

λ_E is the thermal conductivity of the unaltered earth.

Since R_h can only be specified as a constant, the time-dependent function $f(T_D)$ must be approximated. The remaining parameters are illustrated in figure 8.10 and are defined:

r_i is the inner radius of the pipe

r_o is the outer radius of the pipe

r_{ins} is the external radius of the insulation

r_{ci} is the inner radius of the casing

r_{co} is the outer radius of the casing

r_w is the wellbore radius

r_{ea} is the radius of the altered earth zone

The calculated value for the specific thermal resistance, R_h, is entered using the WSEGHEAT keyword.

Heat transfer from the tubing to the casing

Equation 8.102 is used to calculate heat conduction losses from a segment to another segment. This case is only applicable when the multisegment well has been configured with active inner tubing and outer annulus segments. Heat transfer takes place from the bulk flow in one segment across the tubing wall to the bulk flow in the adjoining segment, as illustrated in the following figure.

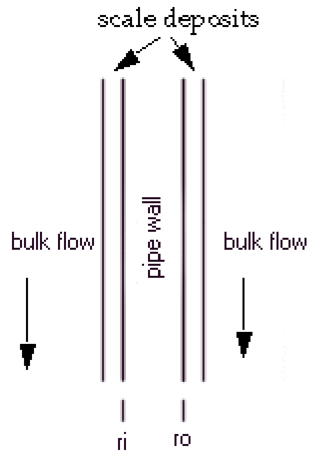

Figure 8.11. Heat conduction between the tubing and the casing

The specific thermal resistance for this case is

$$R_h = \frac{1}{2\pi}\left[\frac{1}{h_f \cdot r_i} + \frac{1}{h_{P_i} \cdot r_i} + \frac{\ln(r_o/r_i)}{\lambda_P} + \frac{1}{h_{P_o} \cdot r_o} + \frac{1}{h_f \cdot r_o}\right]$$

where

r_i, r_o, h_f, h_{P_i} and λ_P retain their previous definitions and:

h_{P_o}	is the heat transfer coefficient across any deposits of dirt or scale on the outer wall.

The calculated value for the specific thermal resistance, R_h, is entered using the WSEGHEAT keyword.

Using the multisegment well model

If the run contains any multisegment wells, the dimensioning data must be supplied in the RUNSPEC section keyword WSEGDIMS. A run cannot contain both multisegment wells and friction wells; any friction wells must be converted to multisegment wells (see below).

Defining multisegment wells

A multisegment well, like any other well, must first be introduced with the WELSPECS keyword. The connection data must also be specified in the usual way with the COMPDAT (or COMPDATL) keyword. The well must then be defined as a multisegment well with the keywords WELSEGS and COMPSEGS (or COMPSEGL; keywords ending in L should be used for wells in local grids).

Eclipse 100

An alternative to using WELSEGS and COMPSEGS is to use the keyword WFRICSEG (or WFRICSGL). The data for these keywords is identical to WFRICTN (and WFRICTNL), and they are primarily intended to provide a means of converting a friction well into a multisegment well having one segment per grid block connection.

The WELSEGS keyword defines the segment structure of a well. The first record defines its top segment, which must be segment number 1. The top segment's node is taken to be the bottom hole reference depth of the well, so that the well's BHP is equal to that segment's pressure. The remaining records define the rest of the well's segments, working down the main stem and, for multilateral wells, outwards along each branch. Branches are identified by their branch number; the main stem must be branch number 1.

The COMPSEGS (or COMPSEGL) keyword defines the location of the grid block connections within the well, by the branch number and the distance down the well to the start and end of the perforations. **All** connections in a multisegment well must have their locations defined with this keyword. Eclipse automatically allocates each grid block connection to the segment whose node lies nearest to the center of the connection. A segment may contain any number of grid block connections. If you wish to define the segments so that there is a single grid block connection within each segment, it is best to position the segment nodes to be at the center of each grid block connection (see figure 8.12), since the flow is modeled as entering the well segment at its node.

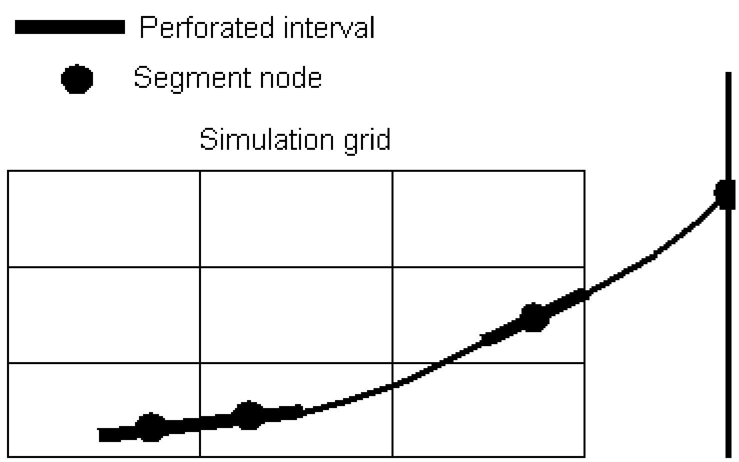

Figure 8.12. Segment node positions with one segment per grid block connection

In multisegment wells the grid block connections are not forced to be at the same depth as the corresponding grid block centers. Since Eclipse knows the depth of the beginning and end of each segment and the position of the center of the connection along the segment, it can work out the depth of the center of the connection. Eclipse sets the connection depths this way by default, although it is possible to redefine the connection depths in the COMPSEGS keyword. The hydrostatic head between the connection depth and the grid block center is calculated explicitly from the fluid contents of the grid block (see "Inflow performance").

Eclipse 100

It is possible to define the initial conditions (the pressure and the fluid contents) in each segment manually using keyword WSEGINIT. This overrides the standard initialization that Eclipse performs, which depends in the type of well and the conditions in the connecting grid blocks.

If required, looped flow paths within multisegment wells can be specified using the WSEGLINK keyword. See "Looped flow paths" for further information.

Defining the flow model

The WELSEGS keyword also defines the default flow model for the well: homogeneous flow or the drift flux model. When the segments are first defined, they adopt the flow model specified in WELSEGS. It is possible to change the flow model for each segment individually using the keyword WSEGFMOD.

If any of the segments in a well are designated to use the drift flux model, that model must be selected as the well's default flow model in WELSEGS item 7, so that the well uses the set of variables required by that model. The drift flux model contains a number of adjustable parameters that you can set (see "The drift flux slip model"). You can change the model definition, which dictates the default values of these parameters for all well segments, using keyword WSEGDFMD. If you wish to give some of the parameters different values from their model defaults, you can do so by **subsequently** entering the WSEGDFPA keyword. You can set selected parameters for individual segments with keyword WSEGFMOD. The drift flux model also requires values of the interfacial tensions; these may either be input as functions of pressure using tables with keywords STOG, STOW and STWG (in Eclipse 100) or calculated from built-in correlations (see "Three-phase flow and oil-water slip").

Individual segments may be designated to have their pressure drops determined from VFP tables, using the keyword WSEGTABL (see "Pressure drops from VFP tables"). The tables must be structured as production well VFP tables, and must have previously been entered with the keyword VFPPROD.

Specific segments may also be designated to represent

- a flow limiting valve by using the keyword WSEGFLIM
- or a general sub-critical valve by using keyword WSEGVALV
- or a 'labyrinth' flow control device by using the keyword WSEGLABY
- or a 'spiral' flow control device by using the keyword WSEGSICD
- or an 'autonomous' flow control device by using the keyword WSEGAICD.

Eclipse uses built-in formulae to model the pressure losses across these devices.

Modifying pressure drops and segment properties

The WSEGMULT keyword applies a scaling factor to the frictional pressure drop along specified segments. This could be used, for example, to reduce the flow from a branch by imposing a high frictional pressure drop in the segment nearest the main stem. It could also be used to account for pressure loss in bends, if their 'equivalent length' is known, as they increase the effective length of the segment for the frictional pressure drop.

Optionally, the pressure drop multiplier can be made to vary with the water-oil ratio and gas-oil ratio of the mixture flowing through the segment. A variable multiplier may be applied, for example, to model the operation of an adjustable flow control device whose purpose is to choke back production from high WOR or GOR regions.

The WSEGPROP keyword can be used to alter the geometric properties of a segment. The segment's length, however, cannot be altered because the grid block connections are located by their overall distance down the tubing.

Workovers

When a workover with an action '+CON' is performed on a multisegment well, that is "close the worst-offending connection and all below it" (see item 7 of the WECON keyword, for example), this is interpreted as meaning “close the connection and all connections **upstream** of it”. Thus instead of selecting the connections to close according to their depth, they are selected according to their length down the tubing - all connections further from the wellhead are closed. In multilateral wells, the branch topology is taken into account so that only branches subordinate to the worst-offending connection are closed off.

Heat loss

Eclipse 300 Thermal

Heat storage in the tubing and casing can be specified using the WELSEGS keyword. This keyword can also be used to specify longitudinal conduction in the tubing and casing between successive segments.

Heat conduction to the reservoir, and heat loss outside the reservoir can be specified using keyword WSEGHEAT. Heat conduction to the reservoir occurs between segments and completion grid cells. Therefore it may be necessary to define completions in each grid cell that the well passes through. These completions can be shut if no flow occurs between the grid cell and the segment. The length of well in each completion cell can be defined using the COMPSEGS keyword.

WSEGHEAT can also be used to model radial heat conduction between non-successive segments, for example between tubing segments and casing segments.

For multisegment well source terms, the keyword WSEGEXSS can be used to specify the temperature of imported gas.

Thermal contact lengths

Heat transfer occurs to or from a segment node. Since each segment node is at the end of a segment, the heat transfer to and from each node represents half the heat transfer in the two segments either side of the node. For heat transfer between a completion and a segment the thermal contact length is the length of well within the completion cell. This should be specified using the COMPSEGS keyword. For heat transfer between segments, or between a segment and an external temperature, the thermal contact length is the length of well closest to the segment node.

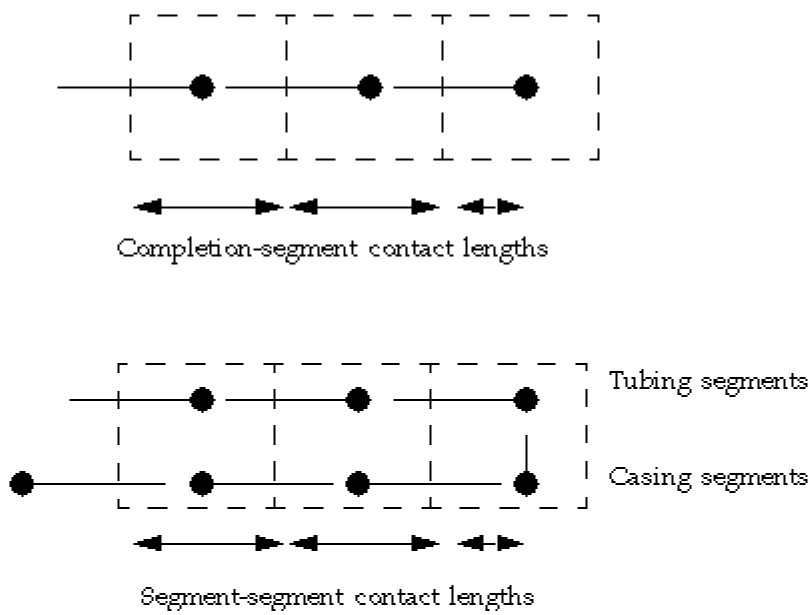

Figure 8.13. Thermal contact lengths

Output

Eclipse 100

The segment structure and connection locations are written to the PRINT file in tabular form if the RPTSCHED mnemonic WELSPECS is set. It is advisable to check your data by examining this output, as an error can easily be made while entering complicated multilateral well data.

At each report time, when well reports are requested (RPTSCHED mnemonic WELLS), an additional table is printed containing information about conditions within each segment.

Information on flows, pressures and pressure drops within individual segments can be written to the SUMMARY file (see the SUMMARY section overview).

More detailed information on the flowing conditions inside multisegment wells can be written to the RFT file at specified times (see the WRFTPLT keyword).

Eclipse 300

The segment structure is written to the PRINT file under the WELLS heading in tabular form if the sixth switch of RPTPRINT is set to 1, which is the default. This occurs whenever the multisegment well is updated.

A completion summary that gives details on connections to the segments is given under the COMPLETIONS heading if the seventh switch of RPTPRINT is set to 1, which is the default. This occurs whenever the multisegment well is updated.

At each report time when well reports are requested (sixth switch of RPTPRINT), an additional table is printed containing information about conditions within each segment.

Information on flows, pressures and pressure drops within individual segments can be written to the SUMMARY file (see the SUMMARY section overview).

Performance

If the multisegment well contains looped flow paths (see the WSEGLINK keyword), then a different solver is used for the well linear equations with each well iteration. The WSEGSOLV keyword can be used to modify the behavior of this looped well linear solver if required.

Eclipse 100

Multisegment wells can require substantially more well iterations to converge than standard wells, especially when being opened or while switching into or out of crossflow. By default, each well is allowed up to MXWSIT iterations to converge, which is set using keyword TUNING (or TUNINGL or TUNINGS). If it cannot converge in this number of iterations, a ‘problem’ message is issued and the latest (unconverged) well solution is used. This may hamper the convergence of the non-linear iterations of the timestep. For multisegment wells, it is recommended to use the WSEGITER keyword to give the well solution more opportunity to converge. The keyword can increase the maximum allowed number of iterations, as an alternative to using TUNING. Furthermore it can invoke a sequence of iteration cycles which traverses ‘difficult’ regions of the residual function by taking smaller steps in the pressure variable. Refer to the WSEGITER keyword documentation for a more detailed explanation of the process.

Other facilities that influence the convergence of the well solution can be controlled using the OPTIONS keyword. Examples include item 30, which governs methods aimed at improving convergence by chopping the increment in all the well’s solution variables and item 63, which can be set if the explicit calculation of the hydrostatic head between the well connections and the corresponding grid block centers causes oscillations.

If there are problems encountered with multisegment wells under THP control, especially when on the point of dying, the wells should be switched to using explicit water cut and GOR values for VFP table lookup (keyword WVFPEXP).

Eclipse 300

Multisegment wells can require more well iterations to converge than standard wells require. There is a CPU expense if the simulator is run in compositional mode with many multisegment wells and many segments per well. To gain the best performance, it is recommended that only a few wells be run as multisegment wells and that the number of active segments not be large. There are tuning parameters available that affect the convergence, but it is not generally recommended to change their values. See keyword CVCRIT for more details.

The complexity of the branches does not significantly affect performance of the multisegment wells. Neither does the number of completions that are connected.

See also "The top segment".

If a multisegment well is crossflowing, there may be multiple solutions and consequent difficulty in converging. If the well exhibits these symptoms, you have the option of setting t item 34 in the OPTIONS3 keyword to be greater than zero. If this argument is set to the number of the particular well in question (wells are numbered in the order they are input), two extra steps are taken to help converge when crossflowing. Firstly, the multisegment well primary variables at the start of each well calculation is initialized to the solution at the end of the last timestep. Secondly, during the solution of the flow control rate constraint, the search for the BHP puts extra weight on bisection rather than Newton. This may result in more iterations being taken to converge the well, but it helps convergence in difficult situations. If item 34 of OPTIONS3 is set to 10000 then all multisegment wells use the above.

Restrictions

Eclipse 100

Multisegment wells cannot at present be used in conjunction with the following facilities:

- The wellbore friction option
- The temperature option
- The solvent model
- The Gi model
- Automatic refinement in the LGR option

In addition, it is strongly recommended that multisegment wells are not used in conjunction with the following facilities:

- The COMPFLSH keyword.

  In both the conventional well and the multisegment well models, the inflowing fluid is transformed to flash conditions when it enters the wellbore through the connections to the reservoir grid. To calculate the fluid properties (for example for the hydrostatic head) a back transformation is applied based on the overall inflow; this is consistent with the treatment of the well as a “mixing tank” and is thus appropriate for the conventional well model. With a multisegment well, this back transformation (based on the overall inflow) is still applied, even though the local fluid properties may vary with position in the well. Thus the local fluid properties are not as accurately represented as they should be for a discretized wellbore model such as a multisegment well. These fluid properties influence the hydrostatic and frictional pressure gradients, the out flowing mixture properties when crossflow occurs, and slip (when using the drift-flux model).

Eclipse 300

Multisegment wells in Eclipse 300 cannot at present be used in conjunction with the following facilities:

- velocity dependent gas relative permeability contribution to D-factor (VDKRG)
- changing well dimensions in the SCHEDULE section (CHANDIMS)
- k-value tables in compositional mode (KVALUES)
- tracers (TRACER)
- flash transformation facility (COMPFLSH).

Converting friction wells to multisegment wells

Eclipse 100

The keywords WFRICSEG and WFRICSGL provide an additional way of defining a multisegment well, as an alternative to the WELSEGS and COMPSEGS (or COMPSEGL) keywords. Their primary purpose is to offer you a very simple way of converting existing data sets that use the wellbore friction model (see "Wellbore friction option") into data sets that use multisegment wells. The data for each keyword is identical to that of the WFRICTN and WFRICTNL keywords, but instead of creating a friction well each keyword creates a multisegment well with the same branch structure and connection locations. Eclipse automatically divides the well into segments such that there is one segment per grid block connection, plus an additional segment at the well's bottom hole reference point. The segments are distributed so that the nodal point of each segment is located at the center of the corresponding grid block connection. The segment lengths are set equal to the tubing length between adjacent connections. The depths of the segment nodes are set equal to the center depths of the connecting grid blocks, as there is no depth data in WFRICTN(WFRICTNL).

The WFRICSEG keyword does not offer the flexibility of WELSEGS and COMPSEGS in defining multisegment wells, so they are not generally recommended for uses other than converting friction well data. The disadvantages of using the WFRICSEG keyword are:

- There is an enforced one-to-one correspondence between segments and grid block connections. This may result in an excessively large number of segments.
- The segment nodes have their depths set equal to the center depths of the connecting grid blocks. Thus, unless the grid is aligned with the well trajectory, a horizontal well may have spurious undulations or deviations from its true trajectory.
- Branch junctions defined in WFRICTN have to be located at a grid block connection, so extra connections have to be defined at branch junctions; if there is no inflow from the formation at these locations the connections should be declared shut in COMPDAT.
- When WFRICSEG is used to define a multilateral well you cannot dictate the branch numbers; Eclipse creates the branches and numbers them itself. However, Eclipse is unable to recognize whether the original main stem continues past a branch junction. Instead, **two** new branches are created at each branch junction, one of these being the continuation of the main stem. It is important to allow for this when setting the maximum number of branches in WSEGDIMS. For example, a well consisting of a main stem and a single lateral branch will be converted to have three branches: the upper part of the main stem, the lateral branch and the section of the main stem below the branch junction.

Converting a data set from using friction wells to using multisegment wells

1. Remove the FRICTION keyword from the RUNSPEC section, replacing it with the WSEGDIMS keyword.
   a. The maximum number of multisegment wells in the run is the same as the maximum number of friction wells.
   b. The maximum number of segments per well is 1 + the maximum number of grid block connections per well.
   c. The maximum number of branches per well should allow for the ‘extra branches’ created by Eclipse in the main stem below lateral branch junctions
2. Change all WFRICTN keywords to WFRICSEG. In LGR runs, any WFRICTNL keywords should be changed to WFRICSGL. The keyword data should remain unchanged.
3. Optionally the SUMMARY section can include the mnemonics associated with multisegment wells (SOFR for example)

When Eclipse converts a friction well into a multisegment well, it prints out its equivalent WELSEGS and COMPSEGS (or COMPSEGL) keyword data in the PRINT file, at the place where WFRICTN (or WFRICTNL) is read. The generated keyword data can be copied and used in subsequent runs in place of the original friction well keyword. It is then possible to edit the data, for example to renumber the whole of the main stem to be branch 1, to remove spurious undulations by adjusting the segment depths to agree with the well trajectory, and to reduce the number of segments if desired.

Summary of keywords

RUNSPEC

- WSEGDIMS Sets array dimensions for multisegment wells.
- WSEGSOLV Sets parameters for the multisegment well iterative linear solver.

PROPS section

Restriction	Keyword	Description
Eclipse 100 (used in the drift flux model).	STOG	Oil-gas surface tension versus pressure
Eclipse 100 (used in the drift flux model).	STOW	Oil-water surface tension versus pressure
Eclipse 100 (used in the drift flux model).	STWG	Water-gas surface tension versus pressure

SUMMARY section

The segment flow rates reported are the flow rates of each phase through the tubing past the segment’s end nearest the well head. Positive values represent flow towards the well head (production), while negative values represent flow away from the well head (injection). The flow rates are reported at surface conditions (the sum of free and dissolved phases in Eclipse 100), equivalent to the reported well rates. The flow velocities refer to the local flow velocities of the free phases through the segment. The flow velocities of all the phases should be equal unless you are using a built-in multiphase flow model that allows slip. The holdup fractions refer to the local volume fraction of the segment occupied by each free phase.

Eclipse 100

The tracer flow rate and concentration keywords should be compounded with the name of the tracer to which they refer (for example: STFROL1 and STFCOL1 output the flow rate and concentration of the tracer OL1

The segment quantity keywords all begin with the letter S. They must be followed by one or more records each containing the well name and the segment number and terminated with a slash. The list of records must be terminated with an empty record just containing a slash. The segment number following a well name may optionally be defaulted, which causes the appropriate quantity to be output for all the segments in the well.

Note: If the segment number is defaulted, as for well PROD2 below, Eclipse reserves a number of SUMMARY vectors equal to the maximum number of segments per well (NSEGMX, set in RUNSPEC section keyword WSEGDIMS). If this number is much greater than the actual number of connections in well PROD2 then defaulting the segment numbers can become quite wasteful

---

For example,

```
SOFR
 PROD1  1 /
 PROD1  4 /
 PROD1  5 /
 PROD2    /
/
```

Refer to the following sections in the *Eclipse Reference Manual.*

- Multisegment wells
- Secondary group and well quantities

SCHEDULE section

Restriction	Keyword	Description
	COMPSEGL	Defines the location of multisegment well completions in a local grid.
	COMPSEGS	Defines the location of completions in a multisegment well.
Eclipse 300	CVCRIT	Items 23 - 26 set convergence criteria for multisegment wells. T
Eclipse 100	OPTIONS	Item 30 controls the use of a binary chop sequence, which restricts the size of the solution increments if the residuals get worse. However, you are recommended to try the WSEGITER keyword before changing this switch. Item 63 controls the calculation of the hydrostatic head between each connection of a multisegment well and its corresponding grid block. Item 77 restores back-compatibility with the pre-2000A drift flux model calculations involving oil-water slip and inclined tubing.
Eclipse 300	OPTIONS3	Item 34 may help convergence in difficult situations involving crossflow.
Eclipse 300	SCONPROD	Secondary well production controls.
Eclipse 300	SCONINJE	Secondary well injection controls
Eclipse 300	STEST	Secondary well reopen test control
	WELSEGS	Defines the segment structure of a multisegment well.

Restriction	Keyword	Description
Eclipse 100	WFRICSEG	Converts WFRICTN keyword data for a friction well into data for a multisegment well.
Eclipse 100	WFRICSGL	Converts WFRICTNL keyword data for a friction well into data for a multisegment well.
	WSEGAICD	Defines segments to represent an ‘autonomous’ inflow control device.
	WSEGDFIN	Inputs a table of inclination scaling factors versus deviation angle for the drift flux slip model
	WSEGDFMD	Defines the drift flux slip model
	WSEGDFPA	Changes parameter values in drift flux slip model.
	WSEGEXSS	Specifies fluid import/removal through an external source/sink.
	WSEGFLIM	Defines segments to represent a flow limiting valve.
	WSEGFMOD	Specifies the multiphase flow model and related parameters.
Thermal	WSEGHEAT	Specifies heat loss from a segment.
Eclipse 100	WSEGINIT	Sets initial pressure and fluid contents of well segments.
Eclipse 100	WSEGITER	Activates the robust iteration scheme for solving multisegment wells.
	WSEGLABY	Defines segments to represent a ‘labyrinth’ inflow control device.
	WSEGLINK	Defines the chord segment links for specifying looped flow paths
	WSEGMULT	Imposes constant or variable multipliers for segment frictional pressure drops.
	WSEGPROP	Modifies the properties of individual well segments.
Eclipse 100	WSEGPULL	Defines a pullthrough pump for a downhole water separator.
Eclipse 100	WSEGSEP	Defines a segment to represent a downhole separator.
	WSEGSICD	Defines segments to represent a ‘spiral’ inflow control device.
	WSEGTABL	Designates segments to have their pressure drops calculated from a VFP table.
	WSEGVALV	Defines segments to represent a sub-critical valve.
Eclipse 300	WSEGWELL	Define the association between segment number and secondary well name.

Wellbore friction option

x	Eclipse 100
	Eclipse 300

The wellbore friction option is a special extension to Eclipse 100; it is not available in Eclipse 300. It models the effects of pressure loss due to friction in the well tubing along the perforated length, and between the perforations and the bottom hole reference point of the well. The facility is primarily intended for use with horizontal and multilateral wells, in which frictional pressure losses may be significant over the horizontal section of the wellbore and in the branches.

In the standard well model, frictional pressure losses between the well connections and the bottom hole reference point are neglected entirely. The difference between the pressure at the connections and the bottom hole pressure is purely the hydrostatic head of the fluid mixture in the wellbore acting over the difference in true vertical depth between these locations. In wells completed vertically through the formation, the frictional pressure drop acting over the relatively short length of the perforated section is generally negligible.

In horizontal wells, however, the perforated section may extend over many hundreds of feet. The frictional pressure drop over this length may have a significant effect on the behavior of the well. The pressure in the wellbore towards the far end of the perforations (that is, away from the wellhead) will be higher than the pressure at the near end, so the drawdown will vary over the perforated length. This may cause the productivity per unit length of perforations to fall off towards the far end. Furthermore, the lower wellbore pressure at the near end of the perforations may cause localized water or gas coning to occur in this region, thus reducing the effectiveness of the horizontal well in overcoming coning problems. It is therefore important to consider the effects of frictional pressure losses when deciding on the optimum length and diameter of a horizontal well.

Calculating the frictional pressure loss

The frictional pressure drop over a length L of tubing is

$$\delta P_f = 2f \cdot \frac{L}{D} \cdot \rho \cdot v^2 \qquad \text{Eq. 8.106}$$

where

f is the Fanning friction factor

D is the tubing inner diameter

ρ is the fluid density

v is the fluid velocity

Transforming the flow velocity into the local volumetric flow rate Q (in rb/day or m3/day) and taking into account the unit conversion factors, the pressure drop becomes:

$$\delta P_f = \frac{C_f \cdot f \cdot L \cdot \rho \cdot Q^2}{D^5} \qquad \text{Eq. 8.107}$$

where values and units for the equation terms are:

C_f values 4.343E-15 (`METRIC`),

2.956E-12 (`FIELD`),

2.469E-13 (LAB)

L units m (METRIC), ft. (FIELD), cm (LAB)

D units m (METRIC), ft. (FIELD), cm (LAB)

ρ units kg/m^3 (METRIC), lb/ft.^{3} (FIELD), gm/cc (LAB)

Q units rm^3/day (METRIC), rb/day (FIELD), rcc/hr (LAB)

The Fanning friction factor depends upon the Reynolds number Re. For laminar flow (Re < 2000),

$$f = \frac{16}{\mathrm{Re}} \quad \text{Eq. 8.108}$$

For Re > 4000 we use Haaland's formula (*Haaland, 1983*),

$$\sqrt{\frac{1}{f}} = -3.6 \log_{10}\left(\frac{6.9}{\mathrm{Re}} + \left(\frac{e}{3.7D}\right)^{10/9}\right) \quad \text{Eq. 8.109}$$

where

e is the absolute roughness of the tubing in the same units as D

In the "uncertain region" (2000 < Re < 4000) we use a linear interpolation between the values at Re = 2000 and Re = 4000. The above scheme has the advantage of providing a direct calculation for values of f that are continuous in Re and are in reasonable agreement with the friction factor chart (see *Welty et al (1984)*, for example). It should be noted, however, that these formulae do not take account of fluid inflow through the perforations. Its influence on the effective roughness of the tubing is largely unknown at present, and the engineer is advised to vary the roughness to gauge the sensitivity of the results to this parameter.

The Reynolds number is

$$\mathrm{Re} = \frac{\rho \upsilon D}{\mu} \quad \text{Eq. 8.110}$$

where

μ is the viscosity of the fluid.

Converting υ to the volumetric flow rate Q and including the unit conversion factors, this becomes

$$\mathrm{Re} = \frac{C_r \rho Q}{D \mu} \quad \text{Eq. 8.111}$$

where values and units for the equation terms are:

C_r values 0.01474 (METRIC),

0.1231 (FIELD),

0.03537 (LAB)

ρ units kg/m^3 (METRIC), lb/ft.^{3} (FIELD), gm/cc (LAB)

Q units rm^3 /day (METRIC), rb/day (FIELD), rcc/hr (LAB)

D units m (METRIC), ft. (FIELD), cm (LAB)

μ units cP (METRIC), cP (FIELD), cP (LAB)

If there is more than one free phase flowing in the well, we assume the flow along the perforated length is homogeneous (that is, no slip between the phases). The homogeneous mixture density (mass flow rate / Q) is used in the equations for Re and δP_f, and the viscosity is the volumetric flow weighted average of the phase viscosities.

Wellbore friction is included in Eclipse by treating the connection head terms as additional variables. These are solved implicitly at the same time as the well solution variables. An implicit solution technique is necessary to avoid numerical instabilities. A pressure balance equation is solved between each pair of adjacent connections, equating the difference in their head terms with the frictional (and hydrostatic, if they are at different depths) pressure losses. The fluid flowing into a connection is treated as entering the connection at the middle of its perforated interval within the grid block. Thus the frictional pressure drop between a connection and its downstream neighbor depends on the production rates of that connection and all connections upstream, and the distance between the central points of the connection and its downstream neighbor.

Using the wellbore friction option

The maximum number of friction wells (NWFRIC) in the problem must be set using the FRICTION keyword in the RUNSPEC section.

You can designate one or more wells to be friction wells, with the keyword WFRICTN. This keyword also sets the data required by the friction calculation, for example, the tubing diameter and roughness, and the locations of the start and end of perforations within each connection. The WFRICTN keyword can only be used on one well at a time, so the keyword must be specified more than once if there is more than one friction well. Other “frictionless” wells can also exist in the run.

An alternative keyword WFRICTNL is provided for designating friction wells that are completed in local refined grids, when using the Local Grid Refinement option. The keyword is similar to WFRICTN, except that the local grid name should be specified in each record of connection data.

Friction wells can have the same set of controls and limits as ordinary wells. If a friction well has an associated VFP table for THP calculations, it is recommended to make the bottom hole reference point of the VFP table the same as that of the well (in WELSPECS item 5). This will ensure that all friction losses upstream of the VFP table’s bottom hole reference point will be included in the well’s connection head terms. A suitable position for the well’s bottom hole reference point is the start of the perforations (that is the end nearest the wellhead).

At present the Wellbore Friction option cannot be run with multisegment wells or the polymer, brine, miscible, solvent or Gi model options in Eclipse 100.

Multilateral wells

Wells with branches emanating from the main stem can also be modeled with the wellbore friction option. The data structure is general enough even to allow for the case where a branch itself may have branches.

To reserve memory for multilateral wells it is necessary to set the second item of the FRICTION keyword in the RUNSPEC section. This item is the maximum number of branches (and for this purpose the main stem counts as a branch) which emanate from any one grid block connection. For a standard well that consists of only a main stem, we could enter

```
FRICTION
 1  1 /
```

as there is a maximum of 1 branch (that is the main stem itself) that emanates from any one connection. Item 2 is set equal to 1 by default. For a "trunk and branch" well as shown in figure 8.14, we would enter

```
FRICTION
 1  2   /
```

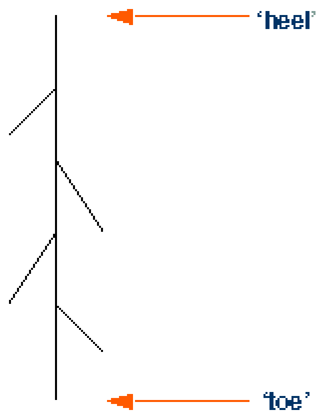

Figure 8.14. A trunk and branch multilateral well

For the "bird's foot" type well shown in figure 8.15, we would enter

```
FRICTION
 1  3   /
```

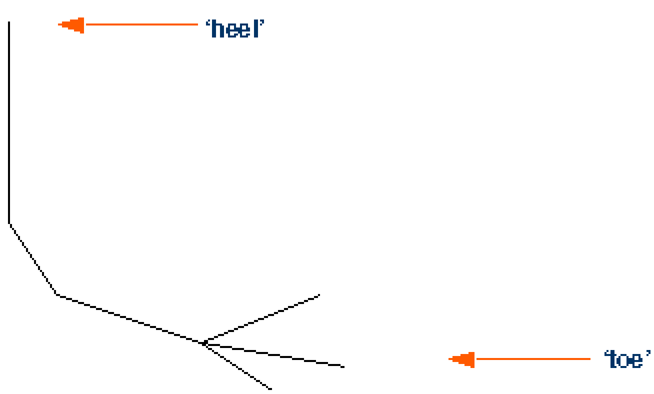

Figure 8.15. A bird's foot multilateral well

For a problem that involved all three of these wells then the maximum overall value should be used for the second argument,

```
FRICTION
 3  3  /
```

The geometry of a multilateral well is defined by the order in which the connections are entered in the keyword WFRICTN or WFRICTNL. WFRICTNL only differs from WFRICTN in that a local grid name is required in addition to the grid block coordinates to define each connection.

The first record of WFRICTN identifies the well, and sets the default tubing diameter, the roughness and the flow scaling factor.

The subsequent records deal with the connections making up the well, and have two purposes. The first is to define certain properties associated with the connections, including the tubing length from the well's bottom hole reference point to the start of the perforations in the grid block, and similarly the tubing length to the end of the perforations. It is possible to give the connection a different tubing diameter to the default value specified in the first record. Connections may be defined singly or in ranges. (A range is a contiguous row or column of grid blocks that are fully penetrated by the well.) Further details can be obtained from the keyword documentation.

The geometry of the well is defined by the order in which the connections are entered. A connection entered for the first time is assumed to be situated upstream (that is further from the heel of the well) of the previously entered connection. So, for a single branched well the connections should simply be ordered from the heel towards the toe.

If a connection is referenced again in WFRICTN after its first entry, either singly or as the **first** connection of a range, then it is taken to indicate that the subsequent connections form a branch that joins the well at this location. The connections in this branch should be ordered from the join out towards its toe. Thus whenever a connection is referenced for a second or subsequent time in WFRICTN, its entry is essentially a “pointer” to where the next connection joins the well. As its properties have already been defined in its first entry, it is only necessary to supply items 1-3 (or items 1-4 when using WFRICTNL) when a connection is entered again as a pointer. If the remaining items in the record are specified (as they must be if the pointer is the first connection in a range describing a new branch), then they will only be applied to the connections being entered for the first time, and will not overwrite the properties of the pointer connection.

Note that there must therefore be a connection defined at each branch point. However, if the well is not actually perforated within these grid blocks the connection should be declared as SHUT in keyword COMPDAT.

As an example of how multilateral well geometry is defined, consider the well shown in figure 8.16. We assume that k = 1 for all the connections. The order of defining this well is by no means unique, but one way is shown as follows:

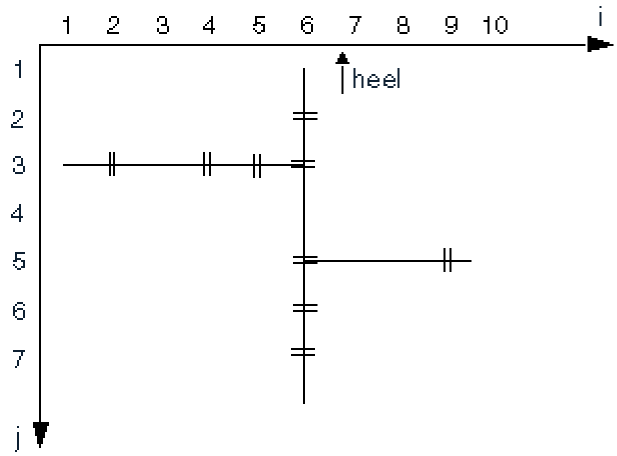

Figure 8.16. Defining multilateral well geometry

```
 WFRICTN
-- well name   tubing diam      Roughness     Scale Fac
   'HORIZ'         0.25            1.0E-3         1.0    /
-- I  J  K   Tlen1   Tlen2  Direction  RangeEnd  Tubing diam
--Define the main stem
   6  2  1    2*               'Y'        3        1*    /
   6  5  1    400     500       3*                       /
   6  6  1    2*               'Y'        7        1*    /
--Pointer back to where first branch leaves main stem.
--(This connection was first entered as part of the range (6, 2-3, 1)
   6  3  1                                               /
--Define this branch
   5  3  1    2*               'X'        4        0.17  /
   2  3  1    700     900       2*                 0.17  /
--Pointer back to where second branch leaves the main stem
   6  5  1                                               /
--Define this branch
   9  5  1    600     800       2*                 0.18  /
/
```

Remember that each of these connections must previously have been defined in keyword COMPDAT (or COMPDATL in a local grid). Note that connections must be defined at the branch points, blocks (6,3,1) and (6,5,1) in this example. If these are not perforated they should be declared as SHUT.

This method of defining multilateral well geometry is flexible enough to allow branches to have sub-branches. As a second example, consider the well shown in the following figure.

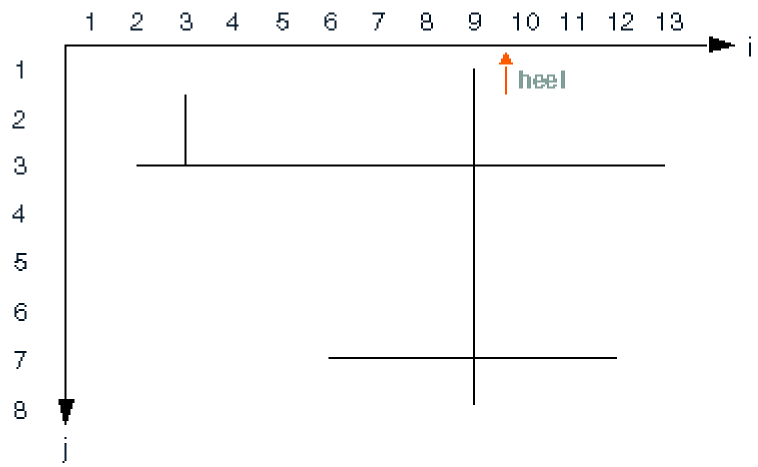

Figure 8.17. A multilateral well with sub-branches

We again assume that k = 1 for all the connections, and we will also assume that the well is completed in every grid block. One way of defining the connections in WFRICTN is shown as follows:

```
 WFRICTN
-- well name   tubing diam      Roughness     Scale Fac
   'HORIZ2'       0.25            1.0E-3         1.0    /
-- I  J  K   Tlen1   Tlen2  Direction  RangeEnd  Tubing diam
--Define the main stem (9, 1-8, 1)
   9  1  1    2*              'J'         8        1*  /
--Pointer to lower branch point on main stem
   9  7  1                                             /
--Define connections in the left branch (8-6, 7, 1)
   8  7  1    2*              'I'         6        1*  /
--and the same for the right branch (10-12, 7, 1)
   9  7  1                                             /
  10  7  1    2*              'I'        12        1*  /
--Pointer to the upper branch point on the main stem
   9  3  1                                             /
--Define the right branch (10-13, 3, 1)
  10  3  1    2*              'I'        13        1*  /
--and the same for the left branch (8-2, 3, 1)
   9  3  1                                             /
   8  3  1    2*              'I'         2        1*  /
--Pointer to the branch point off this branch
   3  3  1                                             /
--Define the single connection in this sub-branch
   3  2  1    1000    1020      3*                     /
/
```

This definition of the well can be shortened by incorporating the pointers into the range of connections:

```
 WFRICTN
-- well name   tubing diam      Roughness     Scale Fac
   'HORIZ2'       0.25            1.0E-3         1.0    /
-- I  J  K   Tlen1   Tlen2  Direction  RangeEnd  Tubing diam
--Define the main stem (9, 1-8, 1)
   9  1  1    2*              'J'         8        1*  /
--Define the lower left branch (9-6, 7, 1)
   9  7  1    2*              'I'         6        1*  /
--and the same for the lower right branch (9-12, 7, 1)
   9  7  1    2*              'I'        12        1*  /
--Define the upper right branch (9-13, 3, 1)
   9  3  1    2*              'I'        13        1*  /
--and the same for the upper left branch (9-2, 3, 1)
   9  3  1    2*              'I'         2        1*  /
--Pointer to the branch point off this branch
```

```
   3  3  1                                              /
--Define the single connection in this sub-branch
   3  2  1    1000    1020      3*                      /
/
```

Occasionally when opening a friction well, and in particular a multilateral friction well, the well solution may have difficulty in converging. This is because the initial "guess" of the pressures used to start off the Newton iteration may be poor. This can often be solved by just allowing the well to take more iterations, that is, increasing `MXWSIT` in the third record of the `TUNING` keyword. However, this may not be effective in all cases, and when this is so the `OPTIONS` keyword item 55 may be tried. This damps the Newton iteration, improving its convergence properties. It should be noted that after the problem has been solved, that is at the next timestep, it is very unlikely that the Newton method will need damping. Now the use of item 55 in `OPTIONS` will hinder convergence rather than help it. Hence the switch should be set to zero as soon as possible after the well has been successfully opened.

Workovers

When a workover with an action ‘ `+CON` ’ is performed on a friction well, that is ‘close the worst-offending connection and all below it’ (see item 7 of the `WECON` keyword, for example), this is interpreted as meaning ‘close the connection and all connections **upstream** of it’. Thus instead of selecting the connections to close according to their depth, they are selected according to their length down the tubing - all connections further from the wellhead are closed. In multilateral wells, the branch topology is taken into account so that only branches subordinate to the worst-offending connection are closed off.

Summary of keywords

RUNSPEC

The keywords are:

- `FRICTION` Enables the Wellbore Friction option and sets array dimensions.

SUMMARY

Refer to Wellbore friction model in the *Eclipse Reference Manual*.

SCHEDULE

Keyword	Description
`OPTIONS`	Item 55 may be used to damp the Newton iterations if well convergence problems arise.
`WFRICTN`	Designates a well as a friction well, and defines its tubing properties and the position in the tubing of the well connections with the grid cells.
`WFRICTNL`	Performs the same task as `WFRICTN`, but should be used if the well is situated in a local grid.

Example

This data set describes the 7th SPE Comparative Solution Project (*Nghiem et al, 1991*) - a horizontal well with friction. The data set describes CASE 4B, having a 2100 ft. horizontal well completed across 7 grid blocks in the x-direction.

RUNSPEC

```
RUNSPEC
TITLE
 7th SPE Benchmark, Case 4b

DIMENS
 9    9    6  /

NONNC
OIL
WATER
GAS
DISGAS
FIELD

TABDIMS
 1    1   15   15    1   15  /

WELLDIMS
 2    9    1    2 /

FRICTION
 1 /

START
1 'JAN' 1990  /

NSTACK
 50 /

FMTOUT
FMTIN
UNIFOUT
UNIFIN
```

GRID

```
GRID        ==========================================================
DXV
  9*300  /

DYV
  620  400  200  100  60  100  200  400  620  /

DZ
  81*20  81*20  81*20  81*20  81*30  81*50  /

PERMX
  486*3000 /

PERMY
  486*3000 /

PERMZ
  486*300 /

PORO
  486*0.2 /

EQUALS
'TOPS'  3590  1 9  1 9  1 1  /
/
```

PROPS

```
PROPS     ==========================================================
-- Use Stone's Method 2 for 3-phase oil relative permeability
```

```
STONE2

SWOF
-- Sw    Krw    Krow     Pcow
  0.22   0.0    1.0000   6.30
  0.3    0.07   0.4000   3.60
  0.4    0.15   0.1250   2.70
  0.5    0.24   0.0649   2.25
  0.6    0.33   0.0048   1.80
  0.8    0.65   0.0      0.90
  0.9    0.83   0.0      0.45
  1.0    1.0    0.0      0.0    /

SGOF
-- Sg    Krg    Krog     Pcgo
  0.0    0.0    1.00     0.0
  0.04   0.0    0.60     0.2
  0.1    0.022  0.33     0.5
  0.2    0.1    0.10     1.0
  0.3    0.24   0.02     1.5
  0.4    0.34   0.0      2.0
  0.5    0.42   0.0      2.5
  0.6    0.5    0.0      3.0
  0.7    0.8125 0.0      3.5
  0.78   1.0    0.0      3.9    /

PVTW
-- Pref    Bw       Cw       Vw    Viscosibility
   3600  1.00329   3.0E-6   0.96          0  /

ROCK
-- Pref     Cr
   3600    4.0D-6   /

DENSITY
--  oil   water   gas
    45.0  62.14   0.0702  /

PVDG
-- Pgas   Bg    Vg
    400  5.9   0.013
    800  2.95  0.0135
   1200  1.96  0.014
   1600  1.47  0.0145
   2000  1.18  0.015
   2400  0.98  0.0155
   2800  0.84  0.016
   3200  0.74  0.0165
   3600  0.65  0.017
   4000  0.59  0.0175
   4400  0.54  0.018
   4800  0.49  0.0185
   5200  0.45  0.019
   5600  0.42  0.0195   /

PVCO
-- Pbub   Rs     Bo     Visco    Co     CVo
    400  0.165  1.012   1.17   1.0E-5   0.0
    800  0.335  1.0255  1.14     2*
   1200  0.500  1.038   1.11     2*
   1600  0.665  1.051   1.08     2*
   2000  0.828  1.063   1.06     2*
   2400  0.985  1.075   1.03     2*
   2800  1.130  1.087   1.00     2*
   3200  1.270  1.0985  0.98     2*
   3600  1.390  1.11    0.95     2*
   4000  1.500  1.12    0.94     2*
   4400  1.600  1.13    0.92     2*
   4800  1.676  1.14    0.91     2*
   5200  1.750  1.148   0.9      2*
   5600  1.810  1.155   0.89     2*
  /

PMAX
 6000  /
```

```
RPTPROPS
 'SOF2'   'SWFN'   'SGFN'   'PVTO'
 'PVTW'   /
```

SOLUTION

```
SOLUTION  =========================================================
EQUIL
-- Dep   Pres  Dwoc  Pcwoc  Dgoc  Pcgoc  Rsvd  Rvvd  Fipc
   3600  3600  3700   0.0   3590   0.0    1      0     0  /

RSVD
-- Dep   Rs
   3500  1.41
   3750  1.41  /

RPTSOL
 'PRES'  'SOIL'  'SWAT'  'SGAS'  'RS'  /
```

SUMMARY

```
SUMMARY   =========================================================
FOPR
FGPR
FWPR
FOPT
FGPT
FWPT
FWCT
FGOR

WBHP
'PROD' /

CPR
'PROD'   6 5 1 /
/

FPR
WIR
FWIT
RUNSUM
```

SCHEDULE

```
SCHEDULE   =========================================================
RPTSCHED
 'WELLS=2'  'SUMMARY=2'  'WELSPECS'  /
```

Specify producer and injector

```
WELSPECS
-- Name  Group  I  J  Dbh   Phase
  'PROD'  'G'   8  5  3600  'OIL'  /
  'INJ'   'G'   8  5  3725  'WAT'  /
/
```

Specify well connections

```
COMPDAT
-- Name   I  J  K1-K2  Op/Sh  SatTab  ConFact  Diam  KH S D  Dirn
  'PROD'  2  5  1  1  'OPEN'   2*              0.375   3*     'X'  /
  'PROD'  3  5  1  1  'OPEN'   2*              0.375   3*     'X'  /
  'PROD'  4  5  1  1  'OPEN'   2*              0.375   3*     'X'  /
```

```
  'PROD'  5  5  1  1  'OPEN'   2*             0.375   3*    'X'  /
  'PROD'  6  5  1  1  'OPEN'   2*             0.375   3*    'X'  /
  'PROD'  7  5  1  1  'OPEN'   2*             0.375   3*    'X'  /
  'PROD'  8  5  1  1  'OPEN'   2*             0.375   3*    'X'  /
  'INJ'   1  5  6  6  'OPEN'   1*      243.4  /
  'INJ'   2  5  6  6  'OPEN'   1*      243.4  /
  'INJ'   3  5  6  6  'OPEN'   1*      243.4  /
  'INJ'   4  5  6  6  'OPEN'   1*      243.4  /
  'INJ'   5  5  6  6  'OPEN'   1*      243.4  /
  'INJ'   6  5  6  6  'OPEN'   1*      243.4  /
  'INJ'   7  5  6  6  'OPEN'   1*      243.4  /
  'INJ'   8  5  6  6  'OPEN'   1*      243.4  /
  'INJ'   9  5  6  6  'OPEN'   1*      243.4  /
/
```

Specify the friction data for the producer.

```
WFRICTN
-- Name   Diam   Roughness
  'PROD'  0.375  3.75E-4  /
-- I J K  Tlen1  Tlen2
   8 5 1   0.0    300  /
   7 5 1   300    600  /
   6 5 1   600    900  /
   5 5 1   900   1200  /
   4 5 1  1200   1500  /
   3 5 1  1500   1800  /
   2 5 1  1800   2100  /
/
```

Specify the production well constraints

```
WCONPROD
-- Name   Op/Sh  CtlMode  Orat  Wrat  Grat  Lrat  Resv  Bhp
  'PROD'  'OPEN'  'LRAT'   3*               9000  1*    1500  /
/
```

Specify the injection well constraints

```
WCONINJE
-- Name  Phase   Op/Sh  CtlMode  Srat  Resv  Bhp
  'INJ'  'WAT'  'OPEN'  'RATE'   6000  1*    5000  /
/
```

Set 50 day maximum timestep and increase linear iteration limit to 40

```
TUNING
1 50 /
/
2* 40 /
```

Timesteps and reports

```
TSTEP
 4*100  /
RPTSCHED
 'PRES'  'SWAT'  'SGAS'  'FIP'  'WELLS=2'
 'SUMMARY=2'  'CPU=2'  'WELSPECS'  /
TSTEP
 100  /
RPTSCHED
 'WELLS=2'  'SUMMARY=2'  'WELSPECS'  /
TUNING
25 25 /
/
2* 50 /
TSTEP
 2*100  /
```

```
TUNING
50 50 /
/
2* 50 /
TSTEP
 2*100  /
RPTSCHED
 'PRES'  'SWAT'  'SGAS'  'FIP'  'WELLS=2'
 'SUMMARY=2'  'CPU=2'  'WELSPECS'  /
TSTEP
 100  /
RPTSCHED
 'WELLS=2'  'SUMMARY=2'  'WELSPECS'  /
TSTEP
 4*100  /
RPTSCHED
 'PRES'  'SWAT'  'SGAS'  'FIP'  'WELLS=2'
 'SUMMARY=2'  'CPU=2'  'WELSPECS'  /
TSTEP
 100  /
END
```